GENERAL PHYSICS 1
(PHY 101)

Produced by Academic Staff

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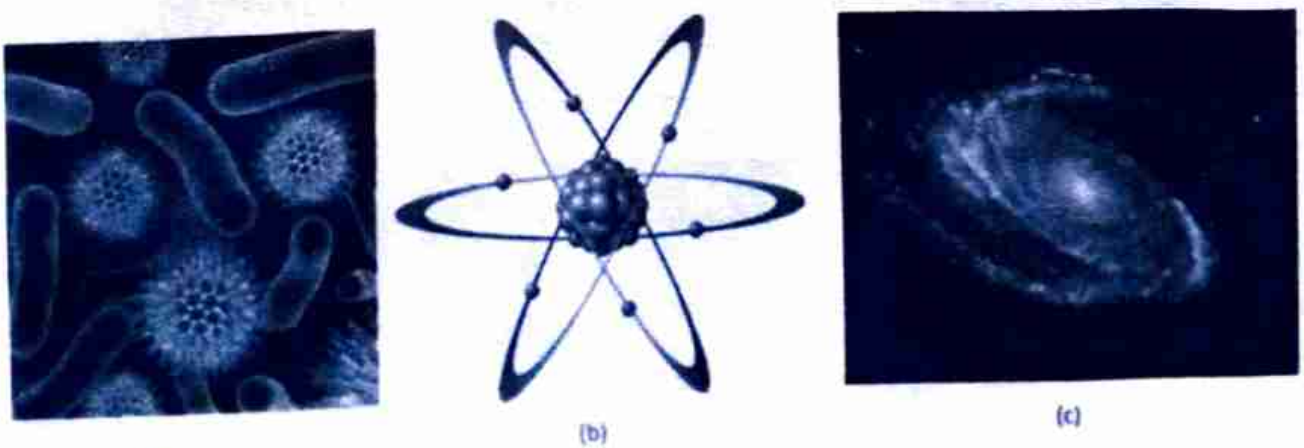
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1. Dimensions, units and conversions

1.1 Units and dimension

Physics in raw terms is the study of everything around us. In your high school, you were taught that Physics is the study of matter and its properties. Matter simply put, is anything that has mass and can occupy space.The goals of Physics or a Physicist is to quantify matter and its characteristics. Physics therefore is based on measurement of physical quantities. In all cases, these measurements are relative and are derived by comparing the output values with know standards.

MEASUREMENT is the determination of the size/magnitude of something. By comparing that unknown quantity with some standard quantity of equal nature, known as measurement unit. Examples are: length, mass, time, electric current, magnetic field, temperature, pressure etc. All physical quantities have dimensions: *dimensions are basic types of quantities that can be measured or computed*. Figure 1.1 shows examples of physical quantities that can be measure.



(a) (b) (c)

Figure 1.1: Picture showing (a) Micro-Organism.(b) an atomic shell with electrons revolving round the nucleolus with dimensions in the range $1.0 \times 10^{-9}m$ and (c) the milky way galaxy consisting of many stars with a dimension of 1 light year $= 9.4 \times 10^{15}m$

Most micro-organism are in the micrometer range 10^{-6}, and all microbes are made of atoms which are in the nanometer range 10^{-9}. The material universes made up of planets, sun and galaxies, and there distances relative to our planet earth can be measured in millions of kilometers or even light years ($9.4 \times 10^{15} m$.

There is a difference between dimensions and units. A dimension is a measure of a physical variable (without numerical values), while a unit is a way to assign a number or measurement to that dimension. For example, length is a dimension, but it is measured in units of feet (ft) or meters (m).

1.2 Basic Dimensions and Units

To quantify matter, it must be dimensioned. In your First year physics program, you will be introduced to three basic dimensions, namely:

Length [L]
Mass [M]
Time [T]

If we are to report the results of a measurement to someone who wishes to reproduce this measurement, a standard must be defined. For example, Figure 1.2 shows a rod of unknown length and a reference length called **Fuoye** that is used as a standard in FUOYE. A likely question is, how long is the length of the unknown rod when measured with FUOYE standard. The unknown rod is $3 - Fuoye$ long. The FUOYE standard mentioned above is used as a unit for measurement. A

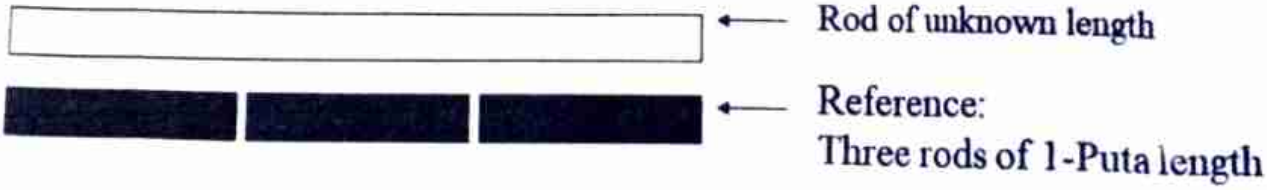


Figure 1.2: Showing the use of standard of length measurement

unit is a quantitative magnitude of a dimension or simply put a standard amount of a dimensional quantity. Unites by themselves don't make sense. To make sense out of any measurement, both number and units must be specified.

1.3 Standards of Length, Mass, and Time

There are seven primary dimensions. Primary (sometimes called basic) dimensions are defined as independent or fundamental dimensions, from which other dimensions can be obtained. The

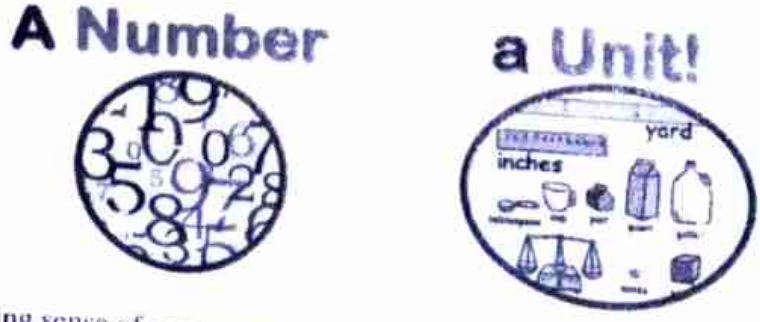


Figure 1.3: Making sense of a measurement requires that yu specify both number and unit

primary dimensions are: mass, length, time, temperature, electric current, amount of light, and amount of matter. For most mechanical and thermal science analysis, however, only the first four of these are required. The others will not be of concern to most mechanical engineering analysis. In order to assign numbers to these primary dimensions, primary units must be assigned. These are listed in the table below for the three unit systems:

Primary Dimension	Symbol	SI unit	BG unit	English unit
mass	m (sometimes M)	kg (kilogram)	slug	lbm (pound-mass)
length	L (sometimes l)	m (meter)	ft (foot)	ft (foot)
time	t (sometimes T)	s (seconds)	s (seconds)	s (seconds)
temperature	T (sometimes °C)	K (Kelvin)	R (degree Rankine)	Rankine
electric current	I (sometimes i)	A (ampere)	A (ampere)	A (ampere)
amount of light	C (sometimes I)	c (candela)	c (candela)	c (candela)
amount of matter	n or N (sometimes mu)	mol (mole)	mol (mole)	mol (mole)

Table 1.1: Table showing basic units used in measurement

For the purpose of these courses we shall focus on the International System of Units (SI units, from Le Systeme International d'Unites, more commonly simply called metric units). All other dimensions can be **derived** as combinations of these seven primary dimensions. These are called *secondary dimensions*, with their corresponding secondary units. A few examples are given in the table below.

The laws of physics are expressed in terms of basic quantities that require a clear definition. In mechanics, the three basic quantities are length (L), mass (M), and time (T). All other quantities in mechanics can be expressed in terms of these three. in the following subsections we discuss briefly these three quantities.

Secondary Dimension	Symbol	SI unit
force	F (sometimes f)	N (Newton $=kgm/s^2$)
acceleration	a	m/s^2
pressure	p or P	N/m^2, i.e. Pa (Pascal)
energy	E (sometimes e)	J (Joule $=N.m$)
Power	P	W (watt $= J/s$)

Table 1.2: Table showing derived units used in measurement

Length

Many systems for measuring length have been developed over the years. In A.D. 1120 the king of England decreed that the standard of length in his country would be named the yard and would be precisely equal to the distance from the tip of his nose to the end of his outstretched arm. In 1960 the length of the meter was defined as the distance between two lines on a specific platinum–iridium bar stored under controlled conditions in France. This standard however has limited accuracy relative to the precision expected to meet the current requirements of science and technology. In October 1983, the basic SI unit the meter (m) was redefined as the distance traveled by light in vacuum during a time of 1/299792458 second. In effect, this latest definition establishes that the speed of light in vacuum is precisely 299792458 m per second.

Mass

The basic SI unit of mass, the kilogram (kg), is defined as the mass of a specific platinum–iridium alloy cylinder kept at the International Bureau of Weights and Measures at Sèvres, France. This mass

standard was established in 1887 and has not been changed since that time because platinum–iridium is an unusually stable alloy

Time

In 1967 the SI unit of time, the second, was redefined using the characteristic frequency of a particular kind of cesium atom as the "reference clock." The basic SI unit of time, the second (s), is defined as 9192631770 times the period of vibration of radiation from the cesium-133 atom.

1.4 Building block of matter

Now that we know how to specify units in a measurement, we like to apply what we have learned. A simple definition of Physics state that it is "the study of matter, energy, and the interaction between them". Physics is concern also about what constitutes matter and how the constituents affect their properties.

Matter is a substance that has inertia and occupies physical space. According to modern physics, matter consists of various types of particles, each with mass and size. The most familiar examples of material particles are the electron, the proton and the neutron. Combinations of these particles form atoms. There are more than 100 different kinds of atoms, each kind constituting a unique chemical element. A combination of atoms forms a molecule. Atoms and/or molecules can join together to form a compound.

Matter can exist in several states, also called phases. The three most common states are known as solid, liquid and gas. A single element or compound of matter might exist in more than one of the three states, depending on the temperature and pressure.

Following the discovery of the nucleus in 1911, a model was developed in which each atom is made up of electrons surrounding a central nucleus. The accumulation of billions of Gold atoms for example through metallic bonding results is the bulk Gold, as shown in the Figure 1.4.

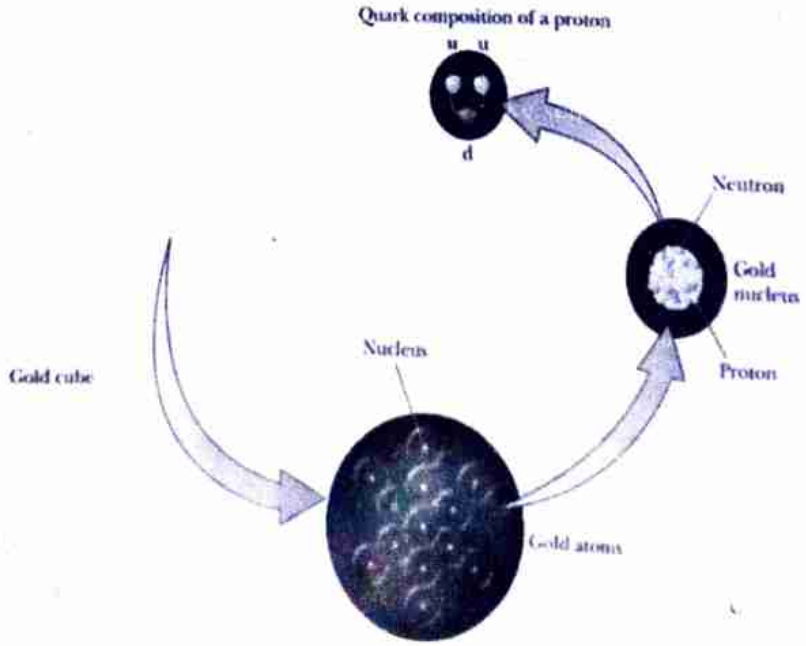


Figure 1.4: Levels of organization in matter. Ordinary matter consists of hundreds of atoms combined together by a combination of forces

1.4.1 Density and Atomic Mass

A property of any substance is its density ρ (Greek letter rho), defined as the amount of mass contained in a unit volume, which we usually express as mass per unit volume:

$$\rho = \frac{M}{V} \tag{1.1}$$

For example, aluminum has a density of $2.70 g/cm^3$, and lead has a density of $11.3 g/cm^3$. Therefore, a piece of aluminum of volume $10.0 cm^3$ has a mass of $27.0 g$, whereas an equivalent volume of lead has a mass of $113 g$. The difference in density between aluminum and lead is due, in part, to their different atomic masses. The atomic mass of an element is the average mass of one atom in a sample of the element that contains all the element's isotopes. The unit for atomic mass is the atomic mass unit (u), where $1u = 1.660540210^{-27} kg$. The atomic mass of lead is $207u$, and that of aluminum is $27.0u$.

The mass of a nucleus is measured relative to the mass of the nucleus of the carbon-12 isotope, often written as $12C$. One mole (mol) of a substance is that amount of the substance that contains as many particles (atoms, molecules, or other particles) as there are atoms in $12g$ of the carbon-12 isotope. One mole of substance A contains the same number of particles as there are in 1 mol of any other substance B. For example, 1 mol of aluminum contains the same number of atoms as 1 mol of lead. Experiments have shown that this number, known as Avogadro's number, A, is $A = 6.0221371023$ particles/mol. Avogadro's number is defined so that 1 mol of carbon-12 atoms has a mass of exactly $12g$. In general, the mass in 1 mol of any element is the element's atomic mass expressed in grams.

▪ **Example 1.1** The density of a cube:

The mass of a solid cube is $856g$, and each edge has a length of $5.35cm$. Determine the density ρ of the solid in SI unit?

Solution

Because $1g = 1 \times 10^{-3} kg$ and $1cm = 1 \times 10^{-2} m$, the mass m and volume V in SI unit are

$m = 856g \times 10^{-3} kg/g = 0.856kg$

$V = L^3 = (5.35cm \times 10^{-2}/cm)^3 = (5.35)^3 \times 10^{-6} m^3 = 1.53 \times 10^3 m^3$

$$\rho = \frac{M}{V} = \frac{0.856kg}{1.53 \times 10^3 m^3} = 5.59 \times 10^3 kg/m^3$$

▪

1.5 Dimensional Analysis

Dimensional analysis is the analysis of the relationships between different physical quantities by identifying their primary quantities (such as length, mass, time, and electric charge) and units of measure (kilometers, kilograms, hour, ampere) and tracking these dimensions as calculations or comparisons are performed. Converting from one dimensional unit to another is often somewhat complex.

Most physical quantities can be expressed in terms of combinations of five basic dimensions. These are mass (M), length (L), time (t), electrical current (I), and temperature, represented by the Greek letter theta (θ). These five dimensions have been chosen as being basic because they are easy to measure in experiments.

In this book to specify the dimensions we shall often use brackets [ ] to denote the dimensions of a physical quantity. For example, the symbol we use for speed is v, and in our notation the dimensions of speed are written $[v] = L/T$. As another example, the dimensions of area A are $[A] = L^2$. Table 1.3 shows the dimensions and units of area, volume, speed, and acceleration

quantity	dimension	SI units
area	$[A] = L^2$	m^2
volume	$[V] = L^3$	m^3
velocity	$[v] = L/T$	m/s
acceleration	$[a] = L/T^2$	m/s^2
mass	$[m] = M$	kg

Table 1.3: Table showing derived units used in measurement

1.6 Dimensional Analysis procedure

How do dimensions behave in mathematical formula?
In carrying out a dimensional analysis procedure, the following rules should be applied to arrive at a dimensionally correct unit.

Theorem 1.6.1 All terms that are added or subtracted must have same dimensions
If $D = A + B + C$ then A, B and C have the same dimension.

▪ **Example 1.2** Consider the addition of two measured values of length $40m + 11cm =$?
The above expression yields
(a) $40.11m$
(b) $4011cm$
(c) A and B
(d) Impossible to evaluate (dimensionally invalid)
Solution
To evaluate the above correctly, you must convert the value into the same units, either in the unit m or cm as both are acceptable unit of length.
(a) $40m + 0.11m = 40.11m$ correct?
or
(b) $4000cm + 11cm = 4011cm$ correct?
▪

Now consider the following exercise and show that they are impossible to evaluate (dimensionally invalid)

Exercise 1.1 Consider the following measurement $1.5m - 3.0kgm/s =$? The above expression yields:
(a) $-1.5m$
(b) $-1.5kgm^2$
(c) $-1.5kg$
(d) Impossible to evaluate (dimensionally invalid)
According to Rule 1 for this evaluation to be dimensionally correct, the value must be in the same units.
▪

Theorem 1.6.2 Dimensions obey rules of multiplication and division

$$D = \frac{A \times B}{C} = \frac{\frac{[M]}{[T^2]}\frac{[T^2]}{[L]}}{\frac{M}{L^2}} = [L].$$

To illustrate this procedure, the equation for the position x of a car at a time t if the car starts

from rest and moves with constant acceleration a is given by $x = \frac{1}{2}at^2$.
For the equation to be dimensionally correct, the dimension of the right hand side must be equal to the left hand side, hence the quantity on the right side must also have the dimension of length. That is, the dimensional form of the equation $\frac{1}{2}at^2$ is

$$L = \frac{L}{T^2} \times T^2 = L \tag{1.2}$$

A more general procedure using dimensional analysis is to set up an expression of the form

$$x \propto a^n t^m \tag{1.3}$$

where n and m are exponents that must be determined and the symbol $\propto$ indicates a proportionality. This relationship is correct only if the dimensions of both sides are the same. That is

$$[a^n t^m] = [L] = [L^1][T^0] \tag{1.4}$$

Because the dimensions of acceleration are $[L/T^2]$ and the dimension of time is $[T]$, we have

$$(L/T^2)^n T^m = L^1 T^0 \tag{1.5}$$

The exponents of L and T must be the same on both sides of the equation. Thus we have

$$n = 1 \qquad\qquad m + 2n = 0 \tag{1.6}$$

which we can solve simultaneously to give $n = 1$ and $m = 2$.

▪ **Example 1.3** Finding products of two measured values $1.5m \times 3.0kg = ?$
The above expression yields:
(a) $4.5mkg$
(b) $4.5gkm$
(c) A or B
(d) Impossible to evaluate (dimensionally invalid)

Solution
From Rule 2 the dimensional analysis will yield
$1.5[L] \times 3.0[M]$
$= 4.5[LM] = 4.5mKg$

▪

Theorem 1.6.3 Rule 3. In scientific equations, the arguments of "transcendental functions" must be dimensionless.
For example the expressions:
$A = ln(x)$; $B = sin(x)$
$C = exp(x)$; $D = 3^x x$ must be dimensionless
Exception - In engineering correlations, the argument may have dimensions.

▪ **Example 1.4** Consider the Volume of the frustum of a right pyramid with a square base shown below and given by
Solution

$$V = \frac{h}{3}B^2 + Bb + b^2$$

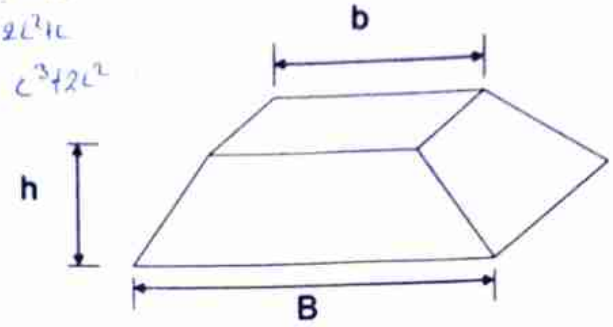
For the expression to be dimensionally correct then the right hand side of the equation should have the same dimension as the left hand side. Now convince yourself that this expression is dimensionally correct.

$$[L^3] = \frac{[L]}{1}[L^2] + [L \times L] + [L^2] = [L^3]$$

$L^3 + L^2 + L^2$

$2L^3 + L$

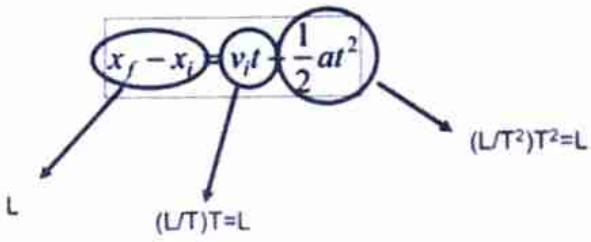
$L^3 + 2L^2$



■ **Example 1.5** Check the equations below with dimensional analysis:

$$x_f - x_i = v_i t - \frac{1}{2}at^2$$

Solution



Procedure

- Each term must have same dimension
- Two variables can not be added if dimensions are different
- Multiplying variables is always fine
- Numbers (e.g. $1/2$ or π) are dimensionless using the above process you should have

Exercise 1.2 Check the equation for dimensional consistency

$$mgh = \frac{mc^2}{1-(v/c)^2} - mc^2$$

Here, m is a mass, g is an acceleration, c is a velocity, h is a length

Exercise 1.3 Consider the equation with dimensional consistency:

$$m\frac{v^2}{r} = G\frac{Mm}{r^2}$$

Where m and M are masses, r is a radius and v is a velocity.
What are the dimensions of G ?
Show that G can be dimensioned with $\frac{L^3}{MT^2}$ ■

Exercise 1.4 Given x has dimensions of distance, u has dimensions of velocity, m has dimensions of mass and g has dimensions of acceleration. Is this equation dimensionally valid?

$$x = \frac{4/3ut}{1 - 2gt^2/x}$$

Similarly consider if the equation below is dimensionally valid?

$$x = \frac{vt}{1 - mgt^2}$$

■

1.7 Unit Conversion

Sometimes it is necessary to convert units from one measurement system to another, or to convert within a system, for example, it is always desired to carry out calculations in SI units, so if your unit of measurement of length is in kilometers you will need to meters. Units can be treated like algebraic quantities that can cancel each other out. One can measure the same quantity in different units. For instance distance can be measured in miles, kilometers, meters etc. Velocity can be measured in km/hour, m/s etc. Tables** to ** shows some basic conversion you must be familiar with in this course as you will need then for various calculations.

Conversion Factors

Length

	m	cm	km
1 meter	1	10^2	10^{-3}
1 centimeter	10^{-2}	1	10^{-5}
1 kilometer	10^3	10^5	1
1 inch	2.540×10^{-2}	2.540	2.540×10^{-5}
1 foot	0.304 8	30.48	3.048×10^{-4}
1 mile	1 609	1.609×10^5	1.609

Mass

	kg	g
1 kilogram	1	10^3
1 gram	10^{-3}	1
1 slug	14.59	1.459×10^4
1 atomic mass unit	1.660×10^{-27}	1.660×10^{-24}

***Note*: 1 metric ton = 1 000 kg.**

Time

	s	min	h	day	yr
1 second	1	1.667×10^{-2}	2.778×10^{-4}	1.157×10^{-5}	3.169×10^{-8}
1 minute	60	1	1.667×10^{-2}	6.994×10^{-4}	1.901×10^{-6}
1 hour	3 600	60	1	4.167×10^{-2}	1.141×10^{-4}
1 day	8.640×10^4	1 440	24	1	2.738×10^{-3}
1 year	3.156×10^7	5.259×10^5	8.766×10^3	365.2	1

Figure 1.5: Showing some common conversion of units

1.8 Tutorial 1

Problem 1.1 How many grams of copper are required to make a hollow spherical shell having an inner radius of $5.70cm$ and an outer radius of $5.75cm$? The density of copper is $8.92g/cm^3$.

Problem 1.2 A small cube of iron is observed under a microscope. The edge of the cube is $5.00 \times 10^{-6}cm$ long. Find (a) the mass of the cube and (b) the number of iron atoms in the cube. The molar mass of iron is $55.9g/\text{mol}$, and its density is $7.86g/cm^3$.

Problem 1.3 The radius r of a circle inscribed in any triangle whose sides are a, b, and c is given by

$$r = [(s-a)(s-b)(s-c)/s]^{(1/2)}$$

where $s = (a+b+c)/2$. Check this formula for dimensional consistency.

Problem 1.4 The displacement of a particle moving under uniform acceleration is some function of the elapsed time and the acceleration. Suppose we write this displacement $s = ka^m t^n$ where k is a dimensionless constant. Show by dimensional analysis that this expression is satisfied if $m = 1$ and $n = 2$. Can this analysis give the value of k?

Problem 1.5 Suppose your hair grows at the rate $1/32cm.$ per day. Find the rate at which it grows in nanometers per second. Since the distance between atoms in a molecule is on the order of $0.1nm$, your answer suggests how rapidly layers of atoms are assembled in this protein synthesis.

Problem 1.6 A solid piece of lead has a mass of $23.94g$ and a volume of $2.10cm^3$. From these data, calculate the density of lead in SI units (kg/m^3).

About Author



Figure 1.6: *Prof. Odo Ekundare Ayodele*

Prof. Odo Ekundare Ayodele holds a B.Sc. (Hons) degree in Engineering Physics from the Obafemi Awolowo University, Ile-Ife and a M.Tech degree in Electronic Measurement and Instrumentation from the Federal University of Technology Akure and a PhD. Degree in Condensed matter/Solid State Physics with expertise in nano-semiconductor and device Physics from the University of Cape Town, South Africa.

Prof. Odo Ayodele was a DAAD scholar and have worked in reputable world facilities for materials research, like the The Advanced Light Source at Lawrence Berkeley National Laboratory USA, the Brazilian Synchrotron Light Laboratory (LNLS) for Research in Energy and Materials (CNPEM), in Campinas (SP), and the iThemba Laboratory for Accelerator Based Sciences (iThemba LABS) South Africa were he worked as a Post-Doctoral Research scientist in Applied Nuclear and Computational Physics.

Prof. Odo Ayodele has also visited the International Center for Theoretical Physics Trieste Italy on several occasion and has participated in more than six program on Advanced Instrumentation and reconfigurable devices using Field Programmable Gate Array (FPGA) and other embedded system devices as well as on Abi-initio Electronic structure calculations.

Prof Odo Ayodele, has two international Patents on the development and applications of semiconductor nano-inks and he is reputed for the development of the **Odo Spherical density algorithm** for the analysis and visualization of A positron emission tomography (**PET**) and positron emission particle tracking (**PEPT**) data.

2. Kinematics, displacement, time and velocity

2.1 Definition of Kinematics

This is the branch of classical mechanics which describes the motion of points, bodies (objects) and systems of bodies (group of objects) without consideration of the causes of motion (i.e. forces). Kinematic as a field of study is often referred to as "geometry of motion". Classical mechanics is the study of motion of bodies (including a special case in which the bodies remain at rest). This is in accordance with the general principles first enunciated by Sir Isaac Newton on the Philosophiae Naturalis Principia mathematical (1687) commonly known as the Principia.
Converting between the Cartesian and Polar Coordinates X and Y are used in representing the Cartesian Coordinates while r and θ values are used for the Polar Coordinates. This can be further explained by the following equations with pictorial representation;

$$sin\theta = \frac{Opposite}{Hypotenuse} = \frac{y}{r} \tag{2.1}$$

$$cos\theta = \frac{Adjacent}{Hypotenuse} = \frac{x}{r} \tag{2.2}$$

$$tan\theta = \frac{Opposite}{Adjacent} = \frac{y}{x} \tag{2.3}$$

$$r^2 = x^2 + y^2 \tag{2.4}$$

$x(t)$ is called displacement and it denotes the position of a body at time. If the displacement is positive then that body is to the right of the chosen origin and if negative, then it is to the left.

▪ **Example 2.1** The Cartesian coordinates of a point xy planes are $(x,y) = (3.50m, 2.50m)$, Find the polar coordinates of the point.

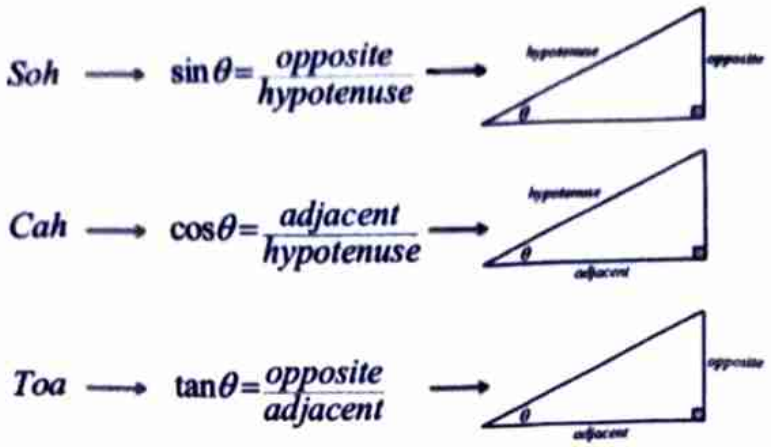


Figure 2.1: Showing the use of trigonometry properties

Solution
From Equation 2.4 $r^2 = x^2 + y^2$

$$r^2 = (3.5)^2 + (2.5)^2$$

$$r^2 = 12.25 + 6.25 = 18.51m$$

$$r = 4.3m$$

For the Angle θ, using

$$Tan\theta = \frac{y}{x} = \frac{2.5}{3.5} = \arctan 0.714 = 35.5^o$$

■

2.2 Motion in one dimension

The study of the causes of motion is called dynamics while the study of motion itself without any concern to the cause is called kinematics. It can be 1- Dimensional, 2- Dimensional or 3 -Dimensional. The displacement of an object is defined as its change in position.

$$\Delta x \equiv x_f - x_i \tag{2.5}$$

x_f denotes the final position of the object
x_i denotes the initial position of the object
Δx denotes change

2.3 Velocity

Average Speed of an object is the length of path it travels divided by the total elapse time. Average Speed = Path Length/elapse time.

$$Averagespeed = \frac{path - length}{elapse - time} \tag{2.6}$$

If a body is moving with average speed v then in time t it will cover a distance $d = vt$. But, in fact, the speed of a car changes from time to time and so one should limit the use of this formula to

small time differences only. So, more accurately, one defines an average speed over the small-time interval Δt as:

$$v = \frac{d}{\Delta t} \tag{2.7}$$

The average velocity of an object is defined as the displacement of an object divided by the total time Average Velocity = displacement/elapse time

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i} \tag{2.8}$$

Where t_f and t_i represents the final and initial time respectively.
Average velocity can either be positive or negative (Average speed is always +ve) because we are in one dimension and the sign tells us the direction. Negative acceleration is called deceleration. The speed of a decelerating body decreases with time.
Some students are puzzled by the fact that a body can have a very large acceleration but can be standing still at a given time. In fact, it can be moving in the opposite direction to its acceleration. There is actually nothing strange here because position, velocity, and acceleration are independent quantities. This means that specifying one does not specify the other. For constant speed and a body that is at $x = 0$, at time $t = 0$, x increases linearly with time,$x \alpha t$ (or $x = vt$). If the body is at position x_0 at time $t = 0$, then at time t, $x = x_o + vt$.

Δt

2.4 Acceleration

The change of an object's velocity is called acceleration.

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i} \tag{2.9}$$

For constant acceleration and a body that starts from rest at $t = 0$, v increases linearly with time. $v \propto t$ (or $v = at$). If the body has speed v_0 at $t = 0$, then at time t, $v = v_o + at$

2.5 Instantaneous Acceleration

This is defined by taking the limit as the end point gets closer and closer.

$$a \equiv \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} \tag{2.10}$$

$$v = v_o + at \tag{2.11}$$

Where $v_i = v_o$ while $v_f = v$

2.6 The Average Velocity

The average velocity could be calculated using the following equations;

$$v = \frac{v_o + v}{2} \tag{2.12}$$

$$v^2 = v_o^2 + 2a\Delta x \tag{2.13}$$

$$v^2 = v_o^2 + 2as \tag{2.14}$$

$$\Delta x = v_o t + \frac{1}{2}at^2 \tag{2.15}$$

$$s = ut + \frac{1}{2}at^2 \tag{2.16}$$

Vectors: a quantity that has a size as well as direction is called a vector. So, for example, the wind blows with some speed and in some direction. So, the wind velocity is a vector.
If we denote a vector $r = (x, y)$ then $r_x = x = r\cos\theta$ and $r_y = y = r\sin\theta$. Note that $x^2 + y^2 = r^2$. Also that $\tan\theta = \frac{y}{x}$

Worked Examples

■ **Example 2.2** An airplane accelerates down a runway at $3.20m/s^2$ for $32.8s$ until is finally lifts off the ground. Determine the distance traveled before takeoff.
Solution
$a = +3.2m/s^2$; $t = 32.8s$; $vi = 0m/s$; d = ?
$d = v_i t + 0.5at^2$
$d = ox32.8 + 0.5 \times 3.20 \times (32.8)^2$
$d = 1720m$
■

■ **Example 2.3** A car starts from rest and accelerates uniformly over a time of 5.21 seconds for a distance of $110m$. Determine the acceleration of the car.
Solution
$d = 110m$; $t = 5.21s$; $v_i = 0m/s$; a = ?
$d = v_i t + 0.5at^2$
$110 = 0 \times 5.21 + 0.5a \times (5.21)^2$
$110 = a \times (13.57)^2$

$a = 110/13.57$
$a = 8.10m/s^2$

■

■ **Example 2.4** A race car accelerates uniformly from $18.5m/s$ to $46.1m/s$ in 2.47 seconds. Determine the acceleration of the car and the distance traveled.
Solution
$vi = 18.5m/s$; $v_f = 46.1m/s$; $t = 2.47s$; $d = ?$; $a = ?$
$a = \delta v/t$
$a = (46.1 - 18.50))/2.47$
$a = 11.2m/s^2$

$$v = u + at$$

$$88.3 = 0 + (2.86)t$$

$$t = \frac{(88.3)}{2.86}$$

$$t = 30.8\ s$$

■

■ **Example 2.8** A bullet is moving at a speed of $367m/s$ when it embeds into a lump of moist clay. The bullet penetrates for a distance of $0.0621m$. Determine the acceleration of the bullet while moving into the clay. (Assume a uniform acceleration.)

Solution

$v_i = 367m/s; \quad v_f = 0m/s; \quad d = 0.0621m; \quad a = ?$

$$v^2 = u^2 + 2ad$$

$$(0)^2 = (367)^2 + 2a(0.0621)$$

$$a = \frac{(-134689)}{0.1242}$$

$$a = -1.08x10^6 m/s^2$$

(The - sign indicates that the bullet slowed down.)

■

■ **Example 2.9** A car traveling at $22.4m/s$ skids to a stop in $2.55s$. Determine the skidding distance of the car (assume uniform acceleration).

Solution

$v_i = 22.4m/s; \quad v_f = 0m/s; \quad t = 2.55s; \quad d = ?$

$$a = \frac{(v+u)}{2}t$$

$$a = \frac{(22.4+0)}{2}X2.55$$

$$a = 28.6m$$

■

■ **Example 2.10** With what speed in miles/hr (1 m/s = 2.23 mi/hr) must an object be thrown to reach a height of $91.5m$ (equivalent to one football field)? Assume negligible air resistance.

Solution

$a = -9.8m/s^2; \quad v_f = 0m/s; \quad d = 91.5m; \quad v_i = ?; \quad t = ?$

First, find speed in units of m/s:

$$v^2 = u^2 + 2ad$$

$$(0)^2 = u^2 + 2(-9.8)(91.5)$$

$$0 \quad u^2 \quad 1793$$

$$u^2 \quad 1793 m/s^2$$

$$u \quad 42.3 m/s$$

Now convert from m/s to mi/hr:

$$u \quad 42.3(\tfrac{m}{s})X2.23(\tfrac{mi}{hr})$$

$$u = 94.4 mi/hr$$

■

■ **Example 2.11** If Michael Jordan has a vertical leap of $1.29m$, then what is his takeoff speed and his hang time (total time to move upwards to the peak and then return to the ground)?

Solution

$a = -9.8m/s^2$; $v_f = 0m/s$; $d = 1.29m$; $v_i = ?$; $t = ?$

$$v^2 = u^2 + 2ad$$

$$(0)^2 = u^2 + 2(-9.8)(1.29)$$

$$0 = u^2 - 25.28$$

$$u^2 = -25.2 \quad u = 5.03 m/s$$

To find hang time, find the time to the peak and then double it.

$$v = u + at$$

$$0 = 5.03 + (-9.8)t$$

$$-5.03 = (-9.8)t$$

$$t = \tfrac{(-5.03)}{(-9.8)}$$

$$t = 0.513s$$

hang time = 1.03 s

■

2.7 Tutorial 2

Problem 2.1 A train starts at rest, accelerates with constant acceleration a for 5 minutes, then travels at constant speed for another 5 minutes, and then decelerates with a. Suppose it travels a distance of 10km in all. Find a.

Problem 2.2 A ball is dropped from a height of $10m$. At the same time, another ball is thrown vertically upwards at an initial speed of $10m/sec$. How high above the ground will the two balls collide?

Problem 2.3 Find the resultant of the two velocity vectors and also, find the angle that the resultant makes with the first vector

About Author



Figure 2.2: *Prof. Olubosede Olusayo*

Prof. Olubosede Olusayo holds a B.Tech. (Hons) in Physics Electronics from Federal University of Technology, Akure. He had his M.Sc. (Theoretical Physics) from the University of Benin and bagged his Ph.D. (Condensed Matter Physics) from The Federal University of Technology, Akure. He started his career as a Graduate Assistant in 2001 and rose through the ranks and is presently a Senior Lecturer. Dr. Olubosede visited the International Center for Theoretical Physics, Trieste, Italy for summer school on Plasma Physics (2007). He has served and he is still serving at the university in various capacities. Notable amongst them was being the Acting Head of Physics Department between 2016-2018. .

3. Motion in One Dimension

3.1 Position, Time and Displacement

The particle's location is specified by its coordinate, which will be denoted by x or y. As the particle moves, its coordinate changes with the time, t. The change in position from x_1 to x_2 of the particle is the displacement Δx, with $\Delta x = x_2 - x_1$.

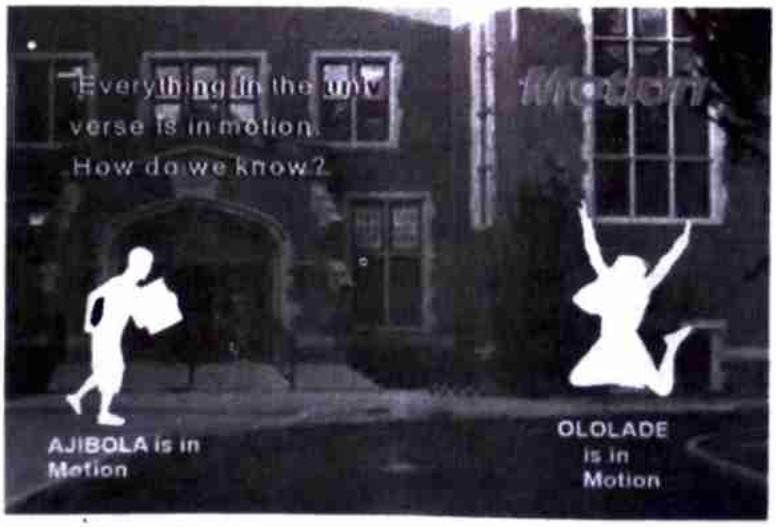


Figure 3.1: Schematic diagram of objects undergoing motion

3.2 Average Velocity and Average Speed

The average speed of the particle is defined as the total distance traveled per total time taken When a particle has a displacement Δx in a change of time Δt, its average velocity for that time interval is:

$$\frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1} \tag{3.1}$$

In general, the value of the average velocity for a moving particle depends on the initial and final times for which we have found the displacements

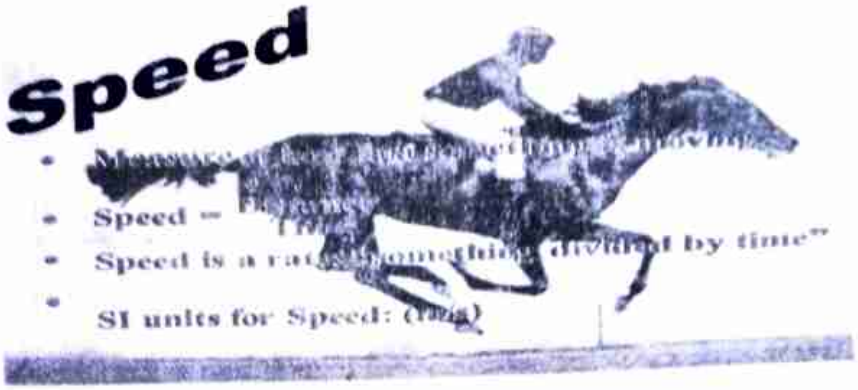


Figure 3.2: Schematic diagram of objects undergoing motion

3.3 Instantaneous Velocity and Speed

Instantaneous velocity or speed is defined as the actual velocity or speed attained at any given moment of time. We can answer the question "how fast is a particle moving at a particular time t?" by finding the instantaneous velocity. This is the limiting case of the average velocity when the time interval Δt include the time t and is as small as we can imagine:

$$v = \lim \frac{\Delta x}{\Delta t} \tag{3.2}$$

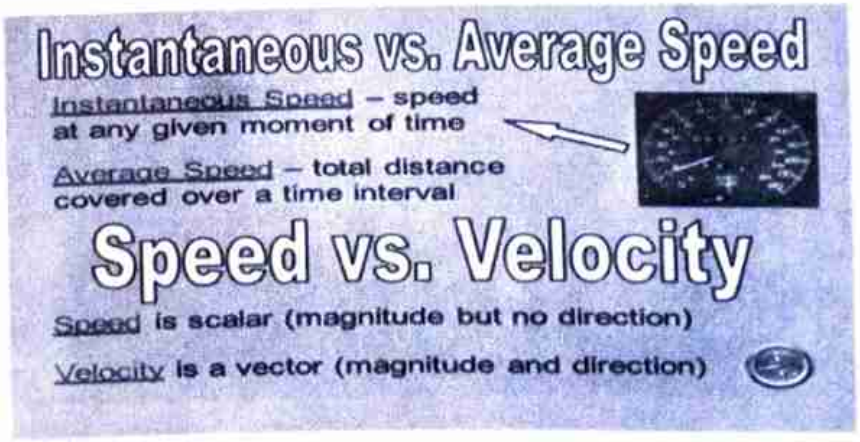


Figure 3.3: Illustration of instantaneous speed and velocity attainment

The instantaneous speed is the absolute value (magnitude) of the instantaneous velocity. If we make a plot of x vs. t for a moving particle, the instantaneous velocity is the slope of the tangent to the curve at any point.

3.4 Acceleration

When a particle velocity changes, then we say that the particle undergoes an acceleration. If a particle's velocity changes from v_1 to V_2 during the time interval t_1 to t_2 then we define the average

acceleration as:

$$\frac{v_2 - v_1}{t_2 - t_1} \tag{3.3}$$

Instantaneous acceleration is given by

$$a = \lim \frac{\Delta v}{\Delta t} \tag{3.4}$$

If the acceleration a is positive it means that the velocity is instantaneously increasing; if a is negative, then v is instantaneously decreasing. At times, we may have retardation or deceleration in a problem. This word is used when the sense of the acceleration is opposite that of the instantaneous velocity (the motion). Then the magnitude of acceleration is given, with its direction being understood.

3.5 Free Fall

An object tossed up or down near the surface of the earth has a constant downward acceleration of magnitude $9.8ms^{-2}$. This value is always denoted by g. Be very careful about the sign. In a coordinate system where the y axis points straight up, the acceleration of a freely-falling object is: $a_y = -9.8ms^{-2} = -g$.
Here we are assuming that the air has no effect on the motion of the falling object. For an object which falls for a long distance, this can be a bad assumption. Remember that an object in free-fall has an acceleration equal to $-9.8ms^{-2}$ while it is moving up and $9.8ms^{-2}$ while it is moving down.

3.6 Worked Examples

▪ **Example 3.1** If Olawale could routinely throw a fastball at an horizontal speed of $160kmhr^{-1}$. How long did the ball take to reach home plate $18.4m$ away?
Solution
We assume that the ball moves in a horizontal straight line with an average speed of 160 km/hr. Of course, in reality this is not quite true for a thrown baseball.
We are given the average velocity of the ball's motion and also a particular displacement, namely

$$\Delta x = 18.4m. \qquad v = \frac{\Delta x}{v}$$

But before using the above equation, it might be convenient to change the units of v. We have:

$$v = \frac{160 \times 1000}{60 \times 60} = 44ms^{-2}$$

Then we find:

$$\Delta t = \frac{\Delta x}{v} = \frac{18.4}{44} = 0.414s$$

The ball takes 0.414 seconds to reach home plate. ▪

▪ **Example 3.2** An electron moving along the x axis has a position given by $x = (16te^{-t})m$, where t is in seconds. How far is the electron from the origin when it momentarily stops?
Solution
To find the velocity of the electron as a function of time, take the first derivative of $x(t)$:

$$v = \frac{dx}{dt} = 16e^{-t} - 16te^{-t} = 16e^{-t}(1 - t)s$$

Now the electron "momentarily stops" when the velocity v is zero.

$$0 == 16e^{-t}(1-t)s$$

$$t = 1s$$

From our expression for v, we see that this occurs at $t = 1s$. At this particular time we can find the value of x:

$$x(1s) = 16(1)e^{-1}m = 5.89m$$

The electron was $5.89m$ from the origin when the velocity was zero. ■

■ **Example 3.3** (a) If the position of a particle is given by $x = 20t?5t^3$, where x is in meters and t is in seconds, when is the particle's velocity zero? (b) When is its acceleration a zero?

Solution

(a) We find $v(t)$ from $x(t)$:

$$v(t) = \frac{dx}{dt} = \frac{d}{dt}(20t - 5t^3) = 20 - 15t^2$$

where, if t is in seconds then v will be in ms^{-1} .The velocity v will be zero when
$20 - 15t^2 = 0$
Solve for t: $15t^2 = 20$

$$t^2 = \frac{20}{15} = 1.3$$

This gives: $t = \pm 1.15s$
(b) We find $a(t)$ from $v(t)$:

$$a(t) = \frac{dv}{dt} = \frac{d}{dt}(20 - 15t^2) = -30t$$

$$a(t) = -30(\pm 1.15)m/s$$

■

■ **Example 3.4** In an arcade video game a spot is programmed to move across the screen according to $x = 9.00t?0.750t^3$, where x is distance in centimeters measured from the left edge of the screen and t is time in seconds. When the spot reaches a screen edge, at either $x = 0$ or $x = 15.0cm$, t is reset to 0 and the spot starts moving again according to $x(t)$.

Solution

- (a) At what time after starting is the spot instantaneously at rest?
- (b) W here does this occur?
- (c) W hat is its acceleration when this occurs?
- (d) In what direction is it moving just prior to coming to rest?
- (e) In what direction is it moving Just after?
- (f) When does it first reach an edge of the screen after t = 0?

(a) This is a question about the instantaneous velocity of the spot.
To find $v(t)$ we calculate:

$$v(t) = \frac{dx}{dt} = \frac{d}{dt}\left(9.00t - 0.750t^3\right) = 9.00 - 2.25t^2$$

where this expression will give the value of v is in cm/s and t is given in seconds. We want to know the value of t for which v is zero, i.e. the spot is instantaneously at rest. We solve:

$$9.00 - 2.25t^2 = 0$$

$$t = \pm 2.00s$$

but since we are only interested in times after the clock starts at $t = 0$, we choose $t = 2.00s$.

(b) In this part, we are to find the value of x at which the instantaneous velocity is zero. In part (a) we found that this occurred at $t = 2.00s$ so we calculate the value of x at $t = 2.00s$:

$$x(2.00s) = 9.00 \times (2.00) - 0.750 \times (2.00)^3 = 12.0cm$$

(where we have filled in the units for x since centimeters are implied by the equation). The dot is located at $x = 12.0cm$ at this time. (And recall that the width of the screen is $15.0cm$.)

(c) To find the (instantaneous) acceleration at all times, we calculate:

$$a(t) = \frac{dv}{dt} = \frac{d}{dt}\left(9.00 - 2.25t^2\right) = -4.50t$$

where we mean that if t is given in seconds, a will be given in m^{-2} . At the time in question ($t = 2.0s$) the acceleration is:

$$a(t = 2.0s) = -4.50 \times (2.00) = -9.0$$

That is, the acceleration at this time is ?$9.0ms^{-2}$

(d) From part (c) we note that at the time that the velocity was instantaneously zero the acceleration was negative. This means that the velocity was decreasing at the time. If the velocity was decreasing yet instantaneously equal to zero then it has to be going from positive to negative values at $t = 2.00s$. So just before this time its velocity was positive.

(e) Likewise, from our answer to part (d) just after $t = 2.00s$ the velocity of particle had to be negative.

$$x = 0 = 9.0t - 0.750t^3$$

Factor out t to solve:

$$(9.0 - 0.750t^2) = 0$$

$$t^2 = \frac{9.00}{0.750} = 12.0$$

$$t = 3.46s$$

■

■ **Example 3.5** A jet plane lands with a velocity of $100ms^{-1}$ and can accelerate at a maximum rate of $-5ms^{-2}$ as it comes to rest. (a) From the instant it touches the runway, what is the minimum time needed before it stops? (b) Can this plane land at a small airport where the runway is $0.80km$ long?

Solution

(a) The minus sign in the acceleration indicates that the sense of the acceleration is opposite that of the motion, that is, the plane is decelerating. The plane will stop as quickly as possible if the acceleration does have the value $-5.0ms^{-2}$, so we use this value in finding the time t in which the velocity changes from $v_0 = 100ms^{-1}$ to $v = 0$

Substituting, we find:

$$t = \frac{v - v_0}{a} = \frac{0 - 100}{-5} = 20s$$

The plane needs $20s$ to come to a halt.

(b) The plane also travels the shortest distance in stopping if its acceleration is $5.0ms^{-2}$. With $x_0 = 0$, the plane's final x coordinate can be determined using $t = 20s$ obtained in part (a) above.

$$x = x + 1/2(v + v)t = 0 + 1/2(100 + 0)(20) = 1000m = 1.0km$$

The plane must have at least $1.0km$ of runway in order to come to a halt safely. Therefore, $0.80km$ is not sufficient.

■

■ **Example 3.6** The value of instantaneous acceleration for the particle whose position is given by $x = 2t^3 - 2t^2 + 3$ at $t = 2s$ is?

Solution

$$a(t) = \frac{dv}{dt} = \frac{dx^2}{dt^2}\left(2t^3 - 2t^2 + 3\right)$$

$$a(t) = 12t - 4$$

$$a(t) = 12(2) - 4 = 20ms^{-2}$$

■

■ **Example 3.7** The position of a particle is given by $x = 3t^3 - 2t^2 + 6$. What is its instantaneous velocity at $t = 3s$?

Solution

$$v(t) = \frac{dx}{dt} = \frac{d}{dt}\left(3t^3 - 2t^2 + 6\right)$$

$$v(t) = 9t^2 - 4t$$

$$v(t) = 9(3^2) - 4(3) = 69ms^{-1}$$

■

▪ **Example 3.8** The displacement of a body is described by the equation $x = Ct + Dt^2$, where $C = 3m/s$ and $D = 2m/s$, while x is in meters and t is in seconds. The displacement and the average velocity between the time interval $t = 3s$ and $t = 5s$ are respectively equal to:

Solution

(i) Displacement, $D = 3t^3 - 2t^2 + 6$

at $t = 1s$;

$$x_i = 3(1)^3 - 2(1)^2 + 6$$
$$= 3 - 2 + 6$$
$$= 5m$$

At $t = 3s$;

$$x_f = 3(3)^3 - 2(3)^2 + 6$$
$$= 81 - 18 + 6$$
$$= 69m$$

Displacement (at interval $t = 1s$ and $t = 3s$) $= x_f - x_i = 69m - 5m = 64m$.

(ii) Average velocity (at interval $t = 1s$ and $t = 3s$)

$$\frac{x_f - x_i}{t_f - t_i} = \frac{69 - 5}{3 - 1} = \frac{64}{2} = 32ms^{-1}$$

▪

▪ **Example 3.9** A car accelerates from $30m/s$ to $50m/s$ with an acceleration of 2.5 meter per second square. The displacement of the car will be:

Solution

$$v^2 = u^2 + 2as$$
$$50^2 = 30^2 + 2x2.5s$$
$$s = \frac{2500 - 900}{5}$$
$$s = \frac{1600}{5} = 320m.$$

▪

▪ **Example 3.10** A ball is dropped from a height of $45m$. The ball will land in what seconds?

Solution

$$s = ut + \frac{1}{2}at^2$$

$u = 0$ (The ball is at rest)

$$s = 0 \times t + \frac{1}{2}gt^2$$

($g = 10m/s^2$ acceleration due to gravity)

$$45 = 0 + \frac{1}{2} \times 10t^2$$
$$t^2 = \frac{90}{10} = 9$$
$$t = 3s$$

▪

3.7 Tutorial 3

Problem 3.1 An object initially was moving at a rate of 3m/s accelerates uniformly for 4 seconds. The displacement for the objects over this time is $24m$. The final velocity of the object is:

Problem 3.2 If a person runs one complete lap of a $400m$ oval track in 2 minutes, then:

A. The person's displacement is $400m$.

B The person's displacement is $0m$

C. The person's velocity is $200m/min$.

D. The person's velocity is $150m/min$.

Problem 3.3 How long will it take a car, starting from rest, accelerating at 2 meters per second square, to travel the same distance that another car traveling at a constant rate of $20m/s$ will travel?

Problem 3.4 A golf ball is released from the top of a very tall building. Neglecting air resistance, what is the position and velocity of the ball after $6.00s$ ($g = 9.8ms^2$)?

Problem 3.5 The position of a particle moving on an x-axis is given by: $x = 13.2 + 8.9t^2 + 1.2t^3$. What is the acceleration of the particle at time $t = 2.6s$?

Problem 3.6 A stone is thrown up from the top of a platform with an initial velocity of $19m/s$. If the top of the platform is taken to be the ground level. What is the time taken to reach the maximum height?

Problem 3.7 A lift is moving down with an acceleration of $3ms^{-2}$. A ball is released $1.7m$ above the lift floor. Assuming $g = 9.8ms^{-2}$, how long will the ball take to hit the floor?

Problem 3.8 An Airplane lands on the runway with a velocity of $50ms^{-1}$ and decelerates at $10ms^{-2}$ to a velocity of $20ms^{-1}$. The distance traveled on the runway is?

Problem 3.9 At what time is the particle at the position: $x = 3t^2 - 12t + 20$ at rest?

Problem 3.10 The position of a particle is given by $x = 3t^3 - 2t^2 + 6$. What is its instantaneous velocity at $t = 3s$?

Problem 3.11 The position of a particle is given by $x = 30t^2 - 5t^3$, where x is in meters and t is in seconds, at what time will the particle's acceleration be zero?

Problem 3.12 A mango fell from a height of $125m$. The mango will reach the ground in what seconds?

Problem 3.13 A bicycle accelerates uniformly from rest. At time t, the velocity of the bicycle is v. At what time will the bicycle have a velocity of $4v$?

Problem 3.14 A runner starts from rest and accelerates uniformly for $2s$. The total displacement for the runner over this time interval is $12m$. The final velocity of the runner is:

Problem 3.15 A car traveling at a velocity of y comes to rest in a displacement d. If a second car traveling at $2y$ wants to come to rest in a displacement $2d$, the ratio of acceleration of the first car to the acceleration of the second would be:

4. Kinematics, Newton's laws of motion

4.1 Introduction

These notes cover the fundamentals of kinematics (the study of motion without forces) and extend into dynamics, focusing on Newton's laws,inertial frames, impulses, forces (including action at a distance),conservation of momentum, and relative motion. The content is structured for clarity, with key concepts, equations, and explanations. Equations are derived step-by-step where relevant.

4.2 Kinematics: Description of Motion

Kinematics deals with the geometry of motion, describing position, velocity, and acceleration without considering causes (forces). It can also be defined as the dynamics dealing with the problems without referring to the forces causing the motion of the body. It uses vectors in one, two, or three dimensions.

The Aristotelian law says that external forces are necessary to keep a body in motion, but external forces are also necessary to overcome frictional forces in solids and viscous forces in fluids. These forces are always present in nature. The mechanics of the rigid bodies that deals with the bodies at rest is termed **Statics** and that which deals with bodies in motion is called ~~dynamics~~ Kinetic. The one referring to the forces causing motion is termed ~~Kinetics~~ dynamics.

Newton studied the Ideas of Galileo regarding the motion of an object. He formulated three fundamental laws that govern the motion of objects. These three laws are known as Newton's laws of motion, which are as given below:

4.3 Newton's First Law of Motion

Any object remains in the state of rest or in a uniform Motion along a straight line until it is compelled to change the state by applying an external force. This law as also known as Law of Inertia.

4.3.1 Applications of Newton's First Law of Motion

The following applications are based on Newton's first law of motion:

1. The passengers in a bus fall backward when it starts suddenly
2. This is because the sudden start of the bus brings motion to the bus as well as to our feet in contact with the floor of the bus but the rest of our body opposes this motion because of inertia, so they fall backwards.
3. When a carpet or a blanket is beaten with a stick, then the dust particles separate out from it.

4.4 Inertial frame

An inertial frame of reference is a framework in physics where an object at rest remains at rest and an object in motion continues in motion with constant velocity in a straight line, unless acted upon by an external force. In other words, Newton's first law of motion is valid within an inertial frame. Frames of reference that are accelerating are non-inertial, while those that are stationary or moving at a constant velocity are considered inertial. Earth's rotation is a non-inertial frame.

An inertial frame of reference is one that does not accelerate. It should be noted that for an object to remain at rest or maintain a constant velocity in an inertial frame, it must not be subjected to any net external force.

Examples include a stationary by stander, a car moving at constant velocity while the Earth's frame of reference is often approximated as an inertial frame. Newton's first law of motion is the law of inertia and it holds in an inertial frame and other fundamental principles of physics. Inertial frame serves as the standard reference point for the laws of physics. It allows scientists to predict and understand the behavior of objects in the universe.

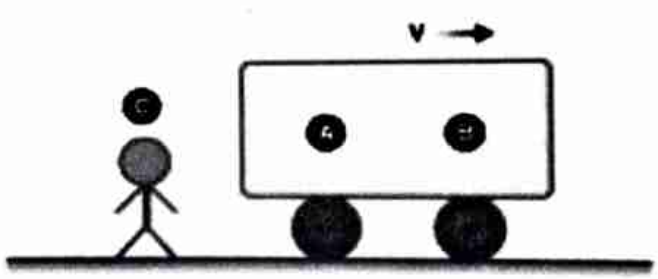


Figure 4.1: Inertia

4.5 Newton's Second Law of Motion

The second law of motion states that the rate of change of momentum of an object is proportional to the applied external force and takes place in the direction in which external force acts.

According to the second law of motion, force, $F\alpha\frac{Changeinmomentum}{Time}$

We know that, change in velocity with respect to time is called acceleration,

$$a = \frac{v-u}{t}$$

And also we know that momentum $(p) = mxv$

Then, $P_2 = Mvandp_1 = Mu$

$$F = K\frac{P_2 - P_1}{t} = K\frac{Mv - Mu}{t} = K\frac{M(v-u)}{t} = KMa \tag{4.1}$$

Where K = constant of proportionality.

4.5.1 Application of Newton's Second Law of Motion

The following applications are based on Newton's second law of motion:

1. A cricket player (or fielder) moves his hands backward on catching a fast cricket ball.

2. During athletics meet, a high jumping athlete is provided either a cushion or a heap of send on the ground to fall upon.

4.6 Law of Conservation of Linear Momentum

When no external force acts on a system of several interacting particles, the total linear momentum of the system is conserved. The total linear momentum is the vector sum of the linear momentum of all the particles of the system.

Consider an isolated system (the system on which no external force acts) of particles. Suppose the m particles have masses, $m_1, m_2, m_3 \ldots\ldots\ldots\ldots m_n$ and are moving with velocities $v_1, v_2\ v_3 \ldots\ldots\ldots\ldots\ldots v_n$ respectively.

Then total linear momentum of the system is

$$P = M_1v_1 + M_2v_2 + M_3v_3 + \ldots\ldots\ldots\ldots\ldots\ldots\ldots M_nv_n$$

$$P = P_1 + P_2 + P_3 + \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots + P_n$$

If is the external force acting on the system, then according to Newton's second law $F = \frac{dp}{dt}$

For an isolated system, F= 0 or $\frac{dp}{dt} = 0$

As the derivative of a constant is zero, so $P_1 + P_2 + P_3 + \ldots\ldots\ldots\ldots P_n$ = constant

Thus in the ambiance of any external force, the total linear momentum of the system is constant. This is the law of conservatism of linear momentum.

4.7 Newton's Third Law of Motion

When two bodies interact, they apply forces to one another that are equal in magnitude and opposite in direction. The third law is also known as the law of action and reaction.

Applications of Newton's Third Law of Motion

Some applications of Newton's third law are as follows:

1. Walking of a person, a person can walk because of Newton's third law of motion. During walking a person pushes the ground in a backward direction and in the reaction the ground also pushes the person with an equal magnitude of force but in the opposite direction. This enables him to move in a forwarding direction against the push.

2. Recoil of the gun when the bullet is fired from a gun, the bullet also pushes the gun in the opposite direction with an equal magnitude of force. This results in the gunman feeling a backward push, i.e., recoil force from the butt of the gun.

4.8 Equilibrium of a Particle:

A particle is in equilibrium if it does not move when acted upon by two or more forces. Thus, the force resultant is zero, and the particle remains stationary.

i.e $\sum F = 0$ Then, $\sum F_x = 0, , \sum F_y = 0, , \sum F_z = 0,$

4.8.1 Stable Equilibrium

When a body is slightly displaced from equilibrium position and tends to regain its original position, it is said to be in stable equilibrium. This means that P.E. is minimum $\frac{d^2u}{dr^2} = +ve$

Therefore, the centre of gravity is at its lowest point.

4.8.2 Unstable Equilibrium

A body is considered to be in unstable equilibrium if it travels in the direction of the displacement after being displaced from the equilibrium position.

P.E. is the maximum in this instance. $\frac{d^2u}{dr^2}$ = -ve

As a result, the centre of gravity is highest.

4.9 Impulse

A large force that acts on an object for a very short time and produces a large change in its momentum is called an impulsive force.

The impulse of a force acting on an object is equal to the product of the large force and small time interval for which it acts on the object.

Impulse (I) = Force x Time

Its unit is Newton second.

4.10 Friction

Friction is a force which opposes the relative motion between the two objects when one object moves or tries to move over the surface of another object.

Types of Friction

There are three major types of friction: Static Friction, Limiting Friction and Kinetic Friction

1. **Static Friction:** It is an opposing force which comes into play when one object tends to move over the surface of another object.
2. **Limiting Friction:** The maximum static frictional force which comes into play when one object just begins to slide over the surface of another object.
3. **Kinetic Friction:** The opposing force that comes into play when one object moves over the Surface of another object. Kinetic friction is of two types which are; Sliding Friction and Rolling Friction

Others include:

Sliding Friction: It comes into play when one object slides over the surface of another object.

Rolling Friction: It comes into play when one object rolls over the surface of another object.

4.11 Force and its Types

Force is the interaction that alters or seeks to alter the body's resting or moving condition after acting on a body.

Six different types of forces rule nature:

1. Conservative force

It is the term for a force whose actions are unaffected by the direction taken. The conservative force's work depends solely on the starting and ending points. Therefore, a conservative force in a closed loop does not work. Examples include the forces of gravity, static electricity, and elastic force.

2. Non-conservative force

The term "non-conservative force" refers to a force whose work depends on the path a particle takes.

Example: Friction force, air drag force, etc.

3. Constant force

Constant force is the term used to describe a force that is constant over time and space. It always stays the same.

4. Variable force

Variable force is defined as a force that varies in time and distance. It depends on various factors, such as time, space, velocity, etc.

5. Contact Force

Contact force requires physical contact between the body and the force's application.

6. Non-Contact Force: It doesn't need the body and the force's agent to make eye contact.

Types of Contact Forces

1. Tension force
2. Friction force
3. Spring force
4. Drag force
5. Pseudo force

1. Tension force

The mass that is connected to the string is balanced by the tension in the string.

2. Friction Force

"The frictional force occurs between the two surfaces when two items are in touch and one of the objects moves or desires to move."

Alternatively stated, "the force preventing the relative motion of solid surfaces is known as friction."

For instance, the car's tyres experience friction as it travels along the road. When we push a big stone and it doesn't move, friction force stops the motion. The friction force is usually denoted by "f".

Friction force (f) = μN

Here N is the normal force and $\mu$is the coefficient of friction.

Types of Friction

Friction is divided into the following categories according to its state of motion and its state of rest:

Static Friction:

Static friction is the friction that exists between the surfaces of two bodies when they are not moving relative to one another but nevertheless have a tendency to move. When the body is at rest, there is friction.

For example: when trying to move a block that is resting on a table while applying force, yet the block does not move. So, in this instance, static friction is acting. In this case, the static friction force balances the applied force.

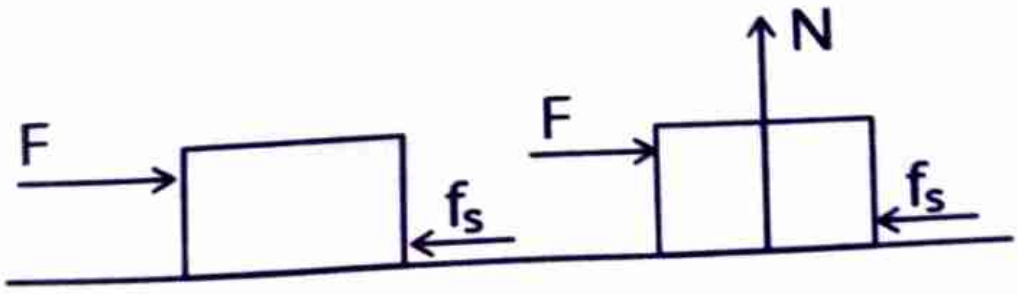


Figure 4.2: Static Friction

Friction is being adjusted via static friction. The amount of static friction grows along with an increase in the applied external force until it reaches a specified maximum value. The limiting value of static friction is referred to as this maximum value.

When an external force is applied to the body, static friction only becomes a factor. Static friction won't exist if a body is held at rest and no force is used to move it.

Static friction has a limit value that is independent of the contact area. The only factors are the surface's composition and the normal force. $(f_s)_{max} = \mu N$

Here μ is the coefficient of static friction and N is the normal force acting on the block.

The force of static friction is typically expressed by: $(f_s)_{max} = \leq \mu N$

If the external force (F) exceeds this limiting friction $(f_s)_{max}$, then the block starts moving on the surface.

Rolling friction is the friction that occurs between a body's surface and a surface when a body is rolling on it.

When a body is rolling, the point of contact with the surface is not moving relative to the surface that the body is rolling on.

A finite area (not a point) of the body comes into contact with the surface as a body rolls because the surfaces in contact temporarily flex a little. The overall result is that motion is opposed by the contact force component that is parallel to the surface.

Rolling friction is denoted by (F_R).

For example: The friction caused by the road on the wheel of a moving vehicle is known as rolling friction i.e friction on the bicycle wheel.

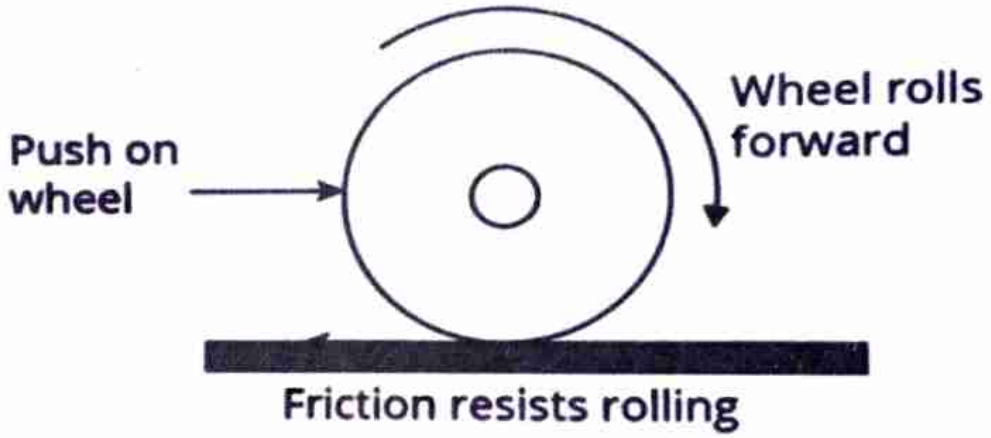


Figure 4.3: Rolling

4.12 Pseudo Forces

There is a force exerted on the object we are witnessing whenever we observe it in relation to an accelerating body. Imaginary force is known as pseudo force.

Pseudo force F_p = ma where a is the acceleration of the body with respect to which we are watching the object, and m is the mass of the object that we are observing. A pseudo force will exert itself on a body outside the car, for instance, if we see it while we are seated inside a speeding vehicle.

Types of Non-Contact Forces

These forces act at a distance.

1. **Gravitational Force:** Gravitational force is the pull of gravity that one object experiences from another due to their respective masses. It is inversely proportional to the square and directly proportional to the product of the masses of those items.

$$F \alpha \frac{M_1 M_2}{r^2}$$

Where, F is the gravitational force between two bodies, M_1 is the mass of the first body, M_2 is the mass of the second body and r is the distance between the centres of two bodies.

The force exerted by two bodies with mass M_1 and M_2 whose centres is r units apart is given as:

$$F = G\frac{M_1 M_2}{r^2}$$

2. **Electrical Force:** Electric force is the interaction of any two charged bodies that is either attractive or repulsive.

The magnitude of the electrostatic interaction between two point charges is inversely proportional to the square of the distance between them and directly proportional to the scalar multiplication of the charges.

Electric force formula can be obtained from Coulomb's law as follows:

$$F = K\frac{q_1 q_2}{r^2}$$

Where F = electric force directed between two charged bodies, K = constant of proportionality q_1 and q_2 = are the amounts of charge on each body and r is the distance between the charged bodies.

3. **Magnetic Force:** The attraction or repulsion that develops between poles of magnet is known as magnetic force.

We know that the formula of magnetic force is $F = qvB \sin\theta$

Where F is the magnetic force

q = charge of moving particle

v = particle velocity

B = magnetic field and

θ = angle between velocity and magnetic field vectors

4.13 Momentum

Momentum is defined as the product of an object's mass (m) and its velocity (v).

The SI unit for momentum is kilogramme meter per second (kgm/s). It is a vector quantity i.e It has both magnitude and direction. It a measure of how difficult it is to stop a moving object.

$$P = m \times v$$

where P = momentum, m = mass and v = velocity.

4.13.1 Factors Affecting Momentum

1. Mass: The larger the mass, the more the momentum, even at the same velocity.
2. Velocity: High velocity also implies high momentum.

Applications of momentum

1. Bullet : A bullet with high velocity, despite its small mass, gives it significant
2. Sports: A fast baseball has more momentum than a slow one, thereby requiring more effort to stop.

Types of Momentum

1. Linear Momentum (P): This is the momentum of an object moving in a straight line.
   $P = m \times v$
2. Angular Momentum (P_α): This describes the momentum of an object moving in a circular path. It is the product of its linear momentum and the radius of its path.

$P_n = mvr$

The rate of change of momentum is directly proportional to the applied force-Newton's second law of motion.

$$F = \frac{\Delta P}{t}$$

Where $\Delta P = Changeinmomentum$ t = time taken (s)
$\Delta P = F \times t = impulse$ (kg m/s or Ns). important.

■ **Example 4.1** A 10 N of force is acting on a body moving on a straight line with initial momentum 20 kgm/s. Calculate the final momentum after 5 s.

Solution

$F = \frac{\Delta P}{t} = \frac{P_f - P_i}{t}$

$10 = \frac{\Delta P}{t} = \frac{P_f - 20}{5}$

$50 = P_f - 20$

$50 + 20 = P_f$

$50 + 20 = P_f$

$P_f = 70kg.m/s$

■

■ **Example 4.2** What happens to the momentum of the object if the velocity is doubled

Solution: $P_1 = mv \ldots\ldots\ldots (i)$

$P_2 = 2mv \ldots\ldots (ii)$

Dividing equation (ii) by (i) $\frac{P_1}{P_2} = \frac{2mv}{mv}$

$\frac{p_1}{p_2} = 2$

$p_1 = 2p_{2(i.edoubled)}$

■

4.14 Conservation of linear momentum

The Law of Conservation of Linear Momentum states that the total momentum of an isolated system remains constant i.e it is conserved. This implies that the momentum before collision is equal to the momentum after collision, in a system of colliding bodies. This principle holds true when the net external force acting on the system is zero. Hence, the total momentum before a collision or interaction within a closed system is conserved.

Isolated System: An isolated system is a system in which no net external force acts. The total momentum within the system is conserved i.e does not change.

Illustrations/examples

1. **System of two colliding objects:**

When two objects collide, the total momentum they possess before the collision is the same as the total momentum they have afterward.

2. **Momentum transfer:**

If one object loses momentum, another object must gain it to keep the total momentum constant. This principle applies to a variety of events, such as two colliding cars or even the recoil of a gun.

For conservation of momentum of a system of two objects,

$m_1u_1(initialmomentumofobject1) + m_2u_2(initialmomentumofobject1) = m_1v_1(finalmomentumofobject1 + m_2v_2(finalmomentumofobject2)$

where

m_1 and m_2 = masses of the two objects.

u_1 and u_2 = initial velocities.

v_1 and v_2 = final velocities after the interaction.

If they stick together after the collision, the two bodies will have common velocity, v i.e

$m_1u_1 + m_2u_2 = (m_1 + m_2).v$

Note: momentum is not conserved when there is a net external force acting on the system. Such an external force could be gravity, friction or any other net force.

▪ **Example 4.3** A 15,000 kg truck is coasting at 2.5 m/s on a straight, frictionless track when a load of rice is dumped directly down into it. The truck's speed after receiving the rice is 2.0 m/s. What is the mass of the rice?

Solution

Let t = truck and r = rice

m_t= 15000 kg, u_t = 2.5 m/s, u_r = 0 m/s, V_{t+r} =2.0 m/s

$m_tu_t + m_ru_r = (m_t + m_r).v_{t+r}$

$15000 \times 2.5 + m_r \times 0 = (15000 + m_r)$ x 2.0

$37500 - 30000 = 2m_r$

$m_r = \frac{7500}{3} = 2500kgi.emassoftherice.$

▪

4.15 Relative motion

Relative motion is the movement of an object that is measured with respect to another object or a frame of reference. Since there is no absolute fixed point in the universe, all motion is relative; for example, as you sit on a moving train, the trees and buildings outside appear to be moving awkward relative to you, just like the observer in the car too. To determine relative motion, it is assumed that the object you are comparing to, is stationary e.g **cars** on a highway, a person on a train etc.

Relative motion explains the fact that our perception of motion depends on our own frame of reference.

▪ **Example 4.4** A motorboat traveling 4 m/s, East encounters a current traveling 3.0 m/s, North. What is its (i) Relative velocity (ii) Resultant velocity?

Solution: velocity due east = V_E, velocity due north = V_N

The relative velocity = $4 - 3m/s$

Using Pythagoras theorem, The resultant velocity

$V_R^2 = V_N^2 + V_E^2$

$V_R = \sqrt{(3^2 + 4^2} = 5m/s$

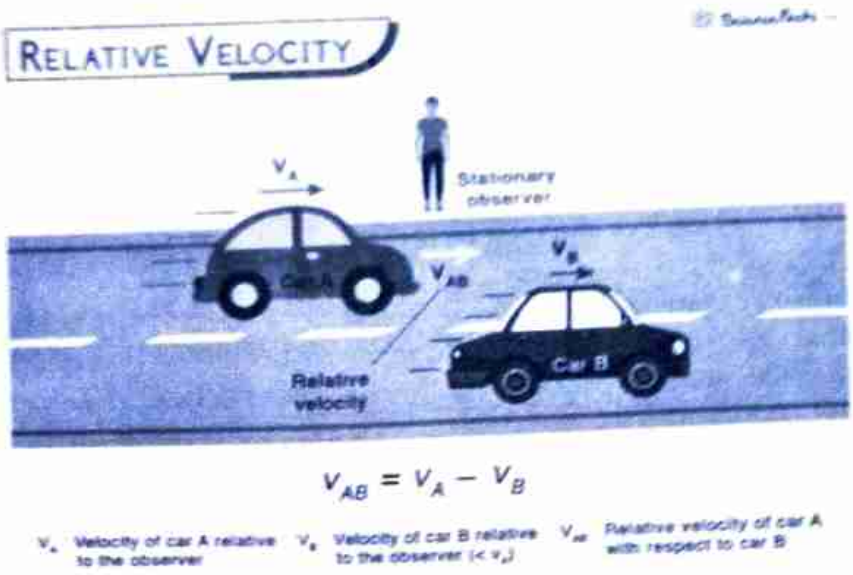


Figure 4.4: Relative motion

$$\tan\theta = \frac{3}{4} = 0.75$$

$$\theta = \tan^{1} 0.75° i.e 36.9°$$

The resultant velocity of the boat is 5 m/s at 36.9 degrees. ■

4.16 Tutorial questions

Problem 4.1 A cyclist starts from rest and accelerates uniformly at 1.5 m/s2 for 8 seconds, then maintains constant velocity for 10 seconds. Calculate: (a) the maximum velocity reached, (b) the total distance traveled during acceleration, and (c) the total displacement for the entire motion.

Problem 4.2 A soccer ball is kicked horizontally from a height of 2 meters with an initial speed of 20 m/s. Ignoring air resistance (g=10 m/s2), determine: (a) the time to hit the ground, (b) the horizontal range, and (c) the final velocity components just before impact.

Problem 4.3 Newton's first law of motion with an example involving a passenger in a car that suddenly stops. Why does the passenger continue moving forward? Discuss how friction plays a role in real-world scenarios versus ideal conditions.

Problem 4.4 A 5 kg box is pushed across a frictionless surface with a horizontal force of 20 N at an angle of 30° to the horizontal. Calculate: (a) the acceleration of the box, (b) the normal force, and (c) the horizontal component of the applied force.

Problem 4.5 A rocket accelerates upward from rest with a constant thrust providing a net acceleration of 15 m/s2 (including gravity, g=10 m/s2). Find: (a) the velocity after 5 seconds using kinematics, (b) the mass if the thrust force is 2000 N, and (c) the height reached in that time.

Problem 4.6 A firefighter of mass 80 kg slides down a 5-meter fire pole with an acceleration of 2 m/s2 downward. Calculate: (a) the frictional force acting on the firefighter, (b) the reaction force from the pole on the firefighter, and (c) the equal and opposite force the firefighter exerts on the pole.

Problem 4.7 Two masses, 2 kg and 5 kg, are placed 1 meter apart in space. Using Newton's law of universal gravitation ($G=6.67\times10^{?11}$ N.m2/kg2), calculate the gravitational force between them. Discuss how this exemplifies action at a distance and why it's negligible compared to Earth's gravity.

Problem 4.8 Two point charges, +3C and -2C, are separated by 0.5 meters in a vacuum. Using Coulomb's law ($k=9\times10^9$ N.m2/C2), find: (a) the magnitude and direction of the electrostatic force between them, (b) the acceleration of the -2 ?C charge if its mass is 0.01 kg, and (c) why this force acts at a distance without contact.

Problem 4.9 A satellite of mass 1000 kg orbits Earth at a height where the gravitational acceleration is 8 m/s2 (action at a distance). Assuming circular orbit, calculate: (a) the orbital speed using centripetal force equaling gravitational force, (b) the period of one orbit, and (c) how Newton's third law applies to the Earth-satellite system.

Problem 4.10 A 0.2 kg ball is thrown upward with an initial velocity of 20 m/s against Earth's gravity (action at a distance, g=10 m/s2). Analyze: (a) the maximum height using kinematics, (b) the net force on the ball at the top (using Newton's second law), and (c) the reaction force Earth exerts on the ball throughout the motion per Newton's third law.

Problem 4.11 A 60 kg skater moving east at 5 m/s collides head-on with an 80 kg skater moving west at 4 m/s. They grab onto each other and move together after the collision. What is their final velocity? (Ans: -0.14 m/s in west direction, after collision).

5. Vectors and Fundamental Laws of Mechanic

5.1 Introduction

In science and engineering, frequently reference is made to physical quantities. For example, we talk of the speed of a vehicle, and the force with which an object is crushed, the length of a wire, etc. Anything that is measurable is termed as 'quantity'. The quantities that we come across in Physics are referred to as physical quantities.
Example: Mass, length, temperature, force, speed, work and time etc. It is useful to separate these physical quantities into two types: the first are called scalars while the second are known as vectors.

5.2 Scalar

A scalar quantity is a physical quantity that can be completely described by its magnitude. Examples of scalar quantities are length, time, density, charge, volume, work etc. The number of students in a class and the cost of a house are familiar scalar quantities. For instance, if the speed of a car is $30m/s$, in this case a quantity having numerical value 30 and the unit of measure m/s are used. The numerical value together with the unit is called the 'magnitude'

5.3 Vector

To describe certain physical quantities like displacement along with the magnitude, the direction is essential. For example, consider a body moving from X to Y. XY is the displacement. On the contrary, if the body moves from Y to X, the displacement is YX. ***'Quantities which require both magnitude and direction for their complete specification or description are called vectors'.*** Examples of vector quantities: momentum, force, torque, magnetic field, acceleration, velocity etc. When an object is moved by applying a force to it we can achieve different effects by applying the force in different directions. In order to specify the force completely we must state not only its magnitude but also the direction in which the force acts. For example we might state that 'a force of 10 newtons is applied vertically from above'. Clearly this force would have a different effect from one applied horizontally or below the object. The direction in which the force acts is very important in this case.

5.4 Vector Representation

Vectors are represented by an arrow pointing in the direction of the vector. The length of the arrow represents the magnitude of the vector, the arrow head indicates its direction. This can be demonstrated as follows: The vector represented by the directed line segment AC in Fig.5.1 is



Figure 5.1: Representation of a vector

denoted by $\overrightarrow{AC}$ or a simple notation as $\vec{a}$ (to be read as vector a). For vector $\overrightarrow{AC}$, A is the initial point and C is the terminal point.
The magnitude of $\overrightarrow{AC}$ (or $\vec{a}$) is denoted by $|\overrightarrow{AC}|$ (read as modulus of vector AC) or $|\vec{a}|$ (or simply AC or a) and is always positive. In printed material, vectors are often represented by bold face type, such as **A** or with designations like A, $\underline{A}$, $\vec{A}$ etc. Alternatively we represent a vector in terms of components in three Cartesian coordinates.

$$\vec{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k} \tag{5.1}$$

Hence a vector can be written as

$$\vec{A} = 8\mathbf{i} + 3\mathbf{j} + 7\mathbf{k} \tag{5.2}$$

where, **i**, **j**, **k**, are unit vectors which are defined as vectors of unit magnitude in the positive x, y, z, directions respectively.

Position vector: To find the position of any point 'P' in a plane or space, generally a fixed point of reference called the origin 'O' is taken. The vector $\overrightarrow{OP}$ is called the position vector of P with respect to O as demonstrated in Fig. 2.

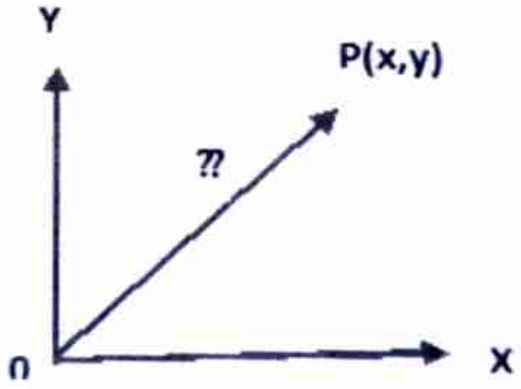


Figure 5.2: Position vector

5.5 Types of vectors

Unit vector: A vector having unit magnitude is called unit vector. Unit vector is defined as a vector divided by its magnitude. That is, if $\vec{A}$ is a vector having magnitude $|\vec{A}| \neq 0$, then $\frac{\vec{A}}{|\vec{A}|}$ is a unit

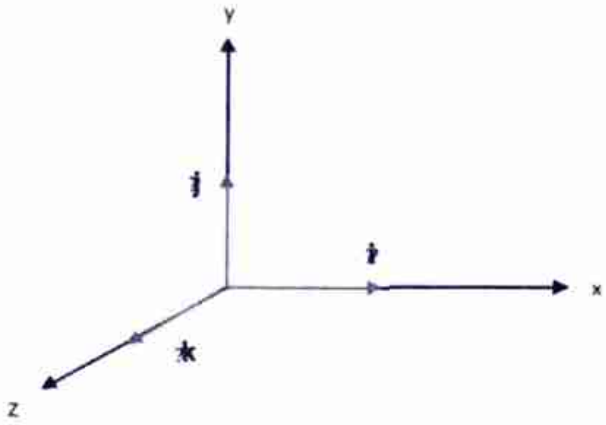
vector having the same direction as $|\vec{A}|$. It is represented as $\hat{A}$ (read as A cap). $|\hat{A}| = 1$. Thus

$$\hat{A} = \frac{|\vec{A}|}{|\vec{A}|} \tag{5.3}$$

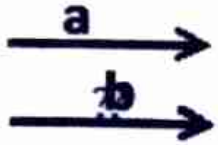
or

$$\vec{A} = |\vec{A}|\hat{A} = A\hat{A} \tag{5.4}$$

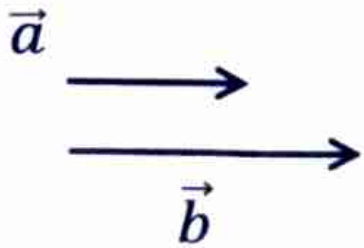
Unit vectors have a magnitude of 1. They only give the direction.
A displacement of $5m$ in the x-direction is written as $\vec{d} = 5m\hat{i}$?



Zero vector or Null vector: A vector having zero magnitude is called a Null vector or Zero vector. It is represented as $\vec{O}$.
Note: (i) the position vector at origin is a zero vector.
(ii) Zero vectors have no specific direction.
Equal vectors: Two vectors are said to be equal if both vectors have same magnitude and direction. Example of equal vectors is as shown in Fig.??



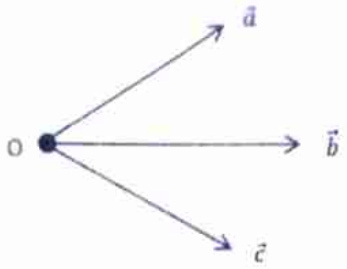
Parallel vectors (Like vectors): Two vectors are said to be parallel if they have the same directions. Their magnitudes may not necessarily be the same.



Note: Two parallel vectors may or may not be equal vectors but, two equal vectors are always parallel.

Anti-parallel vectors (Unlike vectors): Two vectors are said to be anti-parallel if they acts in opposite direction.

Concurrent vectors (Co initial vectors): Vectors are said to be concurrent vectors or co initial vectors if they have the same initial point.

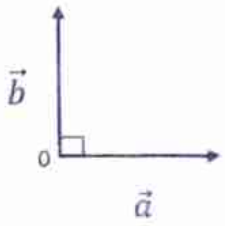


$\vec{a}, \vec{b}$ and $\vec{c}$ are concurrent at point 'O'

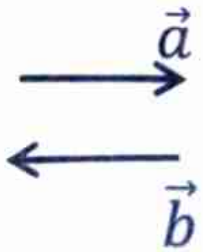
Coplanar vectors: Vectors in the same plane are called coplanar vectors.



Orthogonal vectors: If the angle between two vectors is 90°, then the two vectors are said to be orthogonal to one another. The vectors $\vec{a}$ and $\vec{b}$ are orthogonal to one another.



Negative vectors: The negative vector of any vector is a vector having equal magnitude but acts in opposite direction.



5.6 Operation of vectors

5.6.1 Addition of Vectors.

Addition of scalars involves only the addition of their magnitudes. Whereas in adding two vectors their magnitude as well as direction is considered. **Triangle method:** Triangle law of vector addition states that if two vectors can be represented in magnitude and direction by two sides of a triangle taken in the same order, then their resultant is represented completely by the third side of the triangle taken in opposite order. To add two vectors A and B we follow the steps as below:

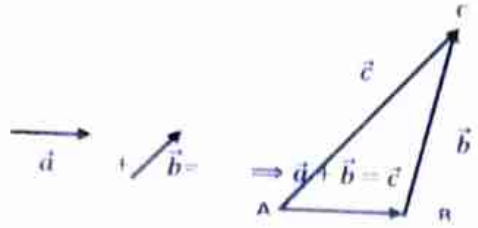


Figure 5.3: Addition of vectors

Draw vector a as the base side, then draw vector b starting from the end of a. Join the beginning of vector a to the end of vector b. The third side c is the sum of vectors a and b.

Parallelogram method of vector addition: *"It states that if two vectors acting at a point can be represented both in magnitude and direction by the two adjacent sides of a parallelogram drawn from that point, the resultant is represented completely by the diagonal of the parallelogram passing through that point"* To add two vectors placed with common initial point, the parallelogram method of vector is used.

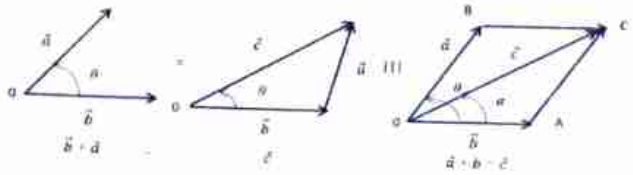
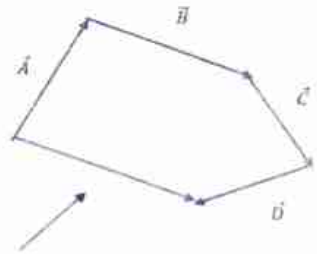


Figure 5.4: Addition of vectors

Law of polygon of vector addition: It states that if a number of vectors are represented both in magnitude and direction by the sides of a polygon taken in the same order, then their sum (resultant) is represented both in magnitude and direction by the closing side of the polygon taken in the opposite order.



$$\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D}$$

Figure 5.5: Addition of vector

In terms of components we can also add vectors

$$A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k} \tag{5.5}$$

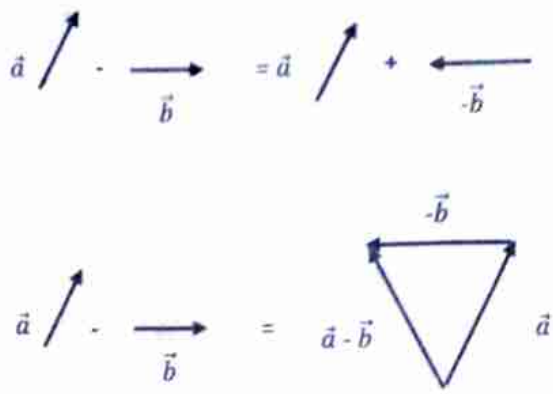
$$B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k} \tag{5.6}$$

as

$$A + B = (A_x + B_x)\mathbf{i} + (A_y + B_y)\mathbf{j} + (A_z + B_z)\mathbf{k} \tag{5.7}$$

5.6.2 Subtraction of Vectors

Subtraction of one vector $\vec{b}$ from another vector $\vec{a}$ can be realized using the definition of a negative of a vector as follows.



In terms of components if

$$A = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k} \tag{5.8}$$

and

$$B = B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k} \tag{5.9}$$

then

$$A - B = (A_x - B_x)\mathbf{i} + (A_y - B_y)\mathbf{j} + (A_z - B_z)\mathbf{k} \tag{5.10}$$

5.6.3 Scalar or Dot Product

Given two vectors

$$\vec{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}$$

and

$$\vec{B} = B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k}$$

we define the scalar or dot product of $\vec{A}$ and $\vec{B}$ as

$$\vec{A} \cdot \vec{B} = A_xB_x + A_yB_y + A_zB_z \tag{5.11}$$

Alternatively, if we have diagrammatic representation of $\vec{A}$ and $\vec{B}$, then we define

$$\vec{A} \cdot \vec{B} = |A|\,|B| \cos\theta = AB\cos\theta \tag{5.12}$$

where θ is the angle between vectors $\vec{A}$ and $\vec{B}$. Note that a scalar product is a scalar quantity and not a vector.

$\hat{i} \cdot \hat{i} = 1$ $\quad$ $\hat{i} \cdot \hat{j} = 0$

$\hat{j} \cdot \hat{j} = 1$ $\quad$ $\hat{j} \cdot \hat{k} = 0$

$\hat{k} \cdot \hat{k} = 1$ $\quad$ $\hat{i} \cdot \hat{k} = 0$

5.6.4 Vector or cross product

Given two vectors,

$$\vec{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k} \cdot$$

and

$$\vec{B} = B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k}$$

then we define a vector or cross product as

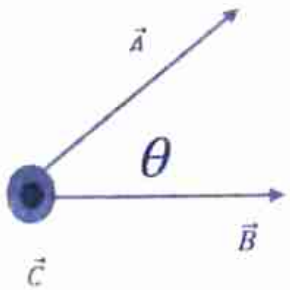
$$\vec{A} \quad \mathbf{x}\,\vec{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= \mathbf{i}(A_yB_z - A_zB_y) + \mathbf{j}(A_zB_x - A_xB_z) + \mathbf{k}(A_xB_y - A_yB_x)..$$

Alternatively, if we are given $\vec{A}$ and $\vec{B}$ diagrammatically

$$\vec{A} \otimes \vec{B} = |\vec{A}||\vec{B}|\sin \qquad (5.13)$$

i.e. $AB\sin\theta$ in magnitude only with a direction defined by the motion of a right handed corkscrew when we rotate $\vec{A}$ into $\vec{B}$. Note also that vector product is a vector quantity.

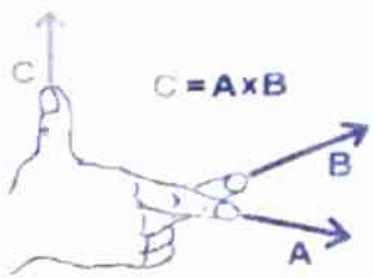


$\hat{i} \times \hat{i} = 0$ $\quad$ $\hat{i} \times \hat{j} = \hat{k}$

$\hat{j} \times \hat{j} = 0$ $\quad$ $\hat{j} \times \hat{k} = \hat{i}$

$\hat{k} \times \hat{k} = 0$ $\quad$ $\hat{k} \times \hat{i} = \hat{j}$

Vector Multiplication II: Right Hand Rule



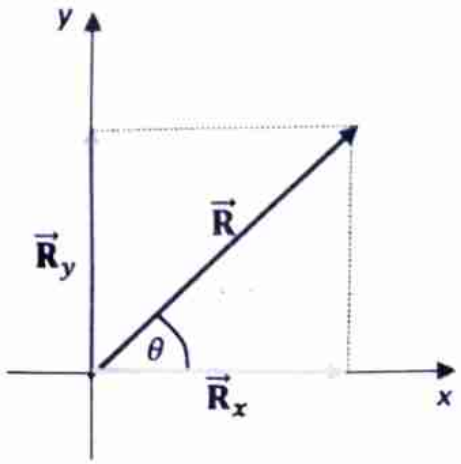
Index finger in the direction of the first vector.

Middle finger in the direction of the second vector.

Thumb points in the direction of the cross product.

5.7 Resolution of Vectors

Resolution of a vector means the process of splitting of a vector into components. If a vector is resolved into two components along the two mutually perpendicular directions, they are called 'rectangular components'. Any vector can be broken down into components along the x and y axes. A vector in two dimensions may be resolved into two component vectors along any two naturally perpendicular directions. Figure 14 shows the vector $\vec{R}$ and its x and y vector components, $\vec{R_x}$ and $\vec{R_y}$ which have magnitudes.



$$|\vec{R}_x| = |\vec{R}|\cos\theta$$

and

$$|\vec{R}_y| = |\vec{R}|\sin\theta$$

Or equivalently,

$$R_x = R\cos\theta$$

and

$$R_y = R\sin\theta$$

▪ **Example 5.1** Given three vectors $\mathbf{A} = 3\mathbf{i} + 7\mathbf{j} + 3\mathbf{k}$; $\mathbf{B} = 5\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ and $\mathbf{C} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$. Find: (a) A.B (b) AxB (c) (AxC).B

Solution:

(a) $A.B = (3i + 7j + 3k).(5j + 2j + 4k)$
$= (3)(5) + (7)(2) + (3)(4)$
$= 15 + 14 + 12 = 41$

(b) $\mathbf{A}\times\mathbf{B}=\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 7 & 3 \\ 5 & 2 & 4 \end{vmatrix}$

$=\mathbf{i}\begin{vmatrix} 7 & 3 \\ 2 & 4 \end{vmatrix}+\mathbf{j}\begin{vmatrix} 3 & 3 \\ 5 & 4 \end{vmatrix}+\mathbf{k}\begin{vmatrix} 3 & 7 \\ 5 & 2 \end{vmatrix}$

$= \mathbf{i}(28-6)+\mathbf{j}(15-12)+\mathbf{k}(6-35)$

$= 22\mathbf{i} + 3\mathbf{j}-29\mathbf{k}$

(c) $(\mathbf{A}\times\mathbf{C}).\mathbf{B}$

$(\mathbf{A}\times\mathbf{C})=\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 7 & 3 \\ 4 & -3 & 2 \end{vmatrix}$

$= \mathbf{i}(14+9)+\mathbf{j}(12-6)+\mathbf{k}(-9-28)$

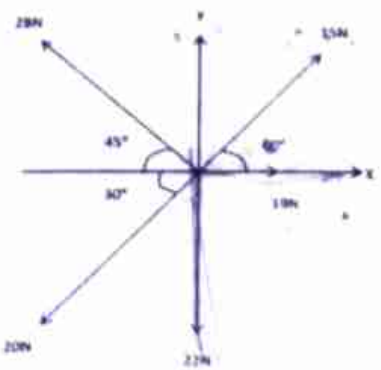
$= 23\mathbf{i} + 6\mathbf{j}-37\mathbf{k}$

$\therefore(\mathbf{A}\times\mathbf{C}).\mathbf{B} = (23\mathbf{i} + 6\mathbf{j}-37\mathbf{k}).(5\mathbf{i}+2\mathbf{j}+4\mathbf{k})$

$=115+12-148$

$= -21$

▪ **Example 5.2** The five coplanar forces shown below, act on an object. Find the resultant.



First we find the x-and y-components of each force. These components are as follows:

Force	x - component	y - component
19N	19N	0N
15N	$15\text{NCos}60deg = 7.5\text{N}$	$15\text{NSin}60deg = 13\text{N}$
28N	$-28\text{NCos}\,45^0 = -19.8\text{N}$	$28\text{NSin}45^0 = 19.8\text{N}$
20N	$-20\text{NCos}30deg = -17.3\text{N}$	$20\text{NSin}30deg = -10.0\text{N}$
22N	0N	-22N

The resultant $\vec{R}$ has components $R_x = \sum F_x$ and $R_y = \sum F_y$
Then we have

$$R_x = 19.0N + 7.5N - 19.8N - 17.3N + 0N = -10.6N$$

$$R_y = 0N + 13.0N + 19.8N - 10N - 22N = 0.8N$$

Magnitude of the resultant is

$$R = \sqrt{R_x^2 + R_y^2}$$

$$= \sqrt{10.6^2 + (0.8)^2} = 10.6N$$

Finally, we sketch the resultant as show in fig.15b.

The angle is obtained by

$$tan\varphi = \frac{0.8N}{10.6N} = 0.075$$

$\varphi = 4.29^o = 4.3^o$.

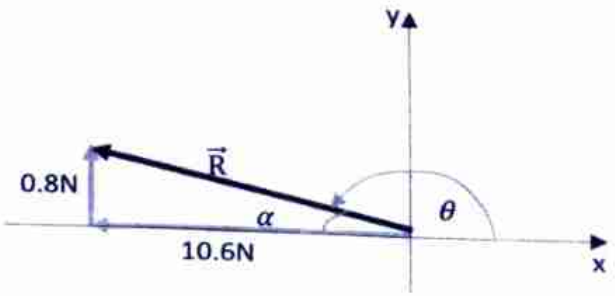
Then $\theta = 180^o - 4.3^o$

$= 175.70^o$.

The resultant is $10.6N$ at 175.70^0

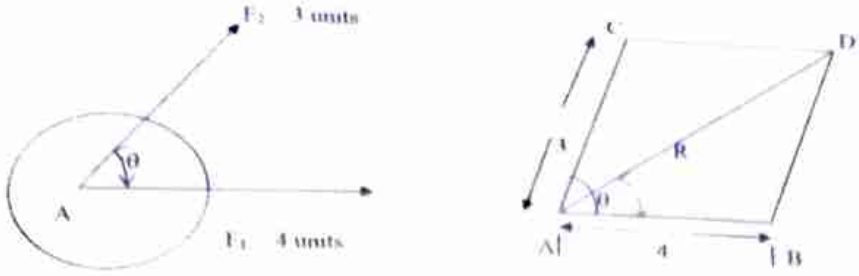
$$\vec{R} = 10.6N - 175.70^o$$

from +x-axis.



■

■ **Example 5.3** In Fig.16a, the force $F_1 = 4$ units and force $F_2 = 3$ units are acting on a body at point A. Thus, to get the resultant of these forces, parallelogram ABCD is constructed such that$AB = 4$ units and $AC = 3$ units. Then according to this law, the diagonal AD represents the resultant in direction and magnitude.



Hence, the single force which can replace the two forces acting at apoint with the same effect as that of the two forces is called the resultant. This can be obtained by measuring AD and the angle. The above problem can also be solved using Sine and Cosine laws.
Hence, using cosine rule, we have that:

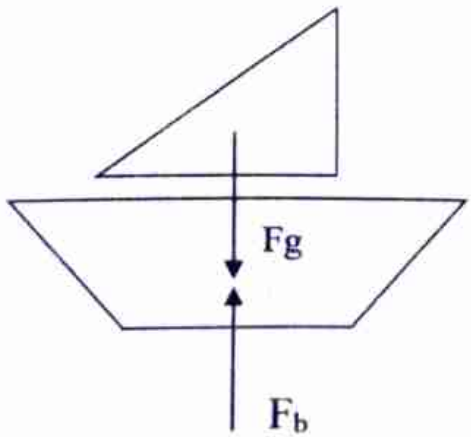
$$R = \sqrt{a^2 + b^2 + 2ab\cos\alpha}$$

5.8 Fundamental Laws of mechanics

Newton's First Law

It states that everybody continues in its state of rest or uniform motion in a straight line unless it is compelled by external agency acting on it. This leads to the definition of force as the external agency which changes or tends to change the state of rest or uniform linear motion of the body.

Demonstration of Newton's first law: Figure 17 shows a floating boat. It can be assumed that there are two forces acting on the boat. The first is the boat's weight Fg. There is also an upward buoyancy force Fb exerted by the water on the boat. If these two forces are equal and opposite, the resultant of these two forces will be zero and therefore the boat will remain at rest (it will not move up or down).



Newton's second law:

It states that the time rate of change of momentum of a body is directly proportional to the impressed force and it takes place in the direction of the force acting on it. Thus, according to this law,

Force α time rate of change of momentum

But momentum = mass $\times$ velocity

As mass does not change,

Force α mass $\times$ rate of change of velocity.

Force α mass × acceleration

$F \alpha ma$

If the water was removed from beneath the boat Fig.17, a non-zero resultant force would act and the boat would accelerate downward.

Newton's Third Law: It states that for every action, there is an equal and opposite reaction. Again considering the boat as above, the water exerts an upward buoyancy force on the boat and the boat exerts an equal and opposite force on the water. This is illustrated below:

Characteristics of Forces:

From Newton's 1st law, we defined the force as the agency which tries to change the state of rest or state of uniform motion of the body. From 2nd law of motion, we arrived at practical definition of unit force as the Force required to produce unit acceleration in a body of 1kg mass. It may be noted that a force is completely specified only when the following four characteristics are specified:

-Magnitude
-Point of application
-Line of action
-Direction

Forces and Interactions

Interaction

*-is between one thing and another

*-requires a pair of forces acting on two objects.

Example: interaction of hand and wall pushing on each other.

Force pair-you push on wall, wall pushes on you.

Newton's Third Law of motion: Interaction

For every action, there is an equal and opposite reaction.

*in all interactions

*regardless of mass or size

The statement means that in every interaction, there is a pair of forces acting on the two interacting objects. The size of the forces on the first object *equals* the size of the force on the second object. The direction of the force on the first object is opposite to the direction of the force on the second object. Forces always come in pairs-equal and opposite action-reaction force pairs.

5.9 Tutorial 5

Problem 5.1 Define the terms vector and scalar quantities. Give two examples of each.

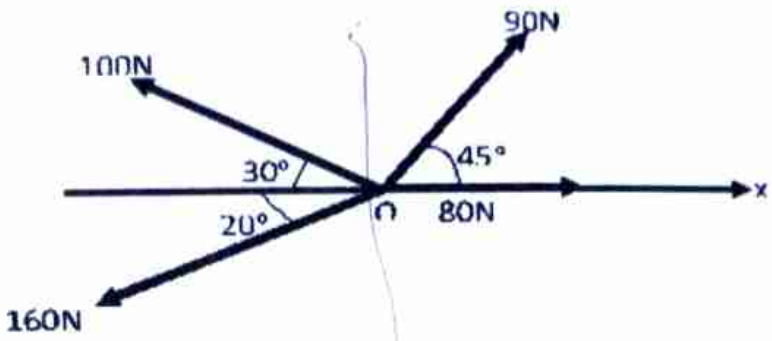
Problem 5.2 which of these are scalars and which are vectors:

- a) The volume of a petrol tank,
- b) A length measured in metres,
- c) A length measured in miles,
- d) The angular velocity of a flywheel,
- e) The relative velocity of two aircraft,
- f) The work done by a force,
- g) Electrostatic potential,

• h) The area of a class room.

Problem 5.3 Given three vectors $A = 3i + 7j + k$; $B = 4i + 2j - 4k$ and $C = 2i - 3j + 2k$. Find: (a)A + C (b)A.B (c) AxB (d) 3A+ C (e) AxB xC.

Problem 5.4 Four coplanar forces act on a body at point O as shown in Fig below.



Use component method to find their resultant. Give your answer for the magnitude correct to two significant figures.

Problem 5.5 Three forces that act on a particle are given by $\vec{F}_1 = (20i - 36j + 73k)N$, $\vec{F}_2 = (-17i + 21j - 46k)N$ and $\vec{F}_3 = (-12k)N$. Find their resultant vector. Also find the magnitude of the resultant to two significant figures.

Problem 5.6 Find (a) $\vec{A} + \vec{B} + \vec{C}$

(b) $\vec{A} - \vec{B}$ and

(c) $\vec{A} - \vec{C}$

if $\vec{A} = 7i - 6j$, $\vec{B} = -3i + 12j$ and $\vec{C} = 4i - 4j$

Problem 5.7 If $\vec{A} = 3i - 2j + k$, find the unit vector C.

1.

[1]This book was written and compiled using LaTeX2.0

6. Rotational motion

Rotational motion, torque, vector product, moment, rotation of coordinate axes, and angular momentum.

6.1 Introduction

Rotational motion refers to the motion of a body around a fixed axis or point. In this motion, every point on the object moves in a circular path around the axis, though at varying distances from the center. It's a fundamental concept in Physics, describing how objects like wheels, planets, or even molecules spin or rotate.

Examples of Rotational Motion:

- The rotation of the blades of an electric fan.
- The rotation of planets around their axes.
- A spinning top.
- A car wheel rotates as it moves forward.
- A car tyre turning as the vehicle moves forward.

6.2 Rotational Motion and Related Terms:

Angular Position or Angular displacement (θ) is usually expressed in radians, in degrees, or in revolutions.

$$1rev = 360^o = 2\pi(rad)$$

or

$$1rad = 57.3^o$$

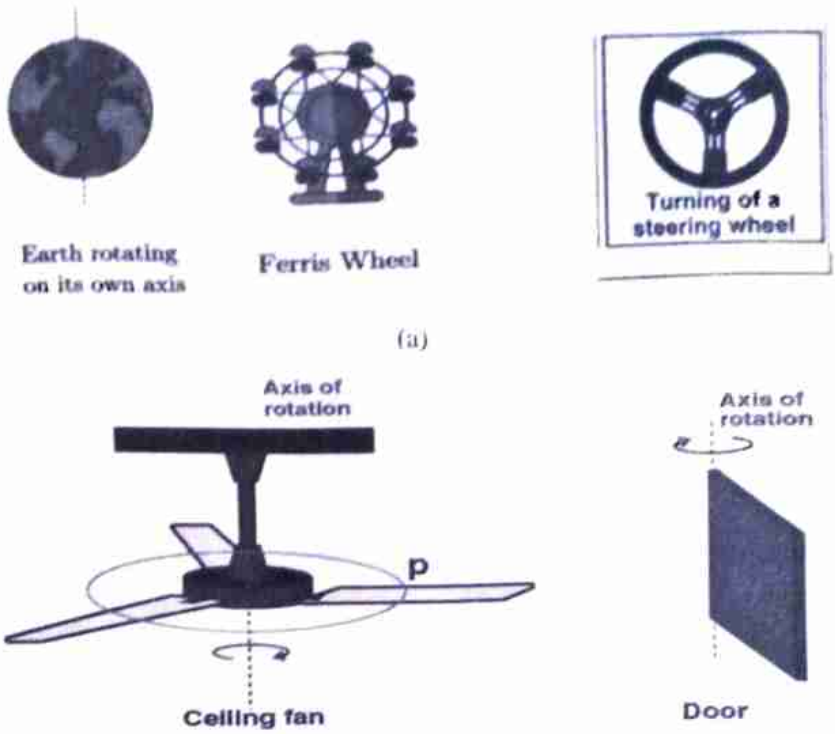


Figure 6.1: Examples of Bodies that undergo Rotational Motion

1 radian is the angle subtended at the centre of a circle by an arc equal in length to the radius of the circle. Hence, an angle θ is given in terms of the arc length l it subtends on a circle of radius r by

$$\theta = \frac{l}{r}$$

The radian of an angle is a dimensionless number.

Angular position describes the orientation of a line with respect to a fixed direction, typically measured in radians. It represents how far an object has rotated from a reference angle (usually taken as zero).

Angular Velocity (ω) This is the rate at which an object rotates around its axis. It tells how fast the angular position changes with time and is measured in radians per second (rad/s).

$$\omega = \frac{d\theta}{dt} = \frac{\theta_f - \theta_i}{t}$$

Angular Acceleration (α) Angular acceleration is the rate of change of angular velocity. It tells how quickly an object's rotation is speeding up or slowing down.

$$\alpha = \frac{d\omega}{dt}$$

The units of α are typically $\frac{rad}{s^2}(rev/sec^2)$

Rotational Kinetic Energy An object in rotational motion also has kinetic energy, which depends on its moment of inertia and angular velocity.

$$K_rot = \frac{1}{2} I \omega^2$$

Relation to Linear Motion

Many concepts in rotational motion have analogs in linear motion:

- **Angular displacement** (θ) $\Longleftrightarrow$ **Linear displacement** (x)
- **Angular velocity** (ω) $\Longleftrightarrow$ **Linear velocity** (v)
- **Angular acceleration** (α) $\Longleftrightarrow$ **Linear acceleration** (a)
- **Torque** (τ) $\Longleftrightarrow$ **Force** (F)
- **Moment of inertia** (I) $\Longleftrightarrow$ **Mass** (m)

6.3 Centripetal and Tangential Forces

In rotational motion, points on the object experience both **centripetal force**, which pulls them toward the axis, and **tangential force**, which pushes them along their circular path. The tangential force causes angular acceleration.

In short, rotational motion deals with how objects rotate and how factors like force and mass distribution affect that motion.

▪ **Example 6.1** A wheel rotates with an angular displacement, $\theta = 5t^2$, where t is in seconds. Find the angular velocity, ω at $t = 4$ seconds.

Solution:

Angular velocity ω is the time derivative of angular displacement θ :

$$\omega = \frac{d\theta}{dt}$$

Given, $\theta = 5t^2$, take the derivative with respect to t:

$$\omega = \frac{d(5t^2)}{dt} = 10t$$

At $t = 4$ seconds:

$$\omega = 10t(4) = 40rads/s$$

So, the angular velocity at $t = 4$ seconds is $40rad/s$. ▪

▪ **Example 6.2** A spinning disc accelerates from rest to an angular velocity of $\omega = 20rad/s$ in 5 seconds. What is the angular acceleration α of the disc?

Solution:

Angular acceleration α is defined as the change in angular velocity over time:

$$\alpha = \frac{d\omega}{dt}$$

Here, the change in angular velocity, $\Delta\omega = 20\frac{rad}{s} - 0\frac{rad}{s} = 20rad/s$, and the time $\Delta t = 5$ So,

$$\alpha = \frac{20rad/s}{5s} = \frac{4rad}{s^2}$$

Thus, the angular acceleration is $4\ rad/s^2$. ▪

▪ **Example 6.3** A solid sphere of mass $m = 2kg$ and radius $r = 0.2m$ is rotating about an axis through its center with an angular velocity $\omega = 10rads/s$. What is the rotational kinetic energy of the sphere?

Solution:

The moment of inertia I of a solid sphere about its center is given by:

$$I = \frac{2}{5}mr^2$$

Substitute the given values:

$$I = \frac{2}{5} \times 2kg \times (0.2)^2 = \frac{2}{5} \times 2 \times 0.4 = 0.03kg.m^2$$

The rotational kinetic energy K is:

$$K = \frac{1}{2}I\omega * 2$$

Substitute the values:

$$K = \frac{1}{2} \times 0.032 \times 10^2 = 1.6J$$

so, the rotational kinetic energy is $1.6j$. ▪

6.4 Torque

Is a concept in physics that describes the tendency of a force to rotate an object around an axis. It plays a key role in understanding how forces cause objects to spin or turn. Torque is essentially the rotational equivalent of force in linear motion and is often referred to as a "rotational force." Hence, torque explains how forces cause objects to rotate, depending on both the force applied and the distance from the axis of rotation.

Definition:

-Torque is the measure of the rotational effect of a force applied to an object.

-It is the product of the applied force and the distance from the point of application to the axis of rotation (also called the lever arm or moment arm).

Formula: $\tau = r \times F \times Sin\theta$

where: τ = torque (measured in Newton-meters (Nm)),

r = distance from the axis of rotation (lever arm),

F = applied force,

θ = the angle between the force vector and the lever arm.

If the force is applied perpendicularly to the lever arm ($\theta = 90^0$), Sin (90^0, so the formula simplifies to:

$$\tau = r \times F$$

Direction of Torque:

-Torque is a **vector quantity**, meaning it has both magnitude and direction.

The direction of torque is determined by the **right-hand rule**: point your right-hand fingers in the direction of the lever arm, curl them toward the direction of the applied force, and your thumb will point in the direction of the torque vector (either into or out of the plane).

Positive and Negative Torque:

Torque is considered **positive** if it causes counterclockwise rotation.

-It is **negative** if it causes clockwise rotation.

Moment Arm (Lever Arm):

-The effectiveness of a force in causing rotational motion depends on the **distance** from the axis (lever arm). The farther the force is applied from the pivot point, the greater the torque.

This is why long wrenches make it easier to turn bolts: they increase the distance (lever arm) from the point of rotation, increasing torque.

Torque and Equilibrium:

For an object to be in rotational equilibrium, the **net torque** acting on it must be zero. This means the clockwise and counterclockwise torques cancel each other out, preventing the object from rotating.

Torque and Angular Acceleration:

Just as force causes linear acceleration in linear motion, torque causes **angular acceleration** in rotational motion. This relationship is given by:

$$\tau = I \times \alpha$$

where: I is the **moment of inertia** (resistance to rotation),

α is the **angular acceleration**.

Examples of Torque in Real Life:

-**Opening a door**: Pushing the door near the handle (far from the hinge) produces more torque than pushing it close to the hinge, making it easier to open.

-**Wrenches**: Longer wrenches generate more torque, making it easier to turn bolts or nuts.

-**See-saw**: A heavier person closer to the center generates less torque than a lighter person farther out, affecting balance.

▪ **Example 6.4** A person applies a force, $F = 20N$ to a door at a distance of $r = 0.8m$ from the hinge, perpendicular to the door. What is the torque applied to the door?

Solution: Torque τ is calculated by the formula:

$$\tau = r \times F \times \sin\theta$$

Since the force is applied perpendicular to the door, $\theta = 90^0$ and $\sin(90^0) = 1$. So,

$$\tau = r.F = 0.8m.20N = 16N.m$$

So, the torque applied to the door is $16N.m$. ▪

▪ **Example 6.5** A wrench is used to tighten a bolt, and a force of 50 N is applied at an angle of 60^0 to the wrench handle, which is 0.3 m long. Find the torque produced by the force.
Solution: The torque is given by:

$$\tau = r \times F \times \sin(\theta)$$

Substitute the values $r = 03m, F = 50N$ and $\theta = 60^0$:

$$\tau = 0.3.50.\sin(60^0)$$

Using Sin (60^0) =0.866:

$$\tau = 0.3.50.0.866 = 12.99N.m$$

So, the torque is approximately 13 N.m.

▪

6.5 The Vector Product (Cross Product)

The **vector product** (also known as the **cross product**) is a mathematical operation between two vectors that results in a third vector, which is perpendicular to the plane formed by the two original vectors. This operation is used in Physics to describe quantities such as torque and angular momentum, where the direction of the resulting vector plays a very important role.

Definition of the Vector Product (Cross Product):
If two vectors, **A** and **B**, are given, their vector product, denoted as $A \times B$, is defined as:
$A \times B = |A||B| = |A||B|\sin(\theta)\ \mathbf{n}$
Where:
|A| and |B| are the magnitudes of the vectors A and B,
θ is the angle between A and B,
$\sin(\theta)$ is the sine of the angle between them,
n is the unit vector perpendicular to the plane of A and B, determined by the **right-hand rule**.

6.5.1 Vector or cross product

Given two vectors,

$$\vec{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}$$

and

$$\vec{B} = B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k}$$

then we define a vector or cross product as

$$\vec{A} \quad \mathbf{x}\ \vec{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= \mathbf{i}(A_yB_z - A_zB_y) + \mathbf{j}(A_zB_x - A_xB_z) + \mathbf{k}(A_xB_y - A_yB_x)..$$

Properties of the Vector Product:

-**Resulting Vector**: The result of a cross product is always a vector perpendicular to the plane formed by the two original vectors.

-**Magnitude**: The magnitude of the resulting vector is: $|A \times B| = |A||B| \sin(\theta)$. This shows that the cross-product depends on the magnitudes of the original vectors and the sine of the angle between them.

-**Right-Hand Rule**: The direction of the resulting vector is given by the right-hand rule. If you point your right hand's fingers in the direction of A and curl them towards B, your thumb will point in the direction of the resulting vector $A \times B$.

-**Commutative Property**: The cross product is not commutative: $A \times B = -(B \times A)$. The order of the vectors matters because reversing the order changes the direction of the resulting vector.

Applications and Examples in Physics:

Torque (τ):

Torque is a vector quantity that describes the rotational effect of a force applied at a distance from a point (axis of rotation). It is given by the cross product of the position vector r and the force vector F:

$$\tau = r \times F$$

Here, r is the distance from the axis of rotation to the point where the force is applied, and F is the applied force. The direction of the torque vector tells us the direction in which the object will rotate (clockwise or counterclockwise).

Example: When you push on a door handle, the position vector points from the hinge to the handle, and the force you apply on the handle creates torque that rotates the door about the hinge.

Angular Momentum (L):

Angular momentum is the rotational equivalent of linear momentum. It is the cross product of the position vector, r and the linear momentum, p of an object:

$L = \mathrm{r} \times \mathrm{p}$

The direction of the angular momentum vector is perpendicular to the plane defined by the object's position and its linear momentum.

Example: The angular momentum of a planet orbiting the sun is determined by the cross-product of its position vector (from the sun) and its linear velocity. The result gives the direction of its rotational motion in space.

Magnetic Force on a Moving Charge:

A moving charge in a magnetic field experiences a force (Lorentz force) given by the cross product of the velocity vector **v** of the charge and the magnetic field vector **B**:

$$F = q(v \times B)$$

where q is the charge. The resulting force is perpendicular to the particle's velocity and the magnetic field's direction.

Example: In a particle accelerator, charged particles like electrons moving through a magnetic field experience forces that cause them to move in circular or helical paths, determined by the vector product of their velocity and the magnetic field.

Area of a Parallelogram:

The magnitude of the vector product of two vectors represents the area of the parallelogram formed by those vectors. If **A** and **B** are two vectors, then:

$$Area = |A \times B|$$

This is often used in physics and geometry to find areas in three-dimensional space.

Example: If you know two adjacent sides of a parallelogram, their cross-product gives you the area of that parallelogram.

■ **Example 6.6** Given three vectors $\mathbf{A}=3\mathbf{i}+7\mathbf{j}+3\mathbf{k}$; $\mathbf{B}= 5\mathbf{i} + 2\mathbf{j}+4\mathbf{k}$ and $\mathbf{C} = 4\mathbf{i} -3\mathbf{j}+2\mathbf{k}$.

Find: (a) $\mathbf{A}.\mathbf{B}$ (b) $\mathbf{A}\times\mathbf{B}$ (c) $(\mathbf{A}\times\mathbf{C}).\mathbf{B}$

Solution:

(a) $\mathbf{A}.\mathbf{B} = (3i+7j+3k).(5j+2j+4k)$

$= (3)(5)+(7)(2)+(3)(4)$

$= 15+14+12 = 41$

(b)

$$A \otimes B = \begin{vmatrix} i & j & k \\ 3 & 7 & 3 \\ 5 & 2 & 4 \end{vmatrix}$$

$$= i\begin{vmatrix} 7 & 3 \\ 2 & 4 \end{vmatrix} + j\begin{vmatrix} 3 & 3 \\ 5 & 4 \end{vmatrix} + k\begin{vmatrix} 3 & 7 \\ 5 & 2 \end{vmatrix}$$

$$= i(28-6) + j(15-12) + k(6--35)$$

$$= 22\mathbf{i} + 3\mathbf{j} - -29\mathbf{k}$$

■

6.6 Moment

There are different types of moments, with the most common ones being the **moment of force** (often just called torque) and the **moment of inertia**.

1. Moment of Force (Torque):

- The moment of force or **torque** is the measure of the turning effect produced by a force acting at some distance from a pivot or axis of rotation.
- It is calculated as:

Torque (τ) = *Force (F)* × *Perpendicular Distance (r)*

where:

F is the magnitude of the force.

r is the perpendicular distance from the pivot point to the line of action of the force.

The unit of torque is the **Newton-meter (N·m)**.

For example, when you push a door, the farther away from the hinges you apply the force, the easier it is to turn the door. This is because you increase the moment arm (distance from the pivot point), which increases the torque.

Moment of Inertia:

The **moment of inertia** refers to the resistance of an object to changes in its rotational motion about an axis. It depends on the mass distribution relative to the axis of rotation.

It is calculated as:

$$I = \sum m_i r_i^2$$

where:

I is the moment of inertia.

m_i is the mass of the i-th particle.

r_i is the distance of the i-th particle from the axis of rotation.

Objects with more mass farther from the axis have larger moments of inertia, meaning they are harder to spin or stop spinning.

The unit of moment of inertia is $Kg.m^2$

▪ **Example 6.7** A force of 50 N is applied perpendicularly to a wrench 0.3 meters away from a pivot point. Calculate the torque produced.

Solution:

We can use the formula for torque:

Torque (τ) = *Force (F)* × *Perpendicular Distance (r)* Given: $F = 50$ N, $r = 0.3$m Substituting these values, one gets

$$\tau = 50N \times 0.3m = 15Nm$$

The torque is $15N \cdot m$ ▪

▪ **Example 6.8** A weightless beam AB sits on two pivots C and D. A load of 20N hangs at 2m from the support at C. Find the value of the reaction forces P and Q at C and D. Take the distances as shown in Figure **??** below.

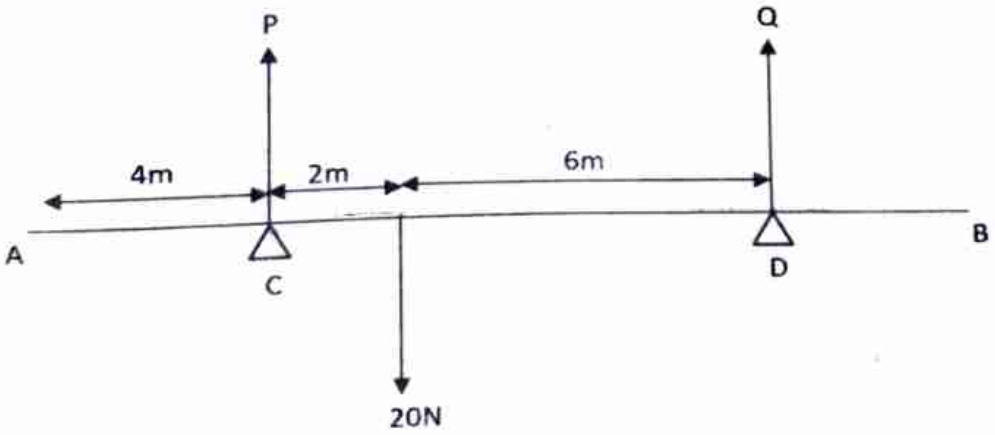


Figure 6.2: Moment

Solution:

From 1st condition of equilibrium, total upward forces = total downward forces, i.e.

$P + Q = 20N$

From the second condition of equilibrium, total clockwise moments = total anticlockwise moments. Therefore, taking moments about C, we have:

$20 \times 2 = Q \times (2 + 6) = 8Q$

$Q = \frac{40}{8} = 5$ N

Hence P = 20 – Q = 20 – 5 = 15N
Alternatively, we can take moments about D
Hence:
$P \times 8 = 20 \times 6$
$P = \frac{20 \times 6}{8} = 15N$
Q = 20 – 15 = 5 N
Thus, we can take moments about C or D and get the same results.

■

6.7 Rotation of Coordinate Axes

The rotation of coordinate axes involves changing the orientation of a coordinate system by rotating it by a certain angle, typically in two-dimensional or three-dimensional space. This is important in many areas of physics, engineering, and mathematics, where it allows you to simplify equations or view problems from a different perspective.

2D Rotation of Axes

In two dimensions, the rotation of the coordinate axes involves rotating the x and y-axes by an angle θ while keeping the origin fixed. The new coordinate system, denoted as (x', y'), is related to the old system (x, y) through the following transformation equations:

$$x' = x\cos\theta + y\sin\theta$$

$$y' = -x\sin\theta + y\cos\theta$$

Geometric Interpretation

When one rotates the coordinate axes counterclockwise by an angle θ:

The new x'-axis is aligned at an angle θ from the original x-axis.

The new y'-axis is aligned at the same angle θ from the original y-axis.

Points that originally had coordinates (x, y) in the old system will now have new coordinates (x', y').

Example of Rotation of Axes (2D):

If we have a point $P(x, y)$, and we want to rotate the coordinate axes by 45° ($\theta = 45^0$) counterclockwise: The transformation equations are:

$$x' = x\cos 45^0 + y\sin 45^0$$

$$y' = -x\sin 45^0 + y\cos 45^0$$

Since, $\cos 45^0 = \sin 45^0 = \frac{1}{\sqrt{2}}$, the equations become:

$$x' = \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}$$

$$y' = \frac{-x}{\sqrt{2}} + \frac{y}{\sqrt{2}}$$

This expresses the new coordinates (x', y'), after the rotation of the axes.

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Alternatively, we can take moments about D
Hence:
$P \times 8 = 20 \times 6$
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This expresses the new coordinates (x', y'), after the rotation of the axes.

Applications of Rotation of Axes

-**Engineering**: When analyzing material stresses and strains, rotating the coordinate system to align with principal directions can simplify the analysis.

-**Physics:** The rotation of coordinate systems is crucial in studying rotating bodies, angular momentum, and analyzing forces and fields.

-**Geometry**: Rotations are used to simplify equations of conic sections, such as transforming the equation of a rotated ellipse or hyperbola back to its standard form.

6.8 Angular Momentum

Linear Momentum vs. Angular Momentum: -Linear momentum is associated with objects moving in a straight line and is defined as the product of mass and velocity. -Angular momentum, on the other hand, is related to objects that are rotating or revolving around a point. It describes the rotational equivalent of linear momentum. -Just as force causes a change in linear momentum (via Newton's second law), **torque** causes a change in angular momentum. The greater the torque applied to a rotating object, the greater the change in its angular momentum.

Definition of Angular Momentum

For a Single Particle For a particle moving in space, the angular momentum $\vec{L}$ about a point (usually the origin) is given by:

$$\vec{L} = \vec{r} \times \vec{p}$$

where:

$\vec{r}$ is the position vector of the particle relative to the point (the distance from the point of rotation to the particle).

$\vec{p} = m\vec{v}$ is the linear momentum of the particle (mass m multiplied by velocity $\vec{v}$.

X represents the cross product, which means angular momentum is a vector quantity.

The direction of $\vec{L}$ is perpendicular to both the position vector $\vec{r}$ and the linear momentum $\vec{p}$. following the right-hand rule.

For a Rigid Body For a rigid body rotating about a fixed axis, the angular momentum is related to the moment of inertia and angular velocity. The angular momentum L of a rigid body is:

$$\vec{L} = I\omega$$

where:

I is the moment of inertia of the body, which depends on the mass distribution relative to the axis of rotation. ω is the angular velocity, the rate at which the object is rotating about the axis.

Units of Angular Momentum

The unit of angular momentum in the SI system is $Kg.m^2/s$.

Conservation of Angular Momentum

One of the most important principles in physics is the conservation of angular momentum. This principle states that if no external torque acts on a system, the total angular momentum of that system remains constant.

Example: Planets in orbit around the Sun have conserved angular momentum. As a planet moves closer to the Sun (perihelion), it speeds up; as it moves farther away (aphelion), it slows down. This is because the angular momentum of the planet about the Sun remains constant.

Angular Momentum and Torque

The rate of change of angular momentum is equal to the applied net torque τ

$$\frac{d\vec{L}}{dt} = \vec{\tau}$$

where:

$\vec{\tau}$ is the torque applied to the system.

$\frac{d\vec{L}}{dt}$ is the time derivative (or rate of change) of angular momentum.

Thus, torque is the cause of a change in angular momentum. If no torque is applied, angular momentum remains constant.

■ **Example 6.9** A particle of mass $m = 2kg$ is moving with a speed of $v = 3$ m/s in a circular path of radius $r = 4m$. Calculate its angular momentum about the center of the circle.

Solution:

For a particle moving in a circular path, the magnitude of angular momentum L about the center of the circle is given by:

$$L = m.r.v$$

Substitute the given values:

$$L = 2kg.4m.3m/s$$

$$L = 24kg.m^2/s$$

■

■ **Example 6.10** Consider a solid disc of mass $M = 5kg$ and radius $R = 0.5$ m rotating about an axis perpendicular to its plane at the center with an angular velocity $\omega = 10$ rad/s. Calculate the angular momentum of the disc.

Solution:

For a solid disc rotating about an axis through its center, the moment of inertia I is:

$$I = \frac{1}{2}MR^2$$

Subtitling the values gives $I = \frac{1}{2}.5kg.(0.5m) = \frac{1}{2}.5kg.(0.25)m^2$

$$I = 0.625kg.m^2$$

But, angular momentum L of the rotating disc is given by $L = I.\omega$

$L = 0.625kg.m^2.10rad/s = 6.25kg.m^2/s$

Angular momentum of the disc = $6.25kg.m^2/s$

■

■ **Example 6.11** A satellite of mass $m = 500kg$ is in a circular orbit around Earth at a radius of $r = 7.0 \times 10^6 m$ from center of Earth. If the satellite's orbital speed is $v = 8000m/s$, calculate its angular momentum about Earth's center.

Solution:

For a satellite in a circular orbit, the angular momentum L about the center of Earth is:

$$L = m.r.v$$

Substituting the values gives

$$L = 500kg.7.0 \times 10^6 m.8000m/s$$

$$L = 2.8 \times 10^{13} kg.m^2/s$$

6.9 Tutorial

Problem 6.1 A solid disc of mass $M = 2kg$ and radius $R = 0.5m$ is rotating about its central axis with angular velocity $\omega = 20rad/s$.

(a) Find the moment of inertia of the disc about the central axis.

(b) Calculate its rotational kinetic energy.

Answer: a 0.25 kgm^2, b 12.5 J

Problem 6.2 A wheel starts from rest and rotates with a uniform angular acceleration of $4rad/s^2$.

(a) Find the angular velocity after $t = 5s$.

(b) How many radians does the wheel rotate during this time?

Answer: a 20 rad/s, b 50 rad

Problem 6.3 A flywheel of mass $10kg$ and radius $0.2m$ is acted upon by a torque of $5Nm$. Find its angular acceleration.

Answer: 25 rad/s^2.

Problem 6.4 A student sits on a rotating stool holding two dumbbells of mass $2kg$ each at a distance of $1m$ from the axis of rotation. The system rotates with angular velocity $\omega = 2rad/s$. If the student pulls the dumbbells inward to a distance of $0.25m$, find the new angular velocity. (Neglect the stool's moment of inertia).

Answer: 32 rad/s

Problem 6.5 Angular displacement is measured in:

(A) degrees only **(B)** radians only **(C)** radians or degrees **(D)** revolutions per minute

Answer: C.

Problem 6.6 In a rigid body rotating about a fixed axis, which quantity is the same for all points?

(A) Tangential speed **(B)** Angular speed **(C)** Centripetal acceleration **(D)** Tangential acceleration magnitude

Answer: B.

Problem 6.7 Moment of inertia depends on:

(A) Mass only **(B)** Shape only **(C)** Mass distribution about the axis **(D)** Angular velocity

Answer: C.

Problem 6.8 If the net external torque on a system is zero, then its:

(A) Angular momentum is conserved **(B)** Angular velocity is constant always **(C)** Rotational kinetic energy is conserved **(D)** Moment of inertia is constant

Answer: A.

Problem 6.9 A door is pushed at two points with equal, opposite, and collinear forces producing no net torque about the hinge. The door's rotational motion:

(A) Must speed up **(B)** Must slow down **(C)** Remains unchanged in ω **(D)** Cannot be determined

Answer: C.

Problem 6.10 The work done by a constant torque τ through angle θ is:
(A) τ/θ **(B)** $\tau\,\theta$ **(C)** $\frac{1}{2}\tau\theta$ **(D)** $I\omega^2$
Answer: B.

7. Newton's Laws of Motion, Statics and Dynamics

7.1 Introduction

Newton's laws of motion are very fundamental to the description of motion of an object and the force causing it. These laws were used to explain and investigate the motion of many physical objects and systems, which laid the foundation for Newtonian mechanics. The concepts of force and mass are suitable to analyze the principles of dynamics. These principles were clearly stated for the first time by Sir Isaac Newton, today we call them Newton's laws of motion. The first law states that when the net force on a body is zero, its motion doesn't change. The second law relates force to acceleration when the net force is not zero. The third law is a relationship between the forces that two interacting bodies exert on each other.

Newton's laws are the foundation of **classical mechanics** (also called **Newtonian mechanics**): using them, we can understand most familiar kinds of motion. Newton's laws need modification only for situations involving extremely high speeds (near the speed of light) or very small sizes (such as within the atom or sub atomic particles or elementary particles, fermions and bosons).

7.2 Static and dynamic

A body is said to be rigid, if the relative position of any two particles in it do not change under the action of the forces. The mechanics of the rigid bodies dealing with the bodies at rest is termed as Statics and that dealing with bodies in motion is called Dynamics. The dynamics dealing with the problems without referring to the forces causing the motion of the body is termed as Kinematics and the one referring to the forces causing motion is called Kinetics.

7.3 Force and Interactions

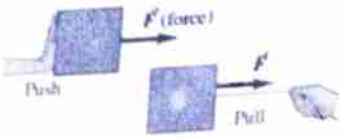
A force is a push or a pull. A better definition is that a force is an interaction between two bodies or between a body and its environment. That's why we always refer to the force that one body exerts on a second body. When you push on a car that is stuck in the snow, you exert a force on the car; a steel cable exerts a force on the beam that is hoisting it at a construction site; and so on. A force is a vector quantity; you can push or pull a body in different directions.

7.3.1 Type of Forces

i. Short-range force or Contact force (pull, push, tension, reaction, frictional force) ii. Long-range force or Force field (gravitation force, magnetic force, electric force)

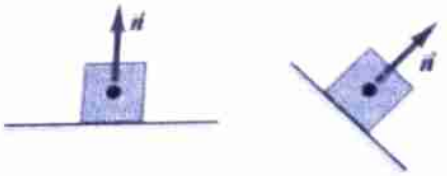
Contact Force:

When a force involves direct contact between two bodies, such as a push or pull that you exert on an object with your hand, we call it a contact force.

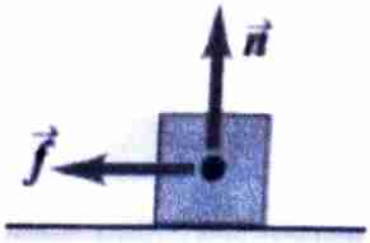


Types of Contact Force

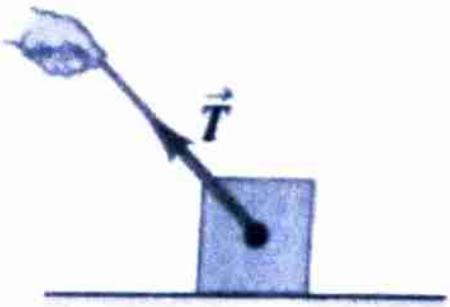
***i.* Normal force (N):** This is the force exerted on an object by any surface with which it is in contact. The normal means that the force always acts perpendicular to the surface of contact, no matter what the angle of that surface may be. see Figure 6.1



***ii.* Friction force (F):** This is the force exerted on an object by a surface acting parallel to the surface, in the direction that opposes sliding or movement. see Figure 6.2



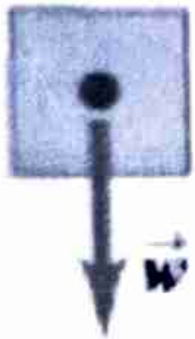
***iii.* Tension force (T):** The pulling force exerted by a stretched rope or cord on an object to which it is attached. see Figure 6.3



Long-Range Forces

There are long-range forces that act even when the bodies are separated by empty space. The force around charge bodies (electric force). The force between two magnets (magnetic force around a magnet) is an example of a long-range force, the force of gravity (force that attract object towards the earth. see Figure 6.4 below); the earth pulls a dropped object toward itself even though there is no direct contact between the object and the earth. The gravitational force that the earth exerts on a body is called the weight. see Figure 6.4

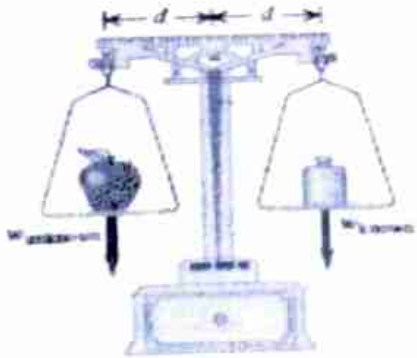
To describe a force vector, we need to describe the direction in which it acts as well as its magnitude, the quantity that describes "how much" or "how hard" the force pushes or pulls. The SI unit of the magnitude of force is the Newton (N).



7.4 Mass and Weight

7.4.1 Mass

The quantity of matter possessed by a body or an object is called a mass. It changes as a result of deformation in materials. It is measured in kilogramme (kg) using beam balance. The mass of an object is constant. Mass is a scalar quantity



An equal-arm balance determines the mass of a body (such as an apple) by comparing its weight to a known weight

7.4.2 Weight

The weight of a body is the gravitational force exerted on it by the earth. Weight is a vector quantity. The magnitude of the weight of a body at any specific location is equal to the product of its mass m and the magnitude of the acceleration due to gravity g at that location. While the weight of

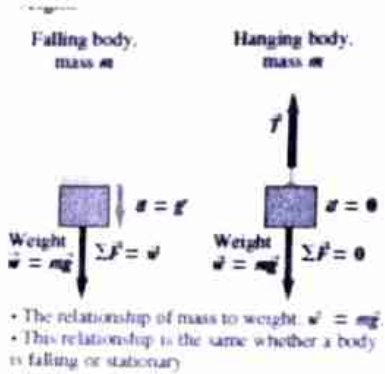
a body depends on its location, the mass is independent of location. The force that makes the body accelerates downward is its weight. The weight of an object may change due to change in gravitational force. Even the body may become weightless when gravitational force vanishes but the mass remains the same. The weight of an object is measured in Newton (N) using spring balance. Mass m and weight w of an object are related by the equation

$$w = mg \tag{7.1}$$

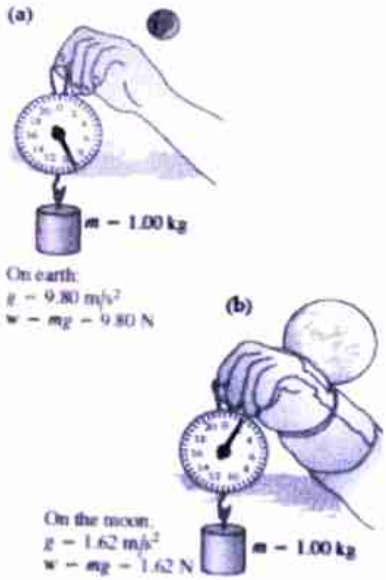
where g is a constant known as the acceleration due to gravity having a magnitude $9.8 m/s^2$

7.4.3 Relationship between mass and weight

Measuring mass and weight



• The relationship of mass to weight: $w = mg$.
• This relationship is the same whether a body is falling or stationary



A 2.49×10^4 N Rolls-Royce Phantom traveling in the +x-direction makes an emergency stop; the x-component of the net force acting on it is -1.83×10^4 N. What is its acceleration?

$$m = \frac{w}{g} = \frac{2.49 \times 10^4 \text{ N}}{9.80 \text{ m/s}^2} = \frac{2.49 \times 10^4 \text{ kg} \cdot \text{m}}{9.80 \text{ m/s}^2}$$
$$= 2540 \text{ kg}$$

Then $\Sigma F_x = ma_x$ gives

$$a_x = \frac{\Sigma F_x}{m} = \frac{-1.83 \times 10^4 \text{ N}}{2540 \text{ kg}} = \frac{-1.83 \times 10^4 \text{ kg} \cdot \text{m/s}^2}{2540 \text{ kg}}$$
$$= -7.20 \text{ m/s}^2$$

7.4.4 Newton's First Law (Law of Inertia)

This first law states that: "If an object is at rest then it will remain at rest or if it is moving along a straight line with uniform speed then it will continue to keep moving unless an external force is applied on it to change its existing state". It states that everybody continues in its state of rest or of uniform motion in a straight line unless it is compelled by an external agency acting on it. This leads to the definition of force as the external agency which changes or tends to change the state of rest or uniform linear motion of the body. It is matter of common experience that a body at rest will remain at rest if the resultant force acting on it is zero. When a body is acted on by zero net force, it moves with constant velocity and zero acceleration. For instance, a boy standing on the field tract will remain unless a force in form of a pull or a push is exerted on him. Mathematically,

$$\sum F = 0 \Longleftrightarrow \frac{dv}{dt} \tag{7.2}$$

where $\sum F$ is the net force on the object, v is the velocity and dv/dt is the derivative of the velocity with respect to time t, known as the acceleration of an object. This law also explains why an un-disturbed body moves in a straight line with constant or uniform velocity. Thus the tendency of a body to keep moving once it is set in motion results from a property called inertia. The more massive a body, the more its inertia and vice versa. This law also explains why a rotating fan refused to stop immediately it is being switched off.

7.4.5 Newton's Second Law (Fundamental Law of Dynamics)

It states that the rate of change of momentum of a body is directly proportional to the applied force and it takes place in the direction of the force acting on it. Mathematically,

$$F = \frac{dP}{dt} = \frac{d(mv)}{dt} = m\frac{dv}{dt} = ma \tag{7.3}$$

where P is the momentum, v is the velocity, m is the mass, a is the acceleration of the body and t is the time taken. There is a proportionality constant called mass, the proportion of the force to the acceleration is always constant for a given object.
This law considers a case when the force is not zero. It is now necessary to take into account the change in the momentum per unit time as the net force applied on a body. Equivalently;

$$F\alpha = \frac{m(v-u)}{t} \tag{7.4}$$

where k is a constant with a unity magnitude. The product $F \times t$ in newton second (Ns) is the impulse of a force.
In a similar way
Newton's Second Law states that if a net external force acts on a body, the body accelerates. The direction of acceleration is the same as the direction of the net force. The mass of the body times the acceleration of the body equals the net force vector. In symbols,

$$\sum \bar{F} = m\bar{a} \tag{7.5}$$

An alternative statement is that the acceleration of a body is in the same direction as the net force acting on the body, and is equal to the net force divided by the body's mass:

$$\bar{a} = \frac{\sum \bar{F}}{m} \tag{7.6}$$

Newton's second law is a fundamental law of nature, the basic relationship between force and motion. The Newton's first and second law is valid only in all inertial frame of reference.

7.4.6 Newton's Third Law (The Law of Reaction)

If body A exerts a force on body B (an "action"), then body B exerts a force on body A (a "reaction"). These two forces have the same magnitude but are opposite in direction. These two forces act on different bodies. To simply put; ***It states that for every action there is an equal and opposite reaction.*** *This third law implies that all forces are interactions between different bodies. If body A exerts a force $\bar{F}_{AonB}$ on body B exerts a force $\bar{F}_{BonA}$ on body A that is equal in magnitude and opposite in direction : $\bar{F}_{AonB} = \bar{F}_{BonA}$*



Note the following
Particle in equilibrium $\sum \bar{F} = 0$
Dynamic of particle $\sum \bar{F} = m\bar{a}$

7.5 Application of Newton's laws of motion:

■ **Example 7.1** If a net horizontal force of $120N$ is applied to a person with mass $60.0kg$ who is resting on the edge of a swimming pool, what horizontal acceleration is produced?
Solution:

$$\sum \bar{F} = m\bar{a}$$

$$\bar{a} = \frac{\sum \bar{F}}{m} = \frac{120N}{60.0kg} = 2.0m/s^2$$

■

■ **Example 7.2** What magnitude of net force is required to give a $165-kg$ refrigerator an acceleration of magnitude $2.40m/s^2$.
Solution:

$$\sum \bar{F} = m\bar{a} = 165kg \times 2.40m/s^2 = 396N$$

■

■ **Example 7.3** A box rests on a frozen pond, which serves as a frictionless horizontal surface. If a fisherman applies a horizontal force with magnitude $64.0N$ to the box and produces an acceleration of magnitude $8.00m/s^2$ what is the mass of the box?
Solution:

$$m = \frac{\sum F}{a} = (64.0N)/(8.00m/s^2) = 8.00kg$$

■

■ **Example 7.4** How much force is needed to give a $20,000kg$ heavy loaded truck on a leveled track an acceleration of $1.5m/s^2$, and what is the force exerted by the track on the truck?
Solution:
Applying Newton's second law in the component form, we find: $\sum F_x = F = ma$ and $\sum F_y =$

7.5 Applications of Newton's Laws of motion:

$N - mg = 0$

From the first and second equations we find F and N as follows:

$$f = ma = (20,000kg)(1.5m/s^2) = 30,000N$$

$$N = mg = (20,000kg)(9.8m/s^2) = 196,000N$$

Example 7.5 Calculate the magnitude of force exerted by each rocket, called its thrust T, for the four rocket propulsion system shown in the Figure 6.8. The sled's initial acceleration is $49m/s^2$, the mass of the system is $2100kg$, and the force of friction opposing the motion is $650N$.



Solution



where F_{net} is the net force along the horizontal direction. We can see from the figure that the engine thrusts add, whereas friction opposes the thrust. In equation form, the net external force is

$$F_{net} = ma$$

$$F_{net} = 4T - f$$

$$F_{net} = ma = 4T - f$$

$$4T = ma + f$$

$$4T = ma + f$$

$$4T = ma + f = (2100kg)(49m/s^2) + 650N$$

Therefore, the total thrust is

$$4T = 1.0 \times 10^5 N,$$

$$T = \frac{(1.0 \times 10^5 N)}{4} = 1.0 \times 10^4 N$$

and the individual thrusts are

Example 7.6 A $0.400kg$ soccer ball is kicked across the field by a player; it undergoes acceleration given by $a? = 3.00i? + 7.00j?m/s^2$. Find (a) the resultant force acting on the ball and (b) the magnitude and direction of the resultant force.

Solution:

The vectors in i ?and j ? format, which indicate force direction along the x-axis and the y-axis.

respectively, are involved, so we apply Newton's second law in vector form.

(a)

$$\vec{F}_{net} = m\vec{a} = (0.400kg)(3.00\hat{i} + 7.00\hat{j})m/s^2) \quad 1.20\hat{i} + 2.80\hat{j}N$$

Magnitude and direction are found using the components of $\vec{F}_{net}$.

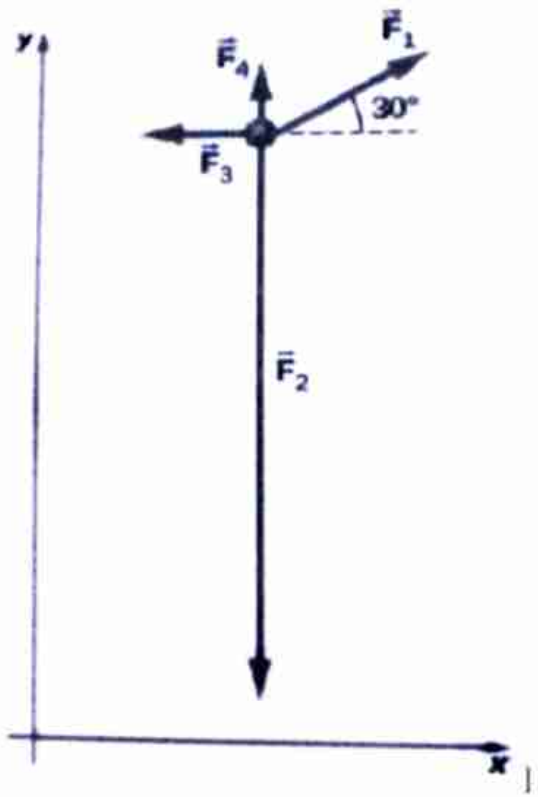
$$F_{net} = \sqrt{(1.20N)^2 + (2.80N)^2} \quad 1.05N$$

and

$$\theta = \tan^{-1}\frac{2.80}{1.20} = 66.8^{\circ}$$

• **Example 7.7** A particle of mass $m = 4.0kg$ is acted upon by four forces of magnitudes, $F_1 = 10.0N$, $F_2 = 40.0N$, $F_3 = 5.0N$, and $F_4 = 2.0N$, with the directions as shown in the free-body diagram in the Figure 6.9. What is the acceleration of the particle?

begin{figure}[h]



We consider all vectors resolved into x- and y-components:

$$\sum F_x = ma_x$$

$$\sum F_y = ma_y$$

$$F_{1x} - F_{3x} = ma_x$$

$$F_{1y} + F_{4y} - F_{2y} = ma_y$$

$$F_1\cos 30^{\circ} - F_{3x} = ma_x$$

$$F_1\sin 30^{\circ} + F_{4y} - F_{2y} = ma_y$$

$$(10.0N)(\cos 30^o) - 5.0N = (4.0kg)a_x$$

$$(10.0N)(\sin 30^o) + 2.0N - 40.0N = (4.0kg)a_y$$

$$a_x = 0.92\frac{m}{s^2}$$

$$a_y = -8.3\frac{m}{s^2}$$

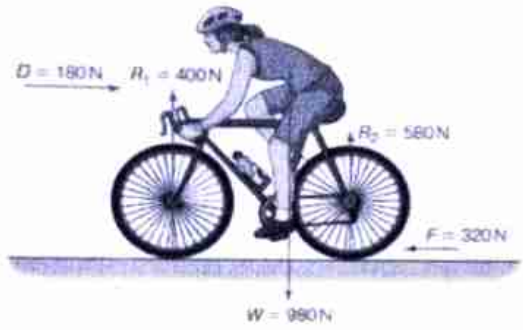
Thus, the net acceleration is

$$\vec{a} = (0.92i - 8.3j)\frac{m}{s^2}$$

which is a vector of magnitude $8.4\frac{m}{s^2}$ directed at 276° to the positive x-axis.

■

■ **Example 7.8** Figure 6.10 shows the forces on a cyclist accelerating along the road. The forces in the vertical direction balance, but the force pushing her along the road, F, is greater than the drag forces, D, acting on her. The mass of the cyclist and the bicycle is $100kg$. Calculate her acceleration.



]

Solution:

$$F - D = ma$$

$$320N - 180N = 100kg \times a$$

$$a = \frac{140}{100}\frac{m}{s^2}$$

$$= 1.4\frac{m}{s^2}$$

■

■ **Example 7.9** A constant force acts on a $5.0kg$ object and reduces its velocity from $7.0\frac{m}{s}$ to $3.0\frac{m}{s}$ in a time of $3.0s$. Find the force.

Solution:

$$a = \frac{v_f - v_i}{t} = \frac{-40\frac{m}{s}}{3.0s} = -1.33\frac{m}{s^2}$$

$$F = ma = (5.0kg)(-1.33\frac{m}{s^2}) = -6.7N$$

The minus sign indicates that the force is a retarding force, directed opposite to the motion. ■

■ **Example 7.10** A passenger travels in a lift that is accelerating upwards at a rate of $1.5\frac{m}{s^2}$. The passenger has a mass of $62kg$. By drawing a free body diagram to show the forces acting on the passenger, calculate the reaction force that the lift exerts on her



$R - W = ma$

Solution:

$$R - W = ma$$
$$R = W + ma$$
$$= 62kg \times 9.8\frac{m}{s^2} + 62kg \times 1.5\frac{m}{s^2}$$
$$= 700N.$$

■

■ **Example 7.11** Compute the least acceleration with which a $45kg$ woman can slide down a rope if the rope can withstand a tension of only $300N$

Solution:

The weight of the woman is

$$mg = (45kg)(9.81\frac{m}{s^2}) = 441N.$$

Because the rope can support only $300N$, the unbalanced downward force F on the woman must be at least

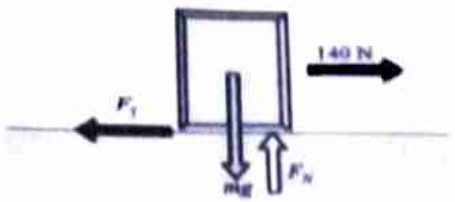
$$441N - 300N = 141N.$$

Her minimum downward acceleration is then

$$a = \frac{F}{m} = \frac{141N}{45kg} = 3.1\frac{m}{s^2}$$

■

■ **Example 7.12** A $70kg$ box is slid along the floor by $400N$ force as shown in Fig. below. The coefficient of friction between the box and the floor is $0.5N$ when the box is sliding. Find the acceleration of the box.



Solution:

Since the y-directed forces must balance,

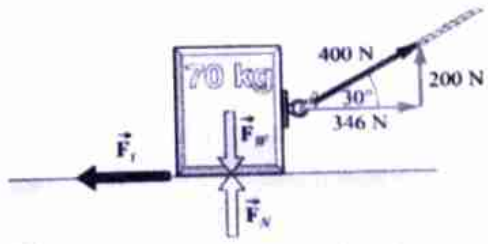
$$F_N = mg = (70kg)(9.81\frac{m}{s^2}) = 687N$$

But the friction force F_f is given by

$$F_f = \mu_k F_N = (0.50)(687N) = 344N$$

Now write $\sum F_x = ma_x$ for the box, taking the direction of motion as positive: $400N - 344N = (70kg)(a)$ or $a = 0.80\frac{m}{s^2}$ ∎

Example 7.13 Suppose, as shown in the Fig. 6.13 below, that a $70kg$ box is pulled by a $400N$ force at an angle of 30° to the horizontal. The coefficient of kinetic friction is 0.50. Find the acceleration of the box



Solution:

Because the box does not move up or down, we have $\sum F_y = ma_y = 0$. From the Fig. above, we see that this equation is

$$F_N + 200N - mg = 0$$

But $mg = (70kg)(9.81\frac{m}{s^2}) = 687N$, and it follows that $F_N = 486N$

We next find the friction force acting on the box:

$$F_f = \mu_k F_N = (0.50)(485N) = 243N$$

$$\sum F_x = ma_x$$

for the box. It is

$$(346 - 243)N = (70kg)(a_x)$$

from which

$$a_x = 1.5\frac{m}{s^2}$$

∎

7.6 Tutorial 6

Problem 7.1 Define the terms vector and scalar quantities. Give two examples of each.

Problem 7.2 At the surface of Jupiter's moon I_o, the acceleration due to gravity is $g = 1.81\frac{m}{s^2}$. A watermelon weighs $44.0N$ at the surface of the earth. (a) What is the watermelon's mass on the earth's surface? (b) What are its mass and weight on the surface of I_o?

Problem 7.3 An astronaut's pack weighs $17.5N$ when she is on earth but only $3.24N$ when she is at the surface of an asteroid. (a) What is the acceleration due to gravity on this asteroid? (b) What is the mass of the pack on the asteroid?

Problem 7.4 A box rests on a frozen pond, which serves as a frictionless horizontal surface. If a fisherman applies a horizontal force with magnitude $48.0N$ to the box and produces an acceleration of magnitude $3.00\frac{cm}{s^2}$ what is the mass of the box?

Problem 7.5 A dock worker applies a constant horizontal force of $80.0N$ to a block of ice on a smooth horizontal floor. The frictional force is negligible. The block starts from rest and moves $11.0m$ in $5.00s$. (a) what is the mass of the block of ice? (b) If the worker stops pushing at the end of $5.00s$, how far does the block move in the next $5.00s$?

Problem 7.6 A crate with mass $32.5kg$ initially at rest on a warehouse floor is acted on by a net horizontal force of $140N$. (a) what acceleration is produced? (b) How far does the crate travel in $10.0s$? (c) What is its speed at the end of $10.0s$?

Problem 7.7 A $4.80kg$ bucket of water is accelerated upward by a cord of negligible mass whose breaking strength is $60.0N$. What is the maximum upward acceleration that can be given to the bucket without breaking the cord?

Problem 7.8 A force of $10.0N$ acts on a body of mass $2.00kg$. What are (a) the body's acceleration (b) its weight in newton, and (c) its acceleration if the force is doubled?

Problem 7.9 If a man weighs $900N$ on the Earth, what would he weigh on Jupiter, where the acceleration due to gravity is $25.9\frac{m}{s^2}$?

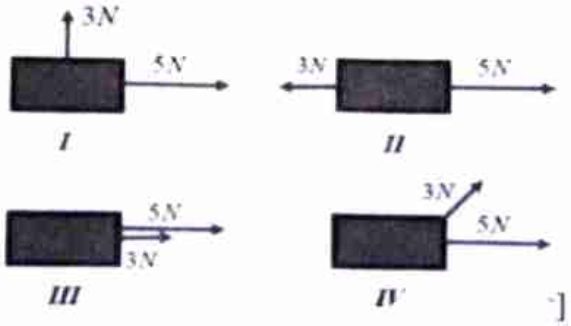
Problem 7.10 A farmer is lifting some moderately heavy rocks from a field to plant crops. He lifts a stone that weighs $40.0lb$ (about $180N$). What force does he apply if the stone accelerates at a rate of $1.5\frac{m}{s^2}$.

Problem 7.11 A physics professor pushes a cart of demonstration equipment to a lecture hall. Her mass is $65.0kg$ the cart's mass is $12.0kg$, and the equipment's mass is $7.0kg$. Calculate the acceleration produced when the professor exerts a backward force of $150N$ on the floor. All forces opposing the motion, such as friction on the cart's wheels and air resistance, total $24.0N$.

Problem 7.12 A $60.0kg$ person weighs $100.0N$ on the Moon. What is the acceleration of gravity on the Moon?

Problem 7.13 Which has greater value, a newton of gold on Earth or a newton of gold on the Moon? (a) The newton of gold on the Earth (b) The newton of gold on the Moon (c) The value is the same, regardless (d) None of the above.

Problem 7.14 Figure 6.14 shows overhead views of four situations in which two forces accelerate the same block across a frictionless surface. Rank the situations below according to the magnitude of the horizontal acceleration of the block, greatest first.



Problem 7.15 If a bird collides with the windshield of a fast moving plane, which experiences an impact force with a larger magnitude? (a) The bird (b) The plane (c) The same force is experienced by both (d) Not enough information is given.

Problem 7.16 From question 14 above, which experiences greater acceleration? (a) The bird (b) The plane (c) The same acceleration is experienced by both the bird and plane (d) Not enough information is given.

Problem 7.17 If a net horizontal force of $130N$ is applied to a person with mass $60.0kg$ who is resting on the edge of a swimming pool, what horizontal acceleration is produced?

Problem 7.18 A superman throws a $3000N$ boulder at an adversary. What horizontal force must superman apply to the boulder to give it a horizontal acceleration of $20.0\frac{m}{s^2}$

Problem 7.19 An object with mass m moves along the x-axis. Its position as a function of time is given by $x(t) = At + Bt^3$, where A and B are constants. Calculate the net force on the object as a function of time.

Problem 7.20 A $3.50kg$ bucket of water is accelerated upward by a cord of negligible mass whose breaking strength is $65.7N$. Find the maximum upward acceleration that can be given to the bucket without breaking the cord?

Problem 7.21 At the surface of Mars the acceleration due to gravity is $g = 3.72m/s^2$. A watermelon weighs $52.0N$ at the surface of the earth. (i) What is its mass on the earth's surface? (ii) What are its mass and weight on the surface of the Mars?

Problem 7.22 What is the mass of a book of weight $3.60N$ at a point $g = 9.8m/s^2$? (b) At the same location, what is the weight of a dog whose mass is $16.0kg$?

Problem 7.23 A superman throws a ~~$2800\ N$~~ boulder at adversary, what horizontal force must superman apply to the boulder to give it a horizontal acceleration of $15.0m/s^2$?

8. Dynamics of particles, frictional forces

8.1 Introduction

The Newton's laws of motion are the foundation of classical mechanics and it can be stated a very simply manner. But applying these laws to situations such as real time situations such as a toboggan sliding down a hill, or an airplane making a steep turn requires analytical skills and problem-solving technique. This chapter will:

- begins with equilibrium problems, in which the forces that act on a body at rest or moving with constant velocity (Particles in Equilibrium) would be analysed;
- then consider bodies that are not in equilibrium, for which we'll have to deal with the relationship between forces and motion (Dynamics of particles);
- learns how to describe and analyse the contact force that acts on a body when it rests on or slides over a surface (Frictional forces); and
- later analyse the forces that act on a body that moves in a circle with constant speed. We shall close the lecture with a brief look at the fundamental nature of force and the classes of forces found in our physical universe (Dynamics of circular motion).

Basic Definition of a particle A 'Particle' is a point mass (body) at some position in space. It can move about, but has no characteristic orientation or rotational inertia. It is characterized by its mass.

Examples of applications where you might choose to idealize part of a system as a particle include:

1. Calculating the orbit of a satellite - for this application, you don't need to know the orientation of the satellite, and you know that the satellite is very small compared with the dimensions of its orbit.

2. A molecular dynamic simulation, where you wish to calculate the motion of individual atoms in a material. Most of the mass of an atom is usually concentrated in a very small region (the nucleus) in comparison to inter-atomic spacing. It has negligible rotational inertia. This approach

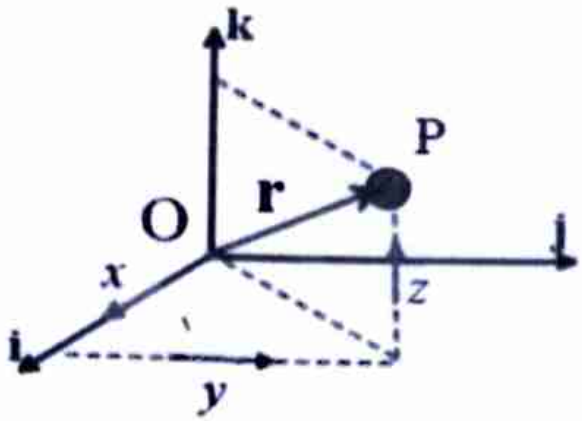


Figure 8.1: A particle P in free space

is also sometimes used to model entire molecules, but rotational inertia can be important in this case.

8.2 Particles in Equilibrium

A body under the influence of forces is called a particle. It is said to be at equilibrium when it is at rest or moving with constant velocity in an inertial frame of reference. Particles or system of particles are said to be in equilibrium when the net force acting on them is zero, according to Newton's first law. This can be expressed as static equilibrium, when the particle is at rest, and as well as dynamic equilibrium, when the particles are in constant velocities. To solve for equilibrium, the forces can be resolved into vertical (z) and horizontal (x, y) components, perpendicular to each other (usually x, y, and z) and set the sum of forces in each direction to zero. Examples of particles in equilibrium are: A hanging lamp, book on a table, a kitchen table, a running water tap, an airplane flying straight and level at a constant speed. Stating Newton's first law is an essential

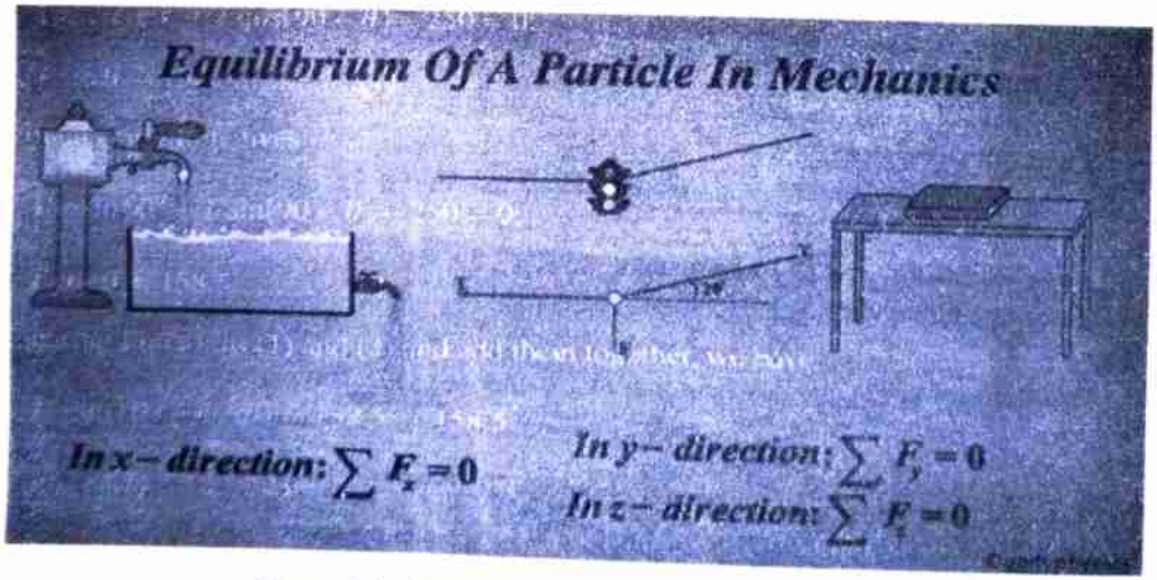


Figure 8.2: Examples of Particles in equilibrium

physical principle i.e when a particle is in equilibrium, the net force acting on it—that is, the vector sum of all the forces acting on it—must be zero:

From Figure 2, $\sum F_x = F_{1x} + F_{2x} + F_{3x} = 0$ and $\sum F_y = F_{1y} + F_{2y} + F_{3y} = 0$

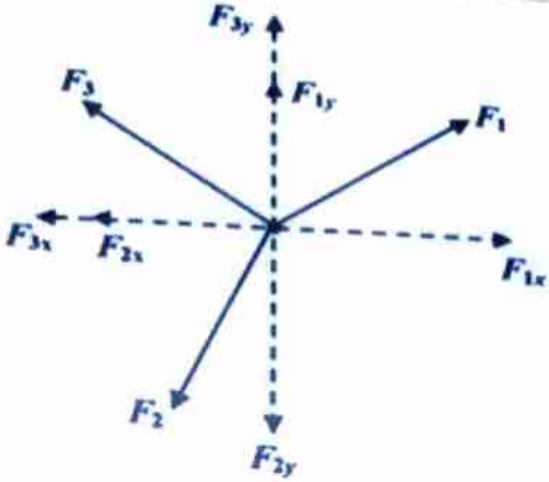


Figure 8.3: A particle is in equilibrium under the action of three forces

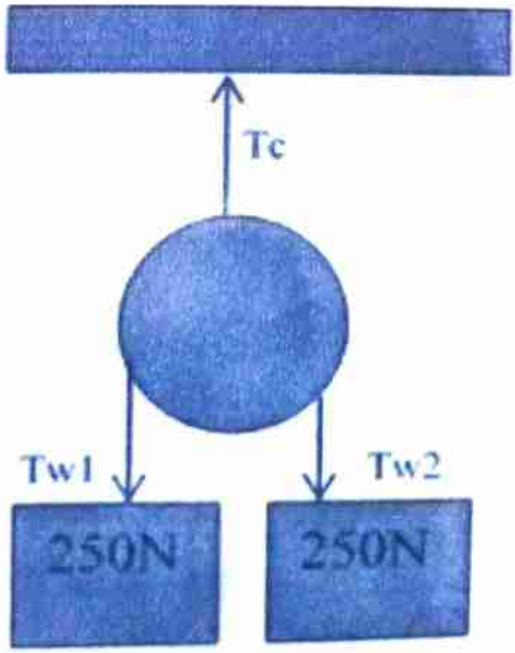
therefore,particle in equilibrium,vector form:

$$\sum_{i=1}^{n} \vec{F}_i = 0 \tag{8.1}$$

particle in equilibrium, component form:

$$\sum F_x = 0, \sum F_y = 0 and \sum F_z = 0 \tag{8.2}$$

Example 8.1 (i) Two 250 N weights are suspended at opposite ends of a rope which passes over a light frictionless pulley is attached to a chain which goes to the ceiling. Determine the tension in the chain.



Solution

Let Tw be the tension cause by each weight and the Tc is the tension on the chain by the ceiling on the pulley.

For the pulley to be in an equilibrium, the upward forces (Tc) and downward forces Tw (1 and 1) must be equal.
Therefore, the net force on the pulley is:

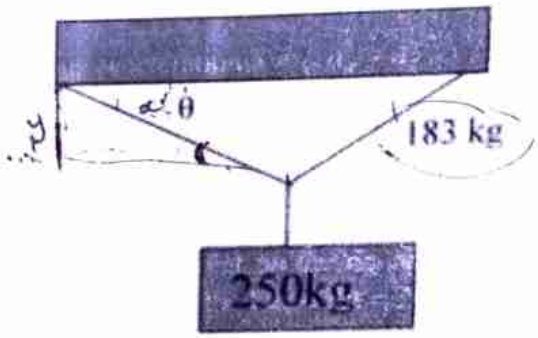
$$Tc - (Tw1 + Tw2) = 0$$

$$Tc = Tw1 + Tw2 = 250 + 250$$

$$Tc = 500N$$

■

■ **Example 8.2** A weight of 250 kg is suspended vertically from two strings as shown below at point O. If the tension in the right string is 183 kg. Determine the tension T.



Solution

The forces acting on the wight are shown in the second figure, For Equilibrium, Vertical resolution of forces gives:

$$183\cos 60^o + T\cos(90-\theta) - 250 = 0;$$

$$T\sin\theta = 158.5 \ldots\ldots\ldots\ldots (1)$$

Horizontal resolution gives

$$183\sin 60^o + T\sin(90-\theta) - 250 = 0;$$

$$T\cos\theta = 158.5 \ldots\ldots\ldots\ldots (2)$$

Squaring Equations (1) and (2) and add them together, we have,

$$T^2(\sin^2\theta + \cos^2\theta) = 158.5^2 + 158.5^2$$

$$T^2 = 158.5^2 + 158.5^2$$

$$T = 224.1kg;$$

and

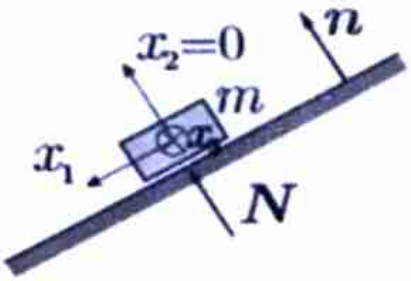
$$\theta = \tan^{-1}(158.8/158.5) = 45^o$$

■

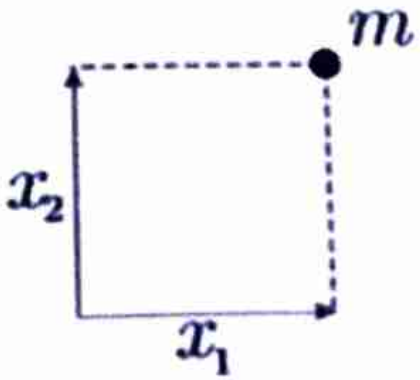
8.3 Dynamics of particles

The study of the motion of particles in terms of the forces associated with that motion is called dynamics of particle. This is fundamental concept of mechanics, crucial for understanding how objects move and respond to forces in difference environment. The dynamics of particles in physics and engineering is concerned with the study of motion of an object, treating them as idealized points with mass, and is divided into kinematics (describing motion without forces) and kinetics (linking forces to motion). However, the aim of dynamics is to find the cause of motion, and knowing the cause to give description of such a motion. The principles that govern particle dynamics differ depending on the scale of the system. In classical mechanics, particles are treated as point masses following deterministic trajectories, while in quantum mechanics, the particles exhibit wave particle duality and their behaviour is described probabilistically. The cause is always with some interaction and there are two types of interaction:

1. body-body interaction, that is when the two bodies are in contact as seen it Figure 3



2. body-field interaction, that is when a body is not in contact with other body. This is called free body (Figure 3).



In another word, a body is said to be dynamics when the net force is NOT equal to zero. This is application of Newton's second law in which the velocity of bodies in motion is not constant. These bodies are not in equilibrium and hence are accelerating. The net force on the body is equal to the mass of the body times its acceleration. There are various forces that can act on particles: gravitational, electromagnetic, frictional and normal forces.

Hints: You have to use Newton's second law for any problem that involves forces acting on an accelerating body, because the force acting on a particle determines its acceleration.

$$\sum_{i=1}^{n} \vec{F}_i \neq 0 \tag{8.3}$$



Figure 8.4: A dynamic system of particles under the action of forces initiating acceleration

In this case, the system is not in static equilibrium anymore but instead will respond to the applied forces and torques with motion, governed by among others - Newton's famous second law.

Newton's second law can be stated mathematically in a vectorized form as

$$\sum \vec{F} = m\vec{a} \tag{8.4}$$

Newton's second law can be stated mathematically in components form as

$$\sum \vec{F}_x = m\vec{a}_x \quad \sum \vec{F}_y = m\vec{a}_y \quad \sum \vec{F}_z = m\vec{a}_z \tag{8.5}$$

8.4 Energy Impulse and Momentum

Understanding the potential and kinetic energy of particles as well as impulse and momentum is essential for analysing their dynamics. Energy is the rate at which work is done on a system or by the system while impulse is the product of the force and the time in which the force acts and the momentum is the product of the mass and the velocity of the body.
A body that increases its velocity from u to v within an interval of time t acquires an acceleration:

$$a = \frac{v-u}{t} \tag{8.6}$$

and the force acting on the body is:

$$F = \frac{m(v-u)}{t} \tag{8.7}$$

Therefore,

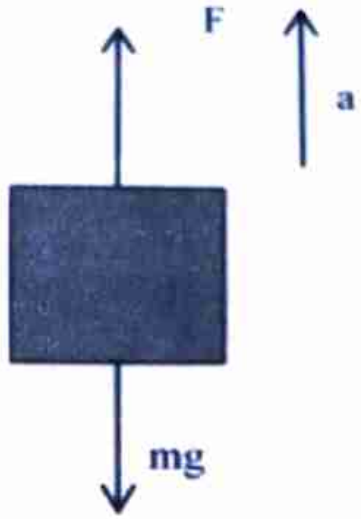
$$Ft = mv - mu \tag{8.8}$$

The above equation states that the impulse on a body is equal to the change in momentum of the body.
Considering a body A of mass m_A with initial velocity u_A and collides with a second body B of mass m_B and initial velocity u_B. if the velocities of A and B after collision are V_A and V_B respectively, then,
Initial momentum of A and B is equal to final momentum of A and B

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B \tag{8.9}$$

This is the principle of conservation of momentum which states that, the total momentum of a system remain constant if no external force act on the system.

Example 8.3 An upward force of $1.2x10^4N$ acts on an elevator of mass $2.0x10^3kg$. Calculate the elevator's acceleration.



Solution

The forces acting on the elevator are its weight, mg, downwards and upward force F. The net force is

$$F - mg = ma$$

$$1.2 \times 104 - (2.0 \times 103 \times 9.8) = (2.0 \times 103)a$$

$$a = -3.8m/s^2$$

8.5 Frictional force

Whenever two bodies interact by direct contact (touching) of their surfaces, we describe the interaction in terms of contact forces. The normal force is one example of a contact force: in this section we'll look in detail at another contact force, the force of friction.

8.5.1 Application of friction

Friction is important in many aspects of everyday life. The oil in a car engine minimizes friction between moving parts, but without friction between the tires and the road we couldn't drive or turn the car. Air drag-the frictional force exerted by the air on a body moving through it-decreases automotive fuel economy but makes parachutes work. Without friction, nails would pull out and light bulbs would unscrew effortlessly.

8.5.2 Kinetic and Static Friction

When you try to slide a heavy box of books across the floor, the box doesn't move at all unless you push with a certain minimum force. Then the box starts moving, and you can usually keep it moving with less force than you needed to Air drag is the frictional force exerted by the air on a body moving through it, this decreases automotive fuel economy but makes parachutes work.

Kinetic Friction

The frictional force that acts when a body slides over a surface is called kinetic friction(see Figure).



Figure 8.5: Figure 6: application of friction

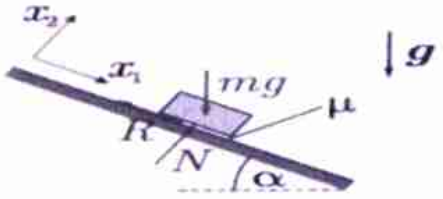


Figure 8.6: Description of kinetic frictional forces

where

$$\sum f_1 = -R + mg\sin\alpha = m\ddot{x}_1 \tag{8.10}$$

and

$$\sum f_2 = N - mg\cos\alpha = 0 \tag{8.11}$$

In dynamics, sliding friction is characterized by the kinetic friction coefficient μ and the relations is given by:

$$f_K \geq \mu_K F_N \tag{8.12}$$

where U_K is the coefficient of kinetic friction and F_N Magnitude of normal force.

Static Friction The frictional force that acts on a body that is resting on a surface is called static Friction. Static friction is what resists the beginning of motion and makes it hard to get a box started sliding across the floor.

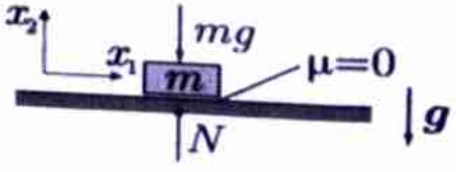


Figure 8.7: Description of static frictional forces

where

$$\sum f_1 = mx_1 = mx_1 = 0 ?x_1 = 0 \tag{8.13}$$

and

$$\sum f_2 = mx_2 = N - mg = 0 ?N = mg \tag{8.14}$$

The magnitude of static friction force is therefore given as:

$$f_s \leq \mu_s F_N \tag{8.15}$$

where μ_s is the coefficient of static friction and F_N Magnitude of normal force

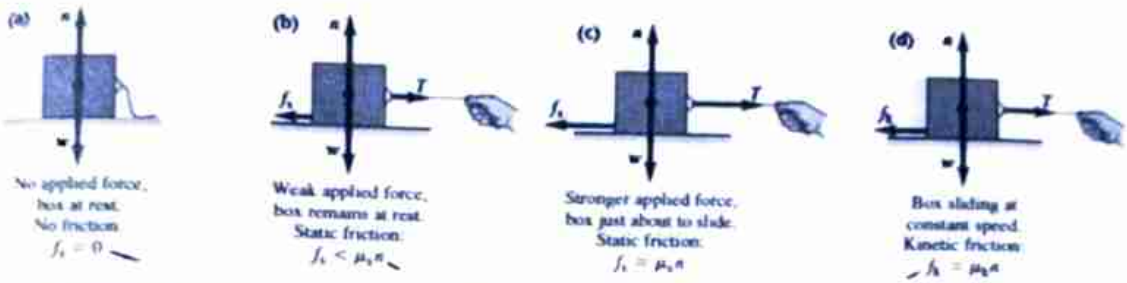


Figure 8.8: Description of coefficient of static and dynamic frictional forces

When there is no relative motion as in (a), (b) and (c), the magnitude of the static friction force f_s is less than or equal to $\mu_s F_N$.
When there is relative motion as in (d), the magnitude of the kinetic friction force f_K equals $\mu_K F_N$..

▪ **Example 8.4** What is the frictional force if the crate is at rest on the surface and a horizontal force of $50N$ is applied to it?
Solution The friction force can prevent motion for any horizontal applied force, since the crate is at rest the friction has not been overcome, therefore, considering horizontal forces

$$\sum f_x = T - f_s$$

$$\Longrightarrow 50N - f_s$$

$$f_s = 50N$$

▪

▪ **Example 8.5** A box of books of mass $2.0kg$ is sliding across a level floor and its retardation is measured to be $4.0m/s^2$. Calculate the coefficient of kinetic friction.

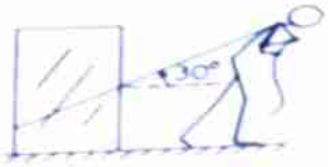
Solution

Resolving vertically gives $N = mg$
The only horizontal force on the book is $u_k N = ma$, $a = 4.0m/s^2$; Dividing equation 1 by 2

$$u_k = \frac{a}{g} = \frac{4.0}{9.8} = 0.408$$

▪

▪ **Example 8.6** Suppose you move the crate by pulling upward on the rope at an angle of 30^oabove the horizontal as shwon below. How hard must you pull to keep it moving with constant velocity? Assume that$\mu_k = 0.40$



Solution

The crate is in equilibrium because its velocity is constant, so we again apply Newton's first law. Since the crate is in motion, the floor exerts a kinetic friction force. The target variable is the magnitude T of the tension force. Figure 5.21 shows our sketch and free-body diagram. The kinetic friction force f_K is still equal to $\mu_K F_N$, from the equilibrium conditions and the equation $f_K = \mu_K F_N$ we have,

$$\sum f_x = T\cos 30^\circ - f_k = 0;$$

$$T\cos 30^\circ = \mu_K F_N \ldots\ldots\ldots(1)$$

$$\sum f_y = T\sin 30^\circ + F_N(-w) = 0;$$

$$F_N = w - T\sin 30^\circ \ldots\ldots\ldots(2)$$

These are two equations for the two unknown quantities T and n. One way to find T is to substitute the expression for n in the second equation into the first equation and then solve the resulting equation.

For T:

$$\Longrightarrow T\cos 30^\circ = \mu_K(w - T\sin 30^\circ)$$

$$T\cos 30^\circ = \mu_K w - \mu_K T\sin 30^\circ$$

$$T(\cos 30^\circ + \mu_K \sin 30^\circ) = \mu_K w$$

$$T = \frac{\mu_K w}{\cos 30^\circ + \mu_K \sin 30^\circ}$$

$$T = 188N$$

But from Eq 2,

$$F_N = w - T\sin 30^\circ$$

$$F_N = w - 188 \times \sin 30^\circ$$

$$F_N = 500 - 188 \times 0.5 = 406N$$

■

8.6 Dynamics of circular motion and gravitation

A particle moving in a circle at a constant speed undergoes uniform circular motion. In this section you will learn how to describe a particle's motion in terms of its angular position, angular velocity and acceleration.

You will also learn that a particle moving in a circle has an acceleration directed toward the center of the circle as at the left hand side.

A particle moving in a circle at a constant speed undergoes uniform circular motion. In this section

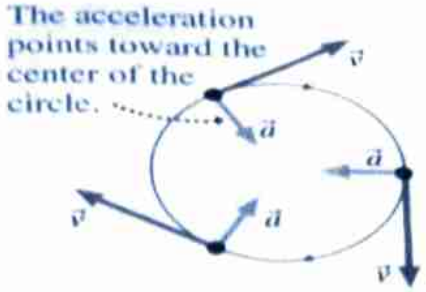


Figure 8.9: Dynamics of circular motion and gravitations

you will learn how to describe a particle's motion in terms of its angular position, angular velocity and acceleration. Uniform circular motion is the motion of an object traveling at a constant speed on a circular path.
A Uniform circular motion is the motion of an object traveling at a constant speed on a circular path.

Periodicity of the uniform acceleration: This is the time (T) taken for the particle to complete one revolution around the circular path of radius r. This is equal to the circumference of the circle divided by the speed. $T = \frac{2\pi r}{v}$

Dynamics of Uniform Circular Motion

Because a particle moving in a circle has an acceleration that points toward the center of the circle, there must be a net force toward the center to cause this acceleration.

▪ **Example 8.7** A Tire-Balancing Machine The wheel of a car has a radius of $0.29m$ and it being rotated at 830 revolutions per minute on a tire-balancing machine. Determine the speed at which the outer edge of the wheel is moving

Solution:

$$T = \frac{1}{830rev/min} = 1.2 \times 10^{-3} min/rev$$

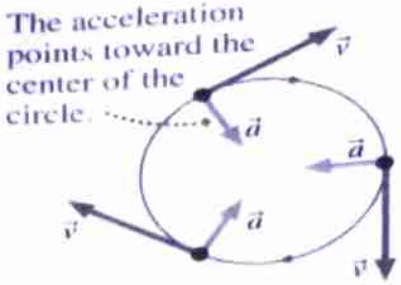
$$T = 0.072sec$$

$$v = \frac{2 \times 0.29\pi}{0.072} = 25m/s$$

▪

Centripetal acceleration:

In uniform circular motion, the speed is constant, but the direction of the velocity vector is not constant. The direction of the centripetal acceleration is towards the center of the circle; in the same direction as the change in velocity. $a_c = \frac{v^2}{r}$



Many rigid bodies move in a circular path such as car negotiating a bend; the revolution of the satellites around the earth and the revolution of the planets around the sun. The acceleration of the circular moving body is directed radially towards the center of the circle and its magnitude is; where r is the radius of the path. From Newton's 2nd law, $F_c = ma_c = \frac{mv^2}{r}$

8.7 Newton's law of gravitation

This states that the force of attraction of two bodies in the universe is proportional to the product of their masses and inversely proportional to the square of the distance separating their centers. $F_g = \frac{Gm_1m_2}{r^2}$

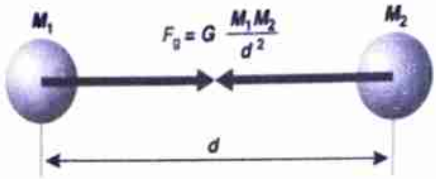


Figure 8.10: Description of Newton's law of gravity

The universal law of gravitation expresses that every mass attracts one another, the gravitational force between two objects increases when either mass increases and that the gravitational force between two objects decreases as the square of the distance between their centers increases.

Gravitational force is stronger between larger masses, the force gets weaker over larger distances and when a planet/star/galaxy is spherical, you can pretend that all the mass is in a single point at the center of the sphere, rather than an extended distribution

8.8 Gravitational Potential Energy V(E):

This is the work done in moving a body initially separated from the center of the earth by distance r is moved further distance dr.

$$dw = F_g dr \tag{8.16}$$

$$\Longrightarrow w_g = \int_{r1}^{r2} F_g dr = \int \frac{Gm_1m_2}{r^2} dr \tag{8.17}$$

$$V = -\frac{Gm_1m_2}{r} \tag{8.18}$$

Thus a body on the surface of the Earth has a gravitational potential energy of $-\frac{Gm_1m_2}{r}$ while one at infinite distance from the earth zero gravitational potential energy.

8.9 Escape Velocity (V_esc)

In physics of the motion (mechanics), escape velocity is the minimum speed a particle needs to overcome a gravitational field of any planetary body or satellite and escape to infinity without any further acceleration. In another word, it is the speed at which the kinetic energy of an object equals the gravitational potential energy biding it to the celestial body. This speed is independent of the escaping object's mass and direction, but it does depend on the mass and radius of the gravitational source.

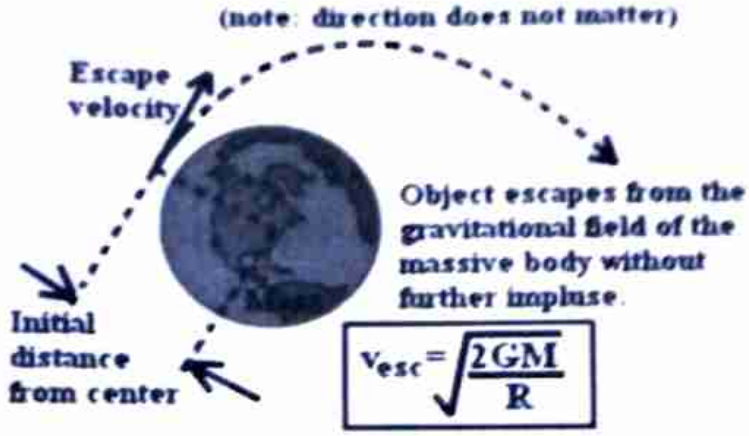


Figure 8.11: Description of a particle overcoming the force of gravitational field

It can be expressed mathematically as:

$$V_esc = \sqrt{\frac{2GN}{r}} \tag{8.19}$$

where G is the gravitational constant, M is the mass of the attracting (celestial) body, and r is the distance from its centre.

The formula shows that escape velocity is proportional to the square root of the mass (M) and inversely proportional to the square root of the distance (r) from the centre of the gravitational body. Implicating that, the higher the mass of the attracting body the higher the escape velocity. The smaller the radius of the attracting body the higher the escape velocity. For Earth, the escape velocity is about 11.2 km per second from the surface while escape velocity for moon is 2.4 km/s. This is the minimum speed required for any spacecraft to leave Earth and never return.

8.10 Satellite Motion and Orbit

Satellite motion describes the physics governing the path of one celestial body, or satellite, as it revolves around a more massive central body due to gravitational attraction. Satellites can be natural, like the moon orbiting Earth, or artificial, like the International Space Station. In physics, a satellite's orbit is a balance between its tangential velocity and the gravitational force from a central body, causing it to move in a curved path (either a circle or ellipse) instead of falling straight to Earth. This gravitational pull acts as a centripetal force, constantly directed towards the centre of the orbit, preventing the satellite from flying off in a straight line. The speed of the satellite determines the path; a certain speed for a given altitude results in a circular orbit, while faster speeds create elliptical paths.

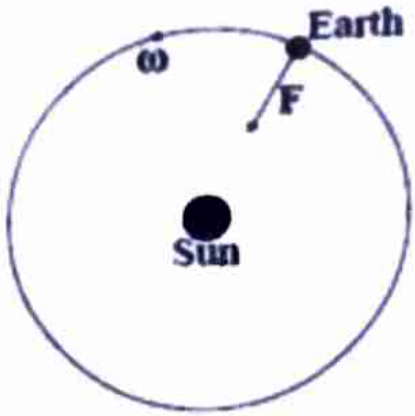


Figure 8.12: Description of satellite orbiting motion. The gravitational force supplies the centripetal acceleration

Consider a planet (Earth) of mass (m) in a circular orbit about the Sun at distance r from the center of Earth (Figure 14). It has centripetal acceleration directed toward the center of Sun. Sun's gravity is the only force acting, so Newton's second law gives

$$\frac{GMm}{r^2} = ma_c = \frac{mv_orbit^2}{r} \tag{8.20}$$

$$v_orbit = \sqrt{\frac{GM}{r}} \tag{8.21}$$

The orbital velocity can be estimated using centripetal and gravitational force. The escape velocity is $\sqrt{2}$ times more than orbital speed, which is about 40%, than the orbital velocity.

Types of orbits

Satellites are placed into specific orbits depending on their function.

- **Low Earth Orbit (LEO):** A relatively close orbit, from 160 to 2,000 km in altitude.
  1. Features: Higher-resolution imaging and shorter communication delays due to its proximity.
  2. Uses: Scientific research, Earth observation, and the International Space Station (ISS).
- **Medium Earth Orbit (MEO):** An intermediate orbit located between LEO and geostationary orbit.
  1. Features: Slower-moving than LEO, providing a longer view of a specific region.
  2. Uses: Global navigation systems like GPS.
- **Geostationary Earth Orbit (GEO):** A high-altitude orbit (35,786 km) directly above the equator.
  1. Features: With an orbital period of 24 hours, the satellite appears stationary relative to a point on Earth's surface.
  2. Uses: Weather monitoring and long-distance telecommunications, like television broadcasting.
- **Polar Orbit:** An orbit that passes over or near the Earth's north and south poles.
  1. Features: As the Earth rotates below, the satellite can view the entire globe over multiple passes.

2. Uses: Earth mapping, environmental monitoring, and reconnaissance.

Rules for Orbital motion

1. The acceleration of an orbiting object only depends on the central mass.
2. If the central mass increases, the velocity of the orbit must increase to stay in orbit.
3. If the separation increases, the velocity decreases.
4. Objects orbit around the center of mass.
5. Objects can escape orbits: If an object gains enough velocity, it may escape (change from a bound to unbound orbit) and escape velocity from Earth $\approx 11.17 km/s$

Recap

- The coefficients of friction, both static and kinetic, have no units.
- The coefficients of friction must be less than 1.
- **When the net force on a particle is zero**
- The particle is in equilibrium.
- It has a constant velocity.
- It has zero acceleration.
- **When the net force on a particle is not zero**
- The particle is dynamic
- The velocity is not constant
- The particle possesses acceleration
- **In uniform circular motion,**
- The acceleration vector is directed toward the center of the circle. The motion is governed by Newton's second law
- Acceleration increases with force and decreases with mass
- Gravity causes attraction between all objects in the universe
- The attraction increases with mass and decreases with distance
- Gravitational acceleration is independent of an object's mass
- Gravitational forces cause objects to orbit in circular motion (or ellipse)
- Objects moving too fast can escape orbit
- The escape velocity is $\sqrt{2}$ times greater than orbital speed, which is about 40%, than the orbital velocity.
- The Earth's escape velocity is about $11.2 km/s$ from the surface while escape velocity for moon is $2.4 km/s$.

8.11 Tutorial 7

Problem 8.1 A $52-N$ sled is pulled across a cement sidewalk at constant speed. A horizontal force of $36N$ is exerted. What is the coefficient of kinetic friction between the sidewalk and the metal runners of the sled?

Problem 8.2 The sled from the example is now on packed snow. The coefficient of friction is 0.12. If a person weighing $650N$ sits on the sled, what force is needed to slide the sled across the snow at a constant speed?

Problem 8.3 A $100-N$ body is shown suspended from a system of cords. What is the tension in the horizontal cord?

Problem 8.4 A skier of mass $65.0kg$ is pulled up a snow-covered slope at constant speed by a tow rope that is parallel to the ground. The ground slopes upward at a constant angle of 26^0 above the horizontal and you can ignore friction. Calculate the tension in the tow rope.

Problem 8.5 A man is dragging a trunk up the loading ramp of a mover's truck. The ramp has a slope angle of 20.0^o and the man pulls upward with force $\boldsymbol{F}$ whose direction makes an angle of

30.0° with the ramp. How large a force $\boldsymbol{F}$ is required in order for the component F_x parallel to the ramp to be 28.0.N? How large will the component F_y then be?

Problem 8.6 A $20-kg$ box rests on a frictionless ramp with a 15° slope. A mover pulls up on a rope attached to the box to pull it up the incline. If the rope makes an angle of 40° with the horizontal, what is the force the mover must exert on the box to give it an acceleration of $1m/s^2$?

Problem 8.7 A 5-kg block is held at rest against a vertical wall by a horizontal force. What is the minimum horizontal force needed to prevent the block from falling if the coefficient of friction between the wall and the block is $m_s = 0.40$?

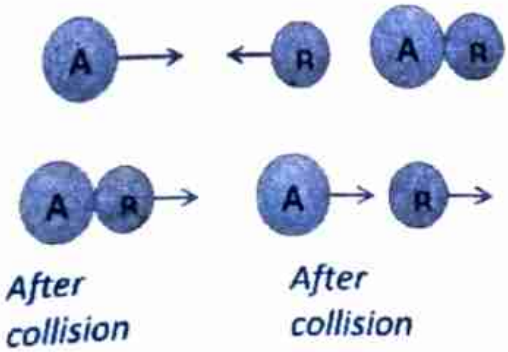
9. Conservation laws

9.1 Introduction

Conservation law also known as law of conservation, in physics, it is a principle that states that a certain physical property (i.e., a measurable quantity) does not change in the course of time within an isolated physical system. In classical physics, laws of this type govern energy, momentum, angular momentum, mass and electric charge. In particle physics, other conservation laws apply to properties of subatomic particles that are invariant during interactions. An important function of conservation laws is that they make it possible to predict the macroscopic behavior of a system without having to consider the microscopic details of the course of a physical process or chemical reaction.

9.2 Conservation of linear momentum

Conservation of linear momentum expresses the fact that a body or system of bodies in motion retains its total momentum, the product of mass and velocity unless an external force is applied to it. In an isolated system, like the universe, there are no external forces, so momentum is always conserved. Because momentum is conserved, its components in any direction will also be conserved. Some applications of the law of conservation of momentum is important in the solution of collision problems. The increased forward momentum of the rocket is equal but opposite in sign to the momentum of the ejected exhaust gases.



Consider two balls A and B of masses m_A and m_B moving in opposite direction along a straight line with velocities u_A and u_B. They collide after time t. After the collision, their velocities become v_A and v_B.

Force exerted by A on B is:

F_{AB}= Rate of change of momentum of

$$A = F_{AB} = \frac{m(V_A - U_A)}{t} \tag{9.1}$$

Force exerted by A on B is: F_{BA} = Rate of change of momentum of

$$B = F_{BA} = \frac{m(V_B - U_B)}{t} \tag{9.2}$$

We assume that no other external unbalanced forces are acting on the balls.

According to Newton's 3rd law action and reaction are equal opposite.

$$F_{AB} = -F_{BA} \tag{9.3}$$

$$\frac{m_A(V_A - U_A)}{t} = \frac{m_B(V_B - U_B)}{t} \tag{9.4}$$

$$m_A(V_A - U_A) = -m_B(V_B - U_B) \tag{9.5}$$

$$m_A U_A + m_B U_B = m_A V_A + m_B V_B) \tag{9.6}$$

Thus, total momentum before collision is equal to total momentum after collision.

■ **Example 9.1** A 1 kg ball moving at $12m/s$ collides head on with $2kg$ ball moving with $24m/s$ in opposite direction. What are the velocities after collision if $e = 2/3$?

Solution

Let v_1 and v_2 be the final velocities of the mass

since the linear momentum is conserved in the collision Momentum before = Momentum after

$$1 \times 12 + 2 \times (-24) = 1 \times v_1 + 2 \times v_2$$

Which becomes

$$-36 = V_1 + 2V_2.$$

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$$\frac{2}{3} = \frac{v_2 - v_1}{12 - (-24)}$$

Solving 1 and 2

$v_2 = 14$ and $v_1 = -28$ ■

■ **Example 9.2** A moving bullet hits a solid target resting on a frictionless surface and gets embedded in it. What is conserved in it? (a) momentum alone (b) Kinetic energy alone (c) both momentum and kinetic energy (d) Neither kinetic energy nor momentum.

Solution

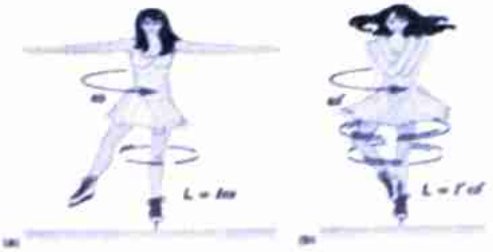
Since no external force is present, Momentum is conserved in the collision Since the collision is in elastic .KE is not conserved. ■

Exercise 9.1 A stationary body of mass 3 kg explodes into three equal parts. Two of the pieces fly off at right angles to each other with the velocities $2im/s$ and $3jm/s$. If the explosion takes place in 10^{-3} sec. Find out the average force action on the third piece in N. (a) $(-2i-3j)10^3$ (b) $(2i+3j)10^3$ (c) $(2i-3j)10^{-3}$ d. none of these ∎

9.3 Conservation of angular momentum

Conservation of angular momentum of rotating bodies is analogous to the conservation of linear momentum. Angular momentum is a vector quantity whose conservation expresses the law that a body or system that is rotating continues to rotate at the same rate unless a twisting force, called a torque, is applied to it. The angular momentum of each bit of matter consists of the product of its mass, its distance from the axis of rotation, and the component of its velocity perpendicular to the line from the axis.

An example of conservation of angular momentum is seen in an ice skater executing a spin, as shown in. The net torque on her is very close to zero, because 1) there is relatively little friction between her skates and the ice, and 2) the friction is exerted very close to the pivot point.



Conservation of Angular Momentum: An ice skater is spinning on the tip of her skate with her arms extended. Her angular momentum is conserved because the net torque on her is negligibly small. In the next image, her rate of spin increases greatly when she pulls in her arms, decreasing her moment of inertia. The work she does to pull in her arms results in an increase in rotational kinetic energy.

$$L = I\omega \tag{9.7}$$

In the above equation, **L** represents angular momentum. The moment of inertia is represented by **I**, and the angular speed is omega. Let's see how this equation works

Shape	*Axis of Rotation*	*Moment of Inertia*
sphere	center of sphere	$I = 2mr^2/5$
point mass	distance r from point mass	$I = mr^2$
solid cylinder	axis of symmetry	$I = mr^2/2$
rod (length L)	center (not axis of symmetry)	$I = mL^2/12$
rod (length L)	one end	$I = mL^2/3$

Example 9.3 What is the moment of inertia of a thin rod rotating about one end that has a mass of 5 kg and is 5 meters long?

Solution

$$I = \frac{1}{3}ML^2$$

$$I = \frac{1}{3}5 \times 5^2$$

$$I = \frac{25}{3} kgm^2$$

■

■ **Example 9.4** What is the angular momentum of the object in number 1 above if it is rotating at $200 rad/s$?
Solution

$$L = I\omega$$

$$L = \frac{25}{3} \times 200$$

$$L = \frac{5000}{3} kgm^2/s$$

■

■ **Example 9.5** A $20kg$ disk rotating about its center of mass has an angular momentum of $520 kgm^2/s$. It is spinning at $2 rad/s$. What is the disk's radius?
Solution

$$L = I\omega$$

$$L = \frac{1}{2}MR^2\omega$$

$$520 = 20 \times R^2 \times 2$$

$$520 = 20 \times R^2$$

$$R = 5.1m$$

■

■ **Example 9.6** A hollow ball rolling downhill reaches an angular velocity of $170 rad/s$. The ball's radius is $0.2m$. The ball's angular momentum is $5 kgm^2/s$. What is the mass of the ball?
Solution

$$L = I\omega$$

$$L = \frac{22}{3}MR^2\omega$$

$$5 = \frac{2}{3}M \times 0.2^2 \times 170$$

$$M = 1.1kg$$

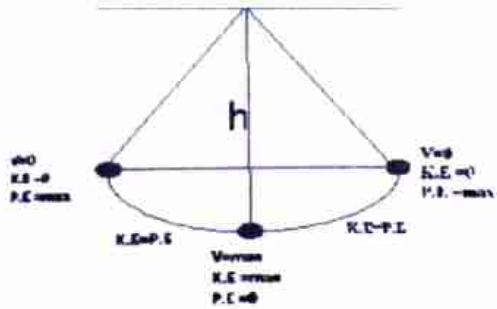
■

9.4 Conservation of Charge

Conservation of Charge states that the total amount of electric charge in a system does not change with time. At a subatomic level, charged particles can be created, but always in pairs with equal positive and negative charge so that the total amount of charge always remains constant. Like Coulomb's law, the principle of charge conservation is a fundamental law of nature. According to this principle, the charge of an isolated system cannot change. If an additional positively charged particle appears within a system, a particle with a negative charge of the same magnitude will be created at the same time; thus, the principle of conservation of charge is maintained. In nature, a pair of oppositely charged particles is created when high energy radiation interacts with matter; an electron and a positron are created in a process known as pair production.

9.5 Conservation of energy

Conservation of energy implies that energy can be neither created nor destroyed, although it can be changed from one form (Mechanical, kinetic, chemical, etc.) into another. In an isolated system the sum of all forms of energy therefore remains constant. Conservation of energy, principle of physics according to which the energy of interacting bodies or particles in a closed system remains constant. The first kind of energy to be recognized was kinetic energy, or energy of motion. In certain particle collisions, called elastic, the sum of the kinetic energy of the particles before collision is equal to the sum of the kinetic energy of the particles after collision. The notion of energy was progressively widened to include other forms. The kinetic energy lost by a body slowing down as it travels upward against the force of gravity was regarded as being converted into potential energy, or stored energy, which in turn is converted back into kinetic energy as the body speeds up during its return to Earth. For example, when a pendulum swings upward, kinetic energy is converted to potential energy. When the pendulum stops briefly at the top of its swing, the kinetic energy is zero, and all the energy of the system is in potential energy. When the pendulum swings back down, the potential energy is converted back into kinetic energy. At all times, the sum of potential and kinetic energy is constant.



9.6 Angular Velocity and Acceleration

The angular velocity describes how fast a body is rotating. The average angular velocity

$$\omega = \frac{angular\ displacement}{change\ in\ time\ of\ rotation} = \frac{d\theta}{dt} = \frac{\theta_2 - \theta_1}{t_2 - t_1} \tag{9.8}$$

ω is measured in radians per seconds (rad/s)

The angular acceleration describes the rate of change of rotational motion. The angular acceleration ? of the particle is define as

$$\alpha = \frac{angular\ velocity}{change\ in\ time\ of\ rotation} = \frac{dv}{dt} = \frac{\omega_2 - \omega_1}{t_2 - t_1} \tag{9.9}$$

For constant angular acceleration

$$\alpha = \frac{\omega - \omega_0}{t - t_0} \tag{9.10}$$

ω_o is the angular velocity at time $t_o = 0$

$$\omega - \omega_o = \alpha t \tag{9.11}$$

$$\omega = \omega_o + \alpha t \ldots\ldots\ldots\ldots\ldots\ldots i \tag{9.12}$$

$$\theta = (\frac{\omega + \omega_0}{2})t \tag{9.13}$$

Substitute in 1

$$\theta = (\frac{\omega_0 + \omega_o + \alpha t}{2})t \tag{9.14}$$

$$\theta = \omega_o t + \frac{1}{2}t^2 \ldots\ldots\ldots\ldots\ldots\ldots ii \tag{9.15}$$

Eliminating t from i and ii

$$\omega^2 = \omega_0^2 + 2\alpha\theta \ldots\ldots\ldots\ldots\ldots\ldots iii \tag{9.16}$$

Period T of revolution is defining as the time taken for a complete revolution or a cycle. $T = 1/f$ Where f is frequency

Frequency F is defining as the number of revolution or cycle completed in one second.

Angular velocity $\omega = 2\pi f$

Tangential velocity of object $V = \omega r$

But, $\omega = 2\pi f$

Therefore, $V = 2\pi f r$

9.7 Moment of inertial

is a quantity expressing a body's tendency to resist angular acceleration, which is the sum of the products of the mass of each particle in the body and the square of its distance from the axis of rotation. The moment of inertial of particle of mass m about a line or axis is define by $I = \sum mr^2$. It is measured in kgm^2. The K.E of particle of mass m under rotational dynamic is

$$K.E = \frac{1}{2}mr^2\omega^2 \tag{9.17}$$

But $I = mr^2$.
Therefore, $K.E = \frac{1}{2}I\omega^2$
Loss in the

$$K.E = \frac{1}{2}I\omega_0^2 - \frac{1}{2}I\omega^2 \tag{9.18}$$

9.7.1 Moment of Inertial of Various Objects

1. The moment of inertial of a thin rod $I = \frac{1}{12}ML^2$
2. The moment of inertial of rectangular plate $I = \frac{1}{12}M(a^2 + b^2)$
3. The moment of inertial of a sphere $I = \frac{2}{5}MR^2$
4. The moment of inertial of a ring with axis through its centre, perpendicular to the plane of the ring $I = MR^2$
5. The moment of inertial of hollow cylinder about its axis of symmetry $I = \frac{1}{2}M(R_1^2 + R_2^2)$
6. The moment of inertial of hollow cylinder with thin wall about its axis of symmetry $I = MR^2$

Angular momentum is the quantity of rotation of a body. It is define as the product of its moment of inertia and its angular velocity. The angular momentum is given by

$$L = \omega \sum mr^2 \tag{9.19}$$

$$L = \omega \sum mr^2 = \omega I \tag{9.20}$$

$$L = \omega I \tag{9.21}$$

It is measured in Kgm^2/s.

▪ **Example 9.7** An object attached to a $0.5m$ string does 4 rotations in one second. Find (a) Period (b) Tangential velocity (c) Angular velocity of the object.
Solution

(a) If the object does 4 rotations in one second, its frequency becomes; $f = 4Hz$ So,

$$T = \frac{1}{f} = \frac{1}{4}s$$

b) Tangential velocity of the object

$$V = \omega r = 2\pi f r = 12m/s$$

c) Angular velocity of the object $\omega = 2\pi f = 24rad/s$. ▪

▪ **Example 9.8** An object having mass m does rotational motion. Its angular velocity is ω and radius of motion path is r. Find kinetic energy of the object in terms of r, ω, and m.
Solution

$$E_K = \frac{1}{2}mV^2$$

But $V = \omega r$
Therefore,

$$E_K = \frac{1}{2}m(\omega r)^2$$

$$E_K = \frac{1}{2}m\omega^2 r^2$$

■ **Example 9.9** Stone having mass $0.5kg$ rotates in horizontal. It is hanged on 1m rope. If the tension on the rope is $80N$, find the frequency of the motion.
Solution

$$F_{net} = 80N = m\omega^2 r$$
$$80 = m4\pi^2 f^2 r$$
$$80 = 0.5 \times 4 \times 3.142^2 \times f^2 \times 1$$
$$f = 2Hz$$

■

■ **Example 9.10** An object in horizontal rotates on a circular road with $10m/s$ velocity. It does 120 revolutions in one minute. Find frequency and period of the object.
Solution
a) 60s x frequency = 120 revolution
f=2 revolution/second

$$T = \frac{1}{f} = \frac{1}{2}s$$

■

■ **Example 9.11** A device does 1800 revolution in 6 minutes. Find angular velocity and period of this device. $(\pi = 3)$.

Solution

$$1800xT = 6x60s$$
$$T = \frac{360}{1800} = \frac{1}{5}s$$
$$\omega = \frac{2\pi}{T} = 30radian/s$$

■

■ **Example 9.12** A marble is thrown with velocity V from point A. If no force is exerted on surface by marble at point B, find the force exerted on point A by marble. (Assume that velocity of marble is constant.)
Solution
Since no force is exerted on point B by marble;

$$mg = \frac{mV^2}{R}$$
$$V^2 = g.R$$

Force exerted by marble on point A;

$$F_A = G + F$$

$$F_A = mg + \frac{mV^2}{R}$$

(we put $V^2 = gR$ into the equation)

$$F_A = mg + \frac{mgR}{R}$$

$$F_A = 2mg$$

Example 9.13 Two objects rotate with same frequency around point O. One of the objects has mass m and other one has mass 9m. If centripetal forces exerted on objects having mass m is F_1, and object having mass 9m is F_2, find ratio of F_1/F

Solution

$$V_1 = 2\pi f r$$

$$V_2 = 2\pi \times 3r \times f$$

$$V_2 = 3V_1$$

$$\frac{f_1}{f_2} = \frac{mV_1^2}{r} \div \frac{9mV_2^2}{3r} = \frac{V_1^2}{r} \times \frac{3r}{9V_2^2}$$

$$\frac{f_1}{f_2} = \frac{1}{27}$$

Example 9.14 A turntable player starting from rest is rotating with a constant angular acceleration of $2.5 rad/s^2$. Find the angular displacement of the turntable player and the angular velocity of the turntable player 5 seconds later.

Solution

From

$$\theta = \omega_o t + \frac{1}{2}\alpha t^2$$

At t = 0, $\omega_o = 0$

$$\theta = 0 + \frac{1}{2} \times 2.5 \times 5^2$$

$= 31.25 rad$ or $4.97 rev$

$$\omega = \omega_o + \alpha t = 0 + 2.5 \times 5 = 12.5 rad/s^2$$

Example 9.15 A body with uniform angular velocity of $2 rev/s$ at $t = 0$ has angular acceleration of $0.5 rev/s^2$. After $10s$, what will be (i) the final angular velocity (ii) the angular displacement?

Solution

At $t = 0$, $\omega_o = 2 rev/s^2 = 2 \times 2\pi = 4\pi rad/s$

$$\alpha = 0.5 \times 2\pi = \pi rad/s$$

using $\omega = \omega_o + \alpha t = 4\pi \times 10\pi = 14\pi rad/s$

$$\theta = \omega_o t + \frac{1}{2}\alpha t^2$$

$$= 4\pi \times 10 + \frac{1}{2}\pi \times 10^2$$

$$= 40\pi + 50\pi = 90\pi rad$$

For a body moving in rotational motion there are two components of acceleration. The centripetal acceleration a_R and tangential acceleration a_T.

$$a_R = \frac{v^2}{R}$$

$$a_T = \frac{dv}{dt} = \frac{d}{dt}\omega r = r\frac{d\omega}{dt} = r\alpha$$

The magnitude of instantaneous acceleration a = $\sqrt{a_R^2 + a_T^2}$ ▪

▪ **Example 9.16** An electric fan blade 3cm in diameter is rotating about a fixed axis with an initial velocity of 5 rev/s. If the angular acceleration is 3 rev/s^2, determine

1. the angular velocity after 2s
2. (ii) the number of revolution completed in this time interval
3. (iii) the tangential velocity of a point on the tip of the blade at t = 2s (iv) the resultant acceleration of a point on the tip of the blade at t = 2s.

Solution

$$\omega_o = 5 \times 2\pi rad/s = 10\pi rad/s$$

$$\alpha = 3 \times 2\pi rad/s^2 = 6\pi rad/s^2$$

i) using $\omega = omega_o + \alpha t$

$$\omega = 10\pi \times 2 + 6\pi \times 2 = 20\pi + 12\pi = 32\pi rad/s$$

ii) using

$$\theta = \omega_o t + \frac{1}{2}?t^2$$

$$\theta = 10\pi \times 2 + \frac{1}{2} \times 6\pi \times 4^2 = 20\pi + 48\pi = 68\pi rad$$

$$2\pi rad = 1rev$$

$$68\pi rad = \frac{68\pi}{2\pi} rev = 34rev$$

iii) $V = \omega r = 22\pi \times 310^{-2} = 2.07m/s$

iv) $a_R = \frac{v^2}{r} = \frac{2.07^2}{3\times 10^{-2}} = 142.8m/s^2$

$$a_T = r\alpha = 3 \times 10^{-2} \times 6\pi = 0.565m/s^2$$

Resultant acceleration =

$$\sqrt{a_R^2 + a_T^2} = (142.8^2 + 0.565^2)^{1/2} = 142.8m/s^2$$

▪

9.8 Tutorial 8

Problem 9.1 A body with angular velocity of $2rev/s$ at $t = 0$ has angular acceleration of $0.5rev/s^2$. After 10s, what will be:

1. The final angular velocity
2. The average angular velocity
3. The angular displacement

Problem 9.2 A wheel is making $360rev/min$ about its center. Calculate the angular velocity at any point on the wheel about the center. Determine also the speed of a particle at a distance 5 cm from the center.

Problem 9.3 The tangential velocity fir machining cast iron is about $0.64m/s$. At how many rev/min should a piece of stock $0.04m$ in diameter be rotated in a lathe machine?

Problem 9.4 A wheel rotates with a constant angular velocity of 15 rad/s. Find

1. The radial acceleration of a point 50cm from the axis
2. The tangential velocity of the point

Problem 9.5 A weight system for an Olympic weight-lifter consists of two spherical masses of $15.0kg$ each and are connected by a light rigid rod which is $1.5m$ long. Determine the moment of inertial of the body about an axis normal to it at its centre.

1.

10. Space and Time; Galilean Invariance

10.1 Introduction

In the beginning of the year 1905, Albert Einstein was an unknown 25 year old clerk in the Swiss patent office. By the end of that year, he had published papers of great importance on Brownian motion, photoelectric effect and special theory of relativity, one of which he was awarded the Nobel Prize in Physics.

The special theory of relativity has made wide-ranging changes regarding our understanding of nature, but Einstein based it on just two simple postulates. The first postulate, principle of relativity, also called Galilean relativity or Galilean Invariance forms the basis of discussion in this chapter. However, before we can have a good understanding on Galilean relativity, we need to understand the concepts of space and time.

10.2 Space and time

- In Physics or physical sciences, space is referred to the three-dimensional coordinate points. In other word, it is an arrangement of all possible point locations where logical statement(s) can be derived to solve a specific problem.
- Time is a one-dimensional quantity that is independent of space.It could be regarded as an origin from which to take duration past or future to any other time instance.
- In classical mechanics, time is separated from space,
- however, in special relativity, time and space are fused together making four-dimensional geometry called space-time

10.3 Frame of reference

The basic idea of motion is *change of position with time*. We say that an object is moving if it occupies different positions in space at different times. If an object does not change its position in space as time flows, we say that the object is at rest.

Our perception tells us that space exists as a stationary, permanent and absolute background in which objects can be placed or moved without any interaction; and that time is also absolute and

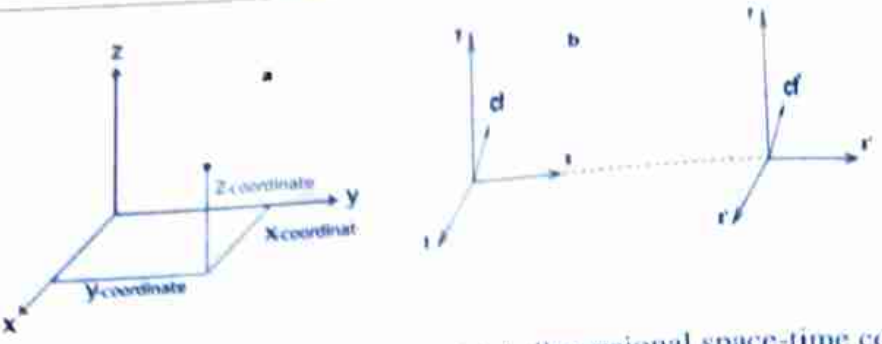


Figure 10.1: (a) 3-dimensional coordinates (b) 4-dimensional space-time coordinate

flows on without relation to the presence of any physical object or event. According to Newtonian absolute space and time, motion of an object is characterized by a change of position in space as time evolves.

However, since this absolute space cannot be seen, how then do we fix the position of an object in this space? Newton regarded the distant stars in the sky at rest, and defined the position of other objects in relation to them. According to astronomers, none of the cosmological objects are at rest in this 'absolute space' with respect to which we can define the position of another object. Therefore, at any instant of time, the position of an object is always defined by the 'observer' with respect to himself. An **observer**, therefore, first defines a local space in which the positions of other objects are then described. This local space to which an observer is attached is called a *frame of reference*.

Hence, in order to describe physical events that occur in space and time such as the motion of bodies, we introduce a coordinate system. Its spatial and temporal coordinates can now specify a space-time event by placing an observer at the origin of the coordinate system. Thus, a **frame of reference** can be defined as *an arbitrary set of coordinates (space-time) that describes the position, velocity and acceleration of objects.*

Generally, we use the rectangular (Cartesian) coordinates (x, y, z) to determine the position of an object in space. Other coordinate systems are spherical polar coordinates (r, θ, ϕ) and cylindrical coordinates (ρ, ϕ, z). The figure below shows the position of a particle P described by the coordinates x, y and z in a frame of reference.

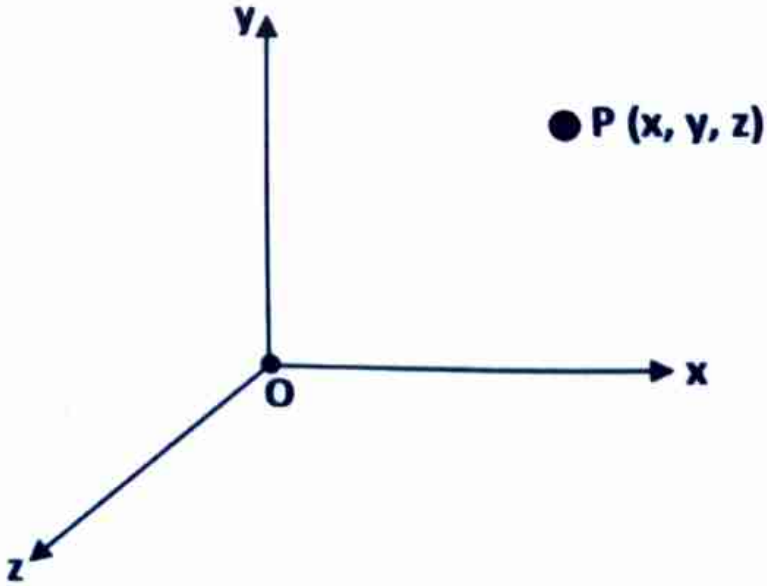


Figure 10.2: Frame of reference

10.4 Galilean Invariance

Galilean invariance or **Galilean relativity** states that *the laws of motion are the same in all inertial frames of reference.* An **inertial frame of reference** is *a three dimensional coordinate system which travel at constant velocity. It is a reference frame in relative motion to absolute space. In such a frame, the object does not accelerate.*

Galilee Galileo first described this principle using the example of a ship traveling at constant velocity, without rocking, on a smooth sea; any observer performing experiments below the deck would not be able to tell whether the ship was moving or stationary.

Among the axioms from Newton's theory are:

- There exists an absolute space in which Newton's laws are true.
- All inertial frames share a universal time.

10.4.1 Galilean Invariance Equation

Suppose we know from experiment that these laws of mechanics are true in one frame of reference, how do they look in another frame, moving with respect to the first frame? To find out, we figure out how to get from position, velocity and acceleration in one frame to these quantities in the second frame.

Note: The two frames must have a constant relative velocity, otherwise the law of inertia would not hold in both of them.

Suppose S' is proceeding relative to S at a speed v along the x-axis:

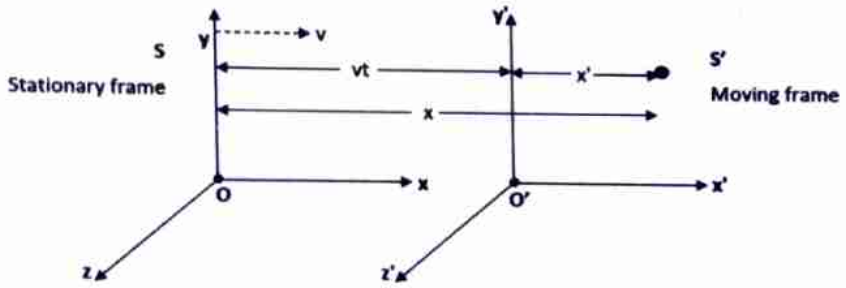


Figure 10.3: Frame of reference

We see from the diagram above that $x = x' + vt$. Also, $y = y'$ and $z = z'$ since there are no movements in these directions and $t = t'$ since time is universal. Thus, (x, y, z, t) in S frame corresponds to (x', y', z', t') in S' frame, where:

$$x' = x - vt$$

$$y' = y$$

$$z' = z$$

$$t' = t$$

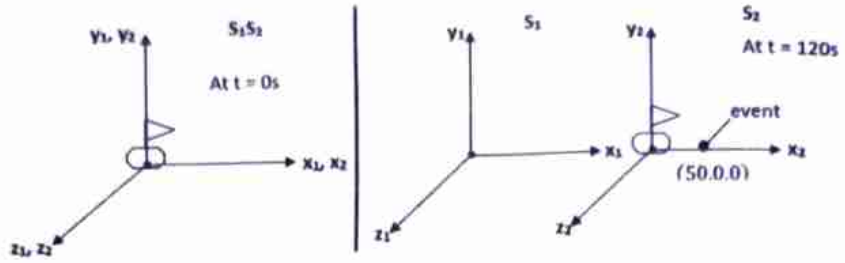
These are the Galilean Transformation Equations

That is how position transforms and is known as Galilean Transformations.

▪ **Example 10.1** A boat that is initially stationary begins to move with a velocity of $10m/s$ with respect to x - axis in the positive direction. Let the water on which it moves be reference frame 1 and the boat represents reference frame 2. Assume that reference frame 1 is always stationary and at t = 0 seconds, the two frames coincide. If we are in reference frame 2 and an event takes place by the boat at point ($x_2 = 50$, $y_2 = 0$, $z_2 = 0$) after the boat traveled for $120s$, find the equivalent

position of this event in reference frame 1.

Solution



Initially, the origin of the two frames (S_1S_2) coincide at $t = 0$ seconds
Since the boat is traveling in the positive x direction,

$$x_1 = x_2 + vt, \quad y_1 = y_2, \quad z_1 = z_2 \quad \text{and} \quad t_1 = t_2$$

However, at $t = 120s$, $x_2 = 50$, $y_2 = 0$, $z_2 = 0$ Therefore,

$$x_1 = x_2 + vt$$

$$x_1 = 50 + 10(120) = 50 + 1200$$

$$x_1 = 1250m$$

Thus, point (50, 0, 0) in frame 2 ≡ point (1250, 0, 0) in frame 1

■

10.4.2 Galilean Velocity Transformation

The velocity u_x' in S' frame in the x' direction as measured by an observer is given as:

$$u_x' = \frac{dx'}{dt'} = \frac{dx'}{dt} \tag{10.1}$$

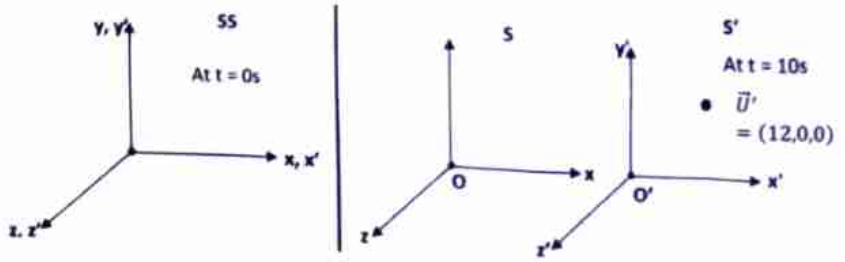
(since $t' = t$)

$$u_x' = \frac{d}{dt}(x - vt) = \frac{dx}{dt} - v \tag{10.2}$$

Therefore, $u_x' = u_x - v$ (Galilean velocity transformation)

■ **Example 10.2** At $t = 0$ seconds, two reference frames consisting of xyz plane coincide such that the origin of one is found at the origin of the second. However, as time progresses, the second reference frame moves in the positive direction along x-axis with velocity of $37m/s$, while the first frame of reference remains stationary. At a time of $t = 10$ seconds, a particle within the second reference frame has a velocity given by vector $\vec{U}' = (12,0,0)m/s$. Find the equivalent velocity U' of the particle with respect to the first reference frame.

Solution



Galilean velocity transformation equations are:

$$\vec{U}_x = \vec{U}'_x + v, \vec{U}_y = \vec{U}'_y, \vec{U}_z = \vec{U}'_z$$

$$\vec{U} = (\vec{U}_x, \vec{U}_y, \vec{U}_z) = (\vec{U}'_x + v, \vec{U}'_y, \vec{U}'_z) = (12 + 37, 0, 0)$$

$$\vec{U} = (49, 0, 0) m/s$$

(Velocity of particle w.r.t. reference frame 1)

■

10.4.3 Galilean Acceleration Transformation

How does acceleration transform?

$$a_x = \frac{du'_x}{dt'} = \frac{du'_x}{dt} \tag{10.3}$$

$$a_x = \frac{d}{dt}(u_x - v) = \frac{du_x}{dt} - \frac{dv}{dt} \tag{10.4}$$

Since v is constant, $\frac{dv}{dt} = 0$

Therefore, $a'_x = a_x$

This means that the acceleration is the same in both frames. Thus, we say that acceleration in both frames is Galilean Invariant. Velocity is not invariant because it differs in both frames by a constant factor i.e. the relative velocity of both frames. We can also show that the law of conservation of momentum (Newton's second law of motion) F = ma, is invariant under Galilean transformation.

In the S' frame of reference, $F' = ma'_x$

$$F' = m\frac{du'_x}{dt} = m\frac{d}{dt}[\frac{dx'}{dt}] \tag{10.5}$$

$$F' = m\frac{d}{dt}[\frac{d}{dt}(x - vt)] = m\frac{d}{dt}[\frac{dx}{dt} - v] \tag{10.6}$$

$$F' = m\frac{d}{dt}[u_x - v] \tag{10.7}$$

$$F' = m[\frac{du_x}{dt} - \frac{dv}{dt}] \tag{10.8}$$

But $\frac{dv}{dt} = 0$

Therefore, $F' = ma_x = F$

10.5 Universal Gravitation

Newton's law of universal gravitation states that a particle attracts every other particle in the universe with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centers. This gravitational force is the same as encountered in Newton's second law of motion, where:

$$F = ma = \frac{GMm}{r^2} \tag{10.9}$$

Where G is gravitational constant and r is the distance of separation between the masses.
Since force, F is Galilean invariant, we can conclude that the law of universal gravitation is also Galilean invariant, supporting the notion that all physical laws are invariant under Galilean transformation.

■ **Example 10.3** Show that the law of conservation of linear momentum for collision is Galilean invariant, given that $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

Solution

From Galilean velocity transformation equation,

$$u_1 = u_1' + v$$

$$u_2 = u_2' + v$$

$$v_1 = v_1' + v$$

$$v_2 = v_2' + v$$

Substituting these equations into the equation for momentum conservation, we have

$$m_1(u_1' + v) + m_2(u_2' + v) = m_1(v_1' + v) + m_2(v_2' + v)$$

$$m_1u_1' + m_1v + m_2u_2' + m_2v = m_1v_1' + m_1v + m_2v_2' + m_2v$$

Collecting like terms and subtracting, we have

$$m_1u_1' + m_2u_2' = m_1v_1' + m_2v_2'$$

This shows that the law of conservation of linear momentum is Galilean Invariant i.e. remain the same in different inertial frames of reference. ■

10.6 Tutorial 9

Exercise 10.1 A girl is riding a bicycle along a straight road at constant speed, and passes a friend standing at a bus stop (event 1). At a time of 60s later, the friend catches a bus (event 2). If the distance separating the events is 126m in the frame of the girl on the bicycle, what is the bicycle's speed?
(Ans: 2.10m/s) ■

Exercise 10.2 A police car traveling on a straight road at a constant speed of 30m/s catches up with a truck traveling at a constant speed of 20m/s. The police car reaches the truck 25s after it is first sighted. How far ahead of the police car was the truck when it was first sighted?
(Ans: 250m) ■

Exercise 10.3 Show that the law of conservation of kinetic energy is Galilean Invariant.
Hint: $(\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_2v_1^2 + \frac{1}{2}m_2v_2^2)$

MULTIPLE CHOICE QUESTIONS

Problem 10.1 Which of the following is NOT Galilean Invariant?
A. Force B. Acceleration C. Velocity D. Time

Problem 10.2 An arbitrary set of coordinates that describes the position, velocity and acceleration of objects is called
A. Observer B. Point C. Origin D. Reference frame

Problem 10.3 The Galilean Transformation equations for an object moving from (x, y, z, t) frame to (x', y', z', t') frame at constant velocity in the positive y axis is given as:
A. $x' = x - vt, y' = y, z' = z, t' = t$ B. $x' = x, y' = y - vt, z' = z, t' = t$
C. $x' = x - vt, y' = y - vt, z' = z, t' = t$ D. $x' = x, y' = y, z' = z, t' = t$

Problem 10.4 All laws of motion are the same in all inertial frame of reference is a statement of
A. Galilean Relativity B. Universal Gravitation C. Heisenberg Principle D. Keplers' Laws

Problem 10.5 An inertial frame of reference is one that
A. moves at different velocities B. is constantly accelerating C. is in relative motion to absolute space D. does not share a universal time

Problem 10.6 In Physics, frames of reference are classified by two main types:
A. True and fictional B. Inertial and non-inertial C. fast and slow D. real and imagined

Problem 10.7 According to theory of relativity by Einstein, what is considered the fourth dimension?
A. Horizontal dimension B. curled dimension C. space dimension D. time dimension

Problem 10.8 Which of the following quantities is Galilean Invariant?
A. Momentum B. Velocity C. Acceleration D. Kinetic Energy

Problem 10.9 Which of the following is the odd one out?
A. An athlete running at a constant speed in a straight line
B. A moving vehicle maintaining a constant velocity
C. A rope whirled round a stone and rotating at constant velocity
D. A stationary ship in a harbor

Problem 10.10 ..... defines the position of an object at any instant in time with respect to himself.
A. Spectator B. Observer C. Passenger D. Spaceman

Problem 10.11 Which of the following can be classified as a frame of reference that is not accelerating?
A. A merry-go round B. Earth C. a rotating fan D. a stationed train

Problem 10.12 The spatial and temporal coordinates of a physical event describes it's ..... event
A. Velocity-time B. Space-temperature C. Space-time D. Speed-temperature

Problem 10.13 Which of the following frames is most inertial?
A. Earth B. Sun C. the center of our galaxy D. an unaccelerated rocket

Problem 10.14 Laws of motion is not valid in a system which is
A. At rest B. inertial C. non-inertial D. moving with uniform velocity

Problem 10.15 15. Which of the following is introduced in order to describe physical events that occur in space and time such as the motion of bodies
A. Coordinate system B. Universal system C. Origin D. Galilean transformation

[1]This book was written and compiled using LaTeX2.0

11. Relativity of Simultaneity, Time Interval and Len

11.1 Introduction

Simultaneity, Time Interval and Length Contraction are among the consequences of the laws postulated by Albert Einstein. Notably, most of these consequences are applicable only when objects are moving at a significant percentage of the speed of light, otherwise they are insignificant. Einstein's theory of relativity is complicated making his work impossible to be fully discussed in a short lecture. But at the end of the class, you will be able to tell your friends you understand the following Learning Objectives:

- Describe relativity of simultaneity with its relevant mathematical expressions
- Describe relativity of time interval with its relevant mathematical expressions
- Describe relativity of length with its relevant mathematical expressions

10.1 Relativity of Simultaneity

11.2 Relativity of Simultaneity

In physics, the concept of relativity of simultaneity is that events which are simultaneous in one reference frame will not be simultaneous in a second inertial frame moving relative to the first frame. Meaning that Simultaneity is not absolute but depend on observers frame of reference.

In figure 11.1, we illustrated how a light signal with speed of light, c sent out from point B at

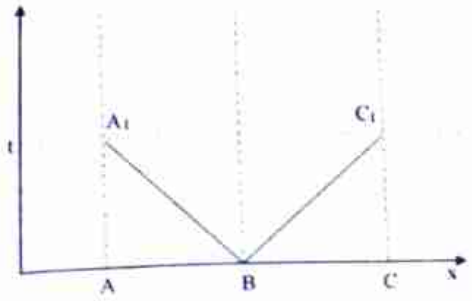


Figure 11.1: : Event happens in a frame at rest, S

t=0 moves forward and backward in the coordinate of $x = x_B + ct$. The two events occur at the intersection of A_1 and C_1 when the light arrives at point A and point C. The simultaneity at A and C as defined by line A_1 & C_1 is parallel to x-axis which joins points possessing the same value of t. Assuming the same event occur along x-axis in which three observers are equally spaced and move in a frame at speed v as depicted in figure 11.2. A light signal as described in figure 11.1, arrived at

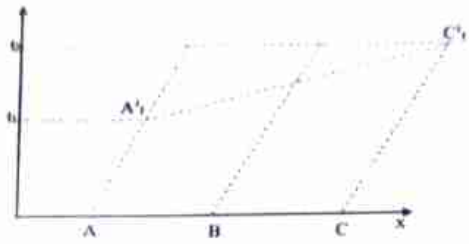


Figure 11.2: Event happens in a moving frame, S'

point A and point C given by the interception of A_1' and C_1'. This arrival is not simultaneous in S' because signal reaches A before C meaning that A is running to meet B, while C is running away. From the two figures 11.1 and 11.2, It can be established that the occurrence of events are functions of time and space.

11.2.1 Mathematical Expression of Simultaneity

Simultaneity can be expressed mathematically by considering the illustration below In figure 11.3,

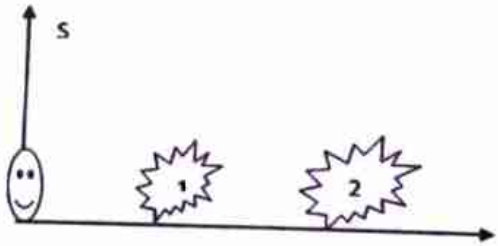


Figure 11.3: Two events happen in a frame at rest, S

two events happen simultaneously in S-frame of reference and the time at which these events are taking place is given as

$$t' = \gamma(t - \frac{v}{C^2}x) \tag{11.1}$$

Equation 11.1 holds provided $t_2 = t_1$
Since we have two events, let's assign time to each of the event.

$$t_1' = \gamma(t - \frac{U}{C^2}x_1) \tag{11.2}$$

$$t_2' = \gamma(t_2 - \frac{v}{C^2}x_2) \tag{11.3}$$

Let's subtract equation 11.2 from equation 11.3, then we have equation ??

$$t_2' - t_1' = \gamma(t_2 - \frac{v}{C^2}x_2) - \gamma(t_1 - \frac{v}{C^2}x_1) \tag{11.4}$$

Equation ?? can be resolved to equation ??

$$t_2^{'} - t_1^{'} = \gamma(t_2 - t_1) - \gamma\frac{v}{C^2}(x_2 - x_1) \tag{11.5}$$

Note: $\Delta t = t_2 - t_1$, $\Delta t^{'} = t_2^{'} - t_1^{'}$ and $\Delta x = x_2 - x_1$

The Substitution of $(t_2^{'} - t_1^{'})$ and $(x_2 - x_1)$ in equation ?? resulting to equation (11.6

$$\Delta t^{'} = \gamma(t_2 - t_1) - \gamma\frac{v}{C^2}(\Delta x) \tag{11.6}$$

Recall $t_2 = t_1$

$$\Delta t^{'} = \gamma(t_1 - t_1) - \gamma\frac{v}{C^2}\Delta x = \gamma(0) - \gamma\frac{v}{C^2}\Delta x \Delta t^{'} = -\gamma\frac{v}{c^2}\Delta x \tag{11.7}$$

Equation 11.7 is a mathematical expression of relativity of simultaneity.

Exercise 11.1 Two events happen simultaneously at space coordinate in metre, $x_1 = 0m$ and $x_2 = 100m$. Supposed S^i moving to the right at $0.9c$, what is the magnitude of time of the events that take place?

Solution

$$\Delta t^{'} = -\gamma\frac{v}{C^2}\Delta_x$$

Recall Lorentz Factor,

$$\gamma = \frac{1}{\sqrt{1 - \frac{U^2}{C^2}}}$$

which is also known as

$$\gamma = \frac{1}{\sqrt{1 - \beta 2}}$$

as

$$\beta^2 = \frac{U^2}{C^2}$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1}{\sqrt{1 - \frac{(0{\cdot}9C)^2}{C^2}}} = 2{\cdot}3$$

$$\Delta t^{'} = -2{\cdot}3x\frac{0.9c}{c^2}x^{1000} = -6{\cdot}9 \times 10^{-6}$$

$$\Delta t^{'} = -6{\cdot}9\mu s$$

■

11.3 Relativity of Time Interval (Time Dilation)

A moving clock ticks slowly in an inertial frame of reference than the clock at rest. That is, Time interval appears longer in a moving frame than the same interval as seen at rest. But as human being, we are used to time being a constant, but special relativity tells us that the passage of time actually varies depending on your frame of reference.

Let's consider a light clock two mirrors (A and B) in which a beam of light is reflecting back and forth between the mirrors. One tick is recorded when the light goes from one to the other back again and the proper time measured is denoted as Δt_0

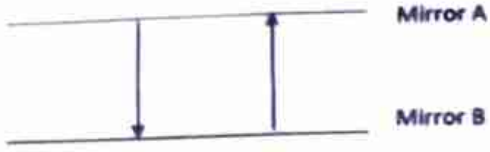


Figure 11.4: Reveals a scenario where the observer and the clock are at the same inertial frame.

Let's separate the mirror A and B by a distance (d) in which the light moves with speed of light called c. The proper time for one tick is given as equation ??

$$\Delta t_0 = \frac{2d}{c} \tag{11.8}$$

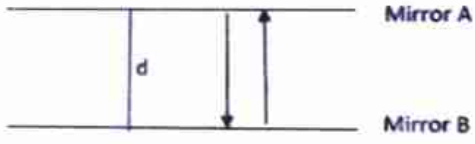


Figure 11.5: Reveals the separation of the two mirrors by a distance 'd'

11.3.1 Mathematical Expression of Time Dilation

Let's consider another scenario in which the observer is not in same inertial frame of reference with the light clock. The light clock is moving with velocity denoted as Um/s in the x-direction. The light takes different paths and the time taken for one tick as an observer time is given as equation ??

$$\Delta t = \frac{2L}{c} \tag{11.9}$$

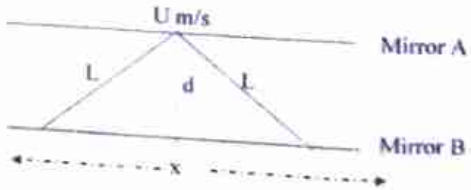
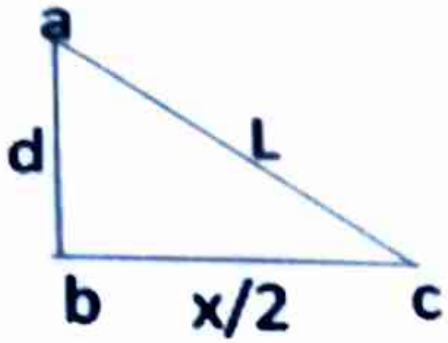


Figure 11.6: Reveals the paths taken by the light.

In figure 11.6, we have two triangles separated by a height, d. let's consider the right-side triangle and apply Pythagorean theorem



$$(ac)^2 = (ab)^2 + (bc)^2$$

$$L^2 = d^2 + (\frac{x}{2})^2$$

$$L^2 = d^2 + \frac{x^2}{4} \tag{11.10}$$

Let's make " d" the subject of the formula in equation ?? and L in equation ??. Having solved for d and L in equation ?? and ??, then substitute for L and d in equation ??.

$$(\frac{C\Delta t}{2})^2 = (\frac{C\Delta t_0}{2})^2 + \frac{(x)^2}{4} \tag{11.11}$$

Recall velocity, U is the distance over time. Therefore, $x = U\Delta t$. then, substitute for x in equation (11.11).

$$\frac{C^2\Delta t^2}{4} = \frac{C^2\Delta t_0^2}{4} + \frac{U^2\Delta t^2}{4} \tag{11.12}$$

Collect the like terms

$$\frac{C^2\Delta t^2}{4} - \frac{U^2\Delta t^2}{4} = \frac{C^2\Delta t^2}{4}$$

$$\frac{\Delta t^2}{4}(C^2 - U^2) = \frac{\Delta t_0^2}{4} \cdot C^2$$

$$\Delta t^2(\frac{C^2 - U^2}{C^2}) = \Delta t_0^2$$

$$\Delta t^2 = \frac{\Delta t_0^2}{(1 - \frac{U^2}{C^2}}$$

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{U^2}{C^2}}} \tag{11.13}$$

$$t' = \frac{t}{\sqrt{1 - \frac{U^2}{C^2}}} \quad \ldots\ldots\ldots\

▪ **Example 11.1** A nearest point π^0, has a lifetime of $8.4\text{x}10^{-17}$ seconds in its own rest frame before decaying into two photons. However, in a laboratory experiment where the pious observed to be moving, their lifetime is measured to be 4.2×10^{-16} seconds. Calculate the speed of the pious light.

Solution

$$t' = 4.2 \times 10^{-16} \& t = 8.4\text{x}10^{-17}$$

Recall time dilation

$$t' = \frac{t}{\sqrt{1 - \frac{U^2}{C^2}}}$$

$$t'\left(\sqrt{1 - \frac{U^2}{C^2}}\right) = t \ldots\ldots\ldots\

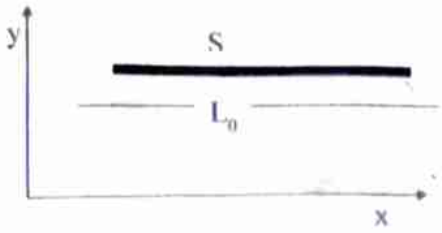


Figure 11.7: Bra frame of reference at rest, S

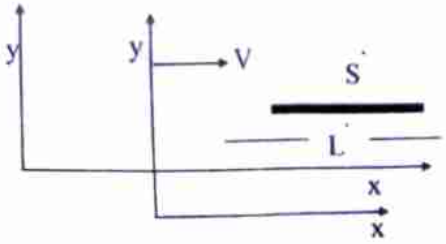


Figure 11.8: Ket at a moving frame of reference

In figure 11.7, Bra is measuring an object using a meter stick of length L in his own rest frame, S and in figure 11.8, Ket is measuring the same object in her own moving frame, S' at the same agreed time. The contracted length is given as equation 11.14

$$\Delta L' = \frac{\Delta L_0}{\sqrt{1-\frac{U^2}{C^2}}} \tag{11.14}$$

Recall Lorentz Factor, $\gamma = \frac{1}{\sqrt{1-\frac{U^2}{C^2}}}$

11.4.1 Mathematical Expression of Length Contraction

$$L' = \frac{L_o}{\gamma} \tag{11.15}$$

where L' is change in length, L_o is original length, u is velocity and c is the speed of light Equation 11.15 is a mathematical expression of length contraction.

▪ **Example 11.2** An observer on earth sees a rocket zoom by at $0.95c$. If the rocket is measured to be $5.5m$ in length, how long is the rocket ship as measured by the astronaut inside the rocket?
Solution

$$L' = 5.5m$$

$$\gamma = \frac{1}{\sqrt{1-\frac{U^2}{C^2}}}$$

Where; $U = 0.95c$

$$L' = \frac{L_o}{\sqrt{1-\frac{U^2}{C^2}}}$$

$$= L_o \times (1-\frac{U^2}{C^2})^{\frac{1}{2}}$$

$$5.5 = L_o \times (1 - \frac{(0.95C)^2}{C^2})^{\frac{1}{2}}$$

$$5.5 = L_o \times 0.312$$

$$L_o = \frac{5.5}{0.312} = 17.6m$$

11.5 Tutorial 10

Problem 11.1 An event is characterized by .....
A. Space and velocity
B. Velocity and time
C. Acceleration and time
D. Space and time
E. Acceleration only.
Ans: D

Problem 11.2 An inertial reference frame can be referred to as
A. Lorentz reference frame
B. Frame of reference
C. Galilean reference frame
D. Inertial space
E. C and D only
Ans: E

Problem 11.3 The following are the consequences of the principles of special relativity EXCEPT
A. Coordinate system
B. Length contraction
C. Galilean transformation
D. Time dilation
E. Velocity transformation
Ans: A

Problem 11.4 Consider an event that occur in a frame at rest in which three observers A, B and C are equally spaced on x-axis. When a light signal is sent out from B, the light arrived A and C. in what time would the arrival of the light signal be simultaneous to observer A and C? when time, t is
A. T = 1
B. T = 0
C. T = 2
D. A and B only
E. B only
Ans: E

Problem 11.5 One of the following equations is a mathematical expression for relativity of simultaneity
A. $\Delta t^{'} = \gamma \frac{V}{C^2} \Delta x$
B. $\Delta t^{'} = -\gamma \frac{V}{C^2} \Delta x$
C. $\Delta t^{'} = -\gamma \frac{V^2}{C^2} \Delta x$
D. $\Delta t = -\gamma \frac{V}{C^2} \Delta x$
E. $\Delta t^{'} = \gamma \frac{V}{C} \Delta x$
Ans: B

Problem 11.6 Lorentz factor as a scaling factor in the mathematical expression for both time dilation and length contraction is known as

11.5 Tutorial 10

A. $\gamma = \frac{1}{\sqrt{1-\beta^2}}$

B. $\gamma = \sqrt{1-\beta^2}$

C. $\gamma = \frac{1}{\sqrt{1+\beta^2}}$

D. $\gamma = \sqrt{1+\beta^2}$

E. $\gamma = \frac{1}{\sqrt{1-u/c}}$

Ans: A

1.



Figure 12.2: Image showing blackhole

12.1.1 Newton's Law of universal gravitation

- Newton's Law of Universal gravitation states that, *every particle in the Universe attracts every other particle with a force that is directly proportional to the product of the masses and inversely proportional to the square of the distance between them.*

Consider two particles of masses m1 and m2 and are separated by distance r. The magnitude of the gravitational force is

$$F = G\frac{m_1 m_2}{r^2} \tag{12.1}$$

G is the universal gravitational constant. Its value in SI unit is

$$G = 6.673 \times 10^{-11} Nm^2/kg^2 \tag{12.2}$$

Isaac Newton recognized that this force is responsible for the falling apple fruit from its tree to the earth. A force attracted the falling apple to the earth. Imagine how we walk around freely and not crawl. Imagine not hitting your head on something when taking steps. This force of attraction communicate between objects in the universe.

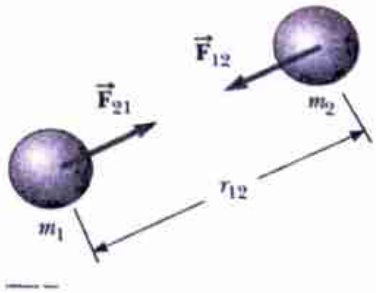


Figure 12.3: Forces on mass

Consider the figure 12.3. The force that mass m_1 exerts on mass m_2 is equal and opposite to the force mass m_2 exerts on mass m_1. The forces form a Newton's third law action-reaction. the force exerted by particle 1 on particle 2 is

$$\vec{F}_{12} = -G\frac{m_1 m_2}{r^2} r_{12} \tag{12.3}$$

The negative sign indicates that particle with m_2 is attracted to particle with mass m_1; hence, the force on m_2 must be directed toward m_1. By Newton's third law, the force exerted by particle with mass m_2 on particle with m_1. The two particles are equal in magnitude to and in the opposite direction. That is, these forces form an action-reaction pair, $\vec{F}_{21} = -\vec{F}_{12}$

12.2 Gravitation and the Principle of Superposition

Given nth group of particles, we find the resultant (net) gravitational force on any one of them from the others by using the principle of superposition. This is a general principle that says a net effect is the sum of the individual effects.

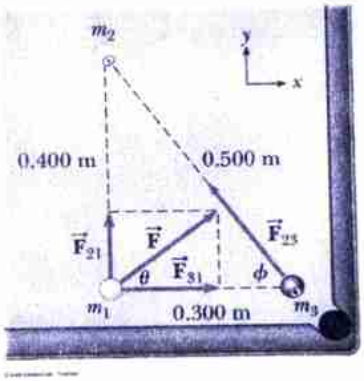
We can write the principle of superposition for the gravitational forces on particle 1 by the nth particles as

$$\vec{F}_{1,net} = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \vec{F}_{15} + \ldots + \vec{F}_{1n} \tag{12.4}$$

The equation can be written in compact vector form as

$$\vec{F}_{1,net} = \sum_{i=2}^{n} \vec{F}_{1n} \tag{12.5}$$

Example 12.1 Three $0.3kg$ billiard balls are placed on a table at the corners of a right triangle as shown in the figure. The sides of the triangle are of lengths $a = 0.4m$, $b = 0.3m$, and $c = 0.5m$. Calculate the gravitational force vector on the cue ball (designated m_1) resulting from the other two balls as well as the magnitude and direction of this force.



Solution

We first find the force exerted by m_2 on the cue ball

$$\vec{F}_{21} = G\frac{m_2 m_1}{r_{21}^2} j$$

$$= (6.67 \times 10^{-11})\frac{0.3 \times 0.3}{0.4^2} j$$

$$= 3.75 \times 10^{-11} jN$$

We then find the second force exerted by m_3 on the cue ball

$$\vec{F}_{31} = G\frac{m_3 m_1}{r_{31}^2} i$$

$$= (6.67 \times 10^{-11})\frac{0.3 \times 0.3}{0.3^2} i$$

$$= 6.67 \times 10^{-11} iN$$

Using the principle of superposition, the net force will be

$$\vec{F} = \vec{F}_{13} + \vec{F}_{12} = 6.67 \times 10^{-11} i + 3.75 \times 10^{-11} j = (6.67i + 3.75j) \times 10^{-11}$$

The magnitude of the force is

$$F = \sqrt{F_{31}^2 + F_{21}^2} = \sqrt{(6.67^2) + (3.75^2)} \times 10^{-11} N$$

$$F = 7.65 \times 10^{-11} N$$

To determine the force, we find the tangent of the angle for net force vector

$$\tan\theta = \frac{F_y}{F_x} = \frac{F_{21}}{F_{31}} = \frac{3.75 \times 10^{-11}}{6.67 \times 10^{-11}} = 0.562$$

$$\theta = \tan^{-1}(0.562) = 29.3^{o}$$

■

Exercise 12.1 Two spherical objects have masses of $200kg$ and $500kg$. Their centers are separated by a distance of 25 m. Find the gravitational attraction between them. ■

Exercise 12.2 Two spherical objects have masses of 1.5×10^5 kg and 8.5×10^2 kg. Their centers are separated by a distance of $2500m$. Find the gravitational attraction between them. ■

12.3 Free fall Acceleration and the Gravitational Force

Let us consider the magnitude of the force acting on a freely falling object of mass $m$near the Earth's surface. The gravitational force is equated to the force described by Newton's second law.

$$F = G\frac{M_E m}{R_E^2} = mg \tag{12.6}$$

M_E represent the mass of earth, R_E is the radius of the earth, m is the mass of the particle g is the free fall acceleration. g from the equation is given as

$$g = G\frac{M_E}{R_E^2} \tag{12.7}$$

In case where the object is located at distance h above the earth surface, the radius will be $(R_E + h)$

$$g = G\frac{M_E}{(R_E + h)^2} \tag{12.8}$$

Therefore, it follows that g decreases with increasing altitude. Values of g at various altitudes are listed in Table below.

Free-Fall Acceleration g at Various Altitudes Above the Earth's Surface

Altitude h (km)	g (m/s^2)
1 000	7.33
2 000	5.68
3 000	4.53
4 000	3.70
5 000	3.08
6 000	2.60
7 000	2.23
8 000	1.93
9 000	1.69
10 000	1.49
50 000	0.13
∞	0

Example 12.2 The International Space Station operates at an altitude of 350 km. Plans for the final construction show that 4.22×10^6 N of material, measured at the Earth's surface, will have been lifted off the surface by various spacecraft. What is the weight of the space station when in orbit?

Solution

We must know that the mass of the space station we will be the same, irrespective of the location. And using

$F = mg$; g on earth is assumed to be 9.8m/s^2

$$m = \frac{F}{g} = \frac{4.22 \times 10^6}{9.8} = 4.31 \times 10^5 kg$$

g at height $320km$ of the orbital location will be

$$g = G\frac{M_E}{(R_E + h)^2}$$

$$= \frac{(6.67 \times 10^{-11}) \times (5.98 \times 10^{24})}{(6.37 \times 10^6 + 0.350 \times 10^6)^2} = 8.83 m/s^2$$

The weight of the space station in the orbit will now be the product of the mass and the acceleration due to gravity in the orbit.

$$weight = mg = 4.31 \times 10^5 \times 8.83 = 3.80 \times 10^6 N$$

■

Exercise 12.3 Calculate the force of gravitation between the earth and the sun. Given that the mass of the earth $= 6 \times 10^{24}$ kg, mass of the sun $= 2 \times 10^{30}$ kg, average distance between the two is $1.5 \times 10^{11} m$ and $G = 6.67 \times 10^{-11} Nm^2 kg^{-2}$. ■

12.4 Kepler's Law and Motion of Planet



Humans regard the Earth as the center of the Universe from many observations of the movements of the planets, stars, and other celestial objects for thousands of years. Johannes Kepler (1571-1630), after a lifetime of study, worked out the empirical laws that govern these motions.
Below is the summary of Kepler's analysis in three statements

1. All planets move in elliptical orbits with the Sun at one focus.

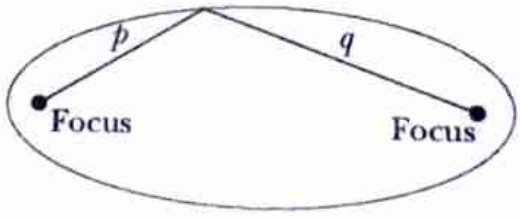
2. The radius vector drawn from the Sun to a planet sweeps out equal areas in equal time intervals.

3. The square of the orbital period of any planet is proportional to the cube of the semi-major axis of the elliptical orbit.

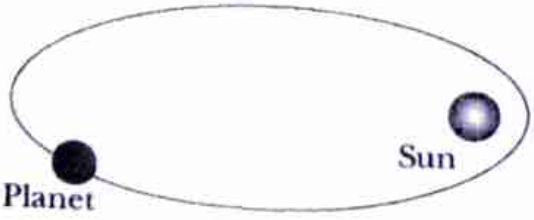
12.4.1 Kepler's First Law

All planets move in elliptical orbits with the Sun at one.

- Any object bound to another by an inverse square law will move in an elliptical path
- Second focus is empty



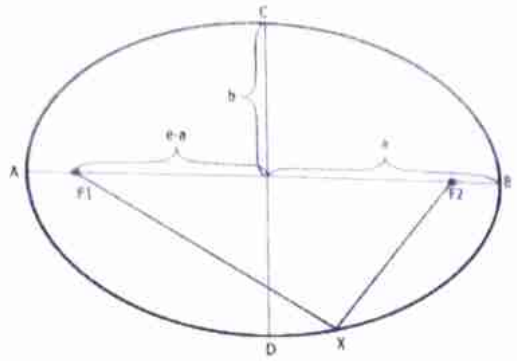
(a)



(b)

Ellipse Parameters

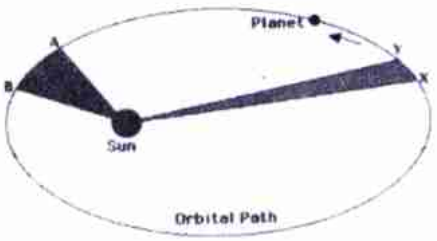
- Distance $a = AB/2$ is the **semi-major axis**
- Distance $b = CD/2$ is the **semi-minor axis**
- Distance from one focus to center of the ellipse is ea, where e is the **eccentricity**.
- Eccentricity is zero for a circular orbit, and gets larger as the ellipse gets more pronounced.



12.4.2 Kepler's Second Law

A line drawn from the Sun to any planet will sweep out equal areas in equal times

- Area from A to B and C to D are the same



12.4.3 Kepler's Third Law

The square of the orbital period of any planet is proportional to cube of the average distance from the Sun to the planet.

$$T^2 = Ka^3 \tag{12.9}$$

$$K_s = \frac{4\pi^2}{GM_s} \tag{12.10}$$

T is the period of the planet a is the average distance from the Sun. Or a is the length of the semi-major axis.
For orbit around the Sun, $K = K_S = 2.97 \times 10^{-19} s^2/m^3$
K is independent of the mass of the planet

Example 12.3 Calculate the mass of the Sun noting that the period of the Earth's orbit around the Sun is $3.156 \times 10^7 s$ and its distance from the Sun is $1.496 \times 10^{11} m$.
Solution

$$T^2 = \frac{4\pi^2}{GM_s} a^3$$

$$M_s = \frac{4\pi^2}{GT^2}a^3 = 1.99 \times 10^{30} kg$$

Exercise 12.4 The planet Mercury travels around the Sun with a mean orbital radius of $5.8 \times 10^{10} m$. The mass of the Sun is $1.99 \times 10^{30} kg$. Use Newton's version of Kepler's third law to determine how long it takes Mercury to orbit the Sun. Give your answer in Earth days 88.17 days

Exercise 12.5 The mass of Earth is 5.97×10^{24} kg, the mass of the Moon is 7.35×10^{22} kg, and the mean distance of the Moon from the center of Earth is 3.84×10^5 km. Use these data to calculate the magnitude of the gravitational force exerted by Earth on the Moon. 1.99×10^{20} N

12.5 Escape Speed

The escape speed is the speed needed for an object to soar off into space and not return.

$$E = K + U = \frac{1}{2}mv^2 - G\frac{M_E m}{r} = 0 \tag{12.11}$$

For the earth, v_{esc} is about $11.2 km/s$
Note, v is independent of the mass of the object

$$v_{esc}\sqrt{\frac{2GM_E}{R_E}} \tag{12.12}$$

TABLE 7.2

Escape Speeds for the Planets and the Moon

Planet	v_e (km/s)
Mercury	4.3
Venus	10.3
Earth	11.2
Moon	2.3
Mars	5.0
Jupiter	60.0
Saturn	36.0
Uranus	22.0
Neptune	24.0
Pluto	1.1

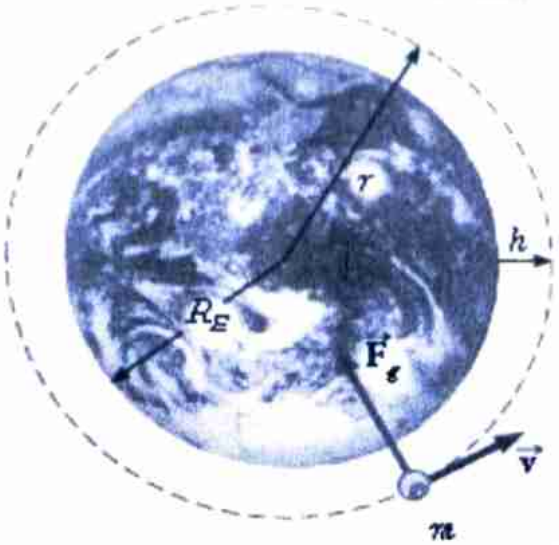
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Practice questions

Exercise 12.6 consider a man of mass $80kg$ standing $10m$ from a woman with a mass of $65kg$. The attractive gravitational force between them would be? 3.47×10^{-9}

Exercise 12.7 The mass of the Earth is $5.98 \times 10^{24} kg$, the mass of the Moon is $7.35 \times 10^{22} kg$, and the Earth and Moon are $3.8 \times 10^{8} m$ apart. The gravitational force between the Earth and Moon is?

Exercise 12.8 Consider a satellite of mass m moving in a circular orbit around the Earth at a constant speed v and at an altitude h above the Earth's surface as illustrated in Figure.



(i) Determine the speed of the satellite in terms of G, h, R_E(the radius of the Earth), and M_E(the mass of the Earth).
(ii) If the satellite is to be *geosynchronous* (that is, appearing to remain over a fixed position on the Earth), how fast is it moving through space?

Exercise 12.9 A planet has two moons of equal mass. Moon 1 is in a circular orbit of radius r. Moon 2 is in a circular orbit of radius 2r. What is the magnitude of the gravitational force exerted by the planet on Moon 2? $F_1 : F_2 = 4 : 1$

Exercise 12.10 Superman stands on top of a very tall mountain and throws a baseball horizontally with a speed such that the baseball goes into a circular orbit around the Earth. While the baseball is in orbit, what is the magnitude of the acceleration of the ball?

Exercise 12.11 Using the known radius of the Earth and that $g = 9.80 m/s^2$ at the Earth's surface, find the average density of the Earth.

Exercise 12.12 Calculate the escape speed from the Earth for a $5.0 kg$ spacecraft and determine the kinetic energy it must have at the Earth's surface to move infinitely far away from the Earth.

12.6 Work and energy

The concepts of work and energy are closely tied to the concept of force because an applied force can do work on an object and cause a change in energy.

12.6.1 Work

Changes in an object's motion depend both on force and how long the force acts. "How long" meant time. The quantity is called ***impulse*** "force x time". But "how long" does not mean time. It can mean distance also. When the concept of force x distance is consider, we are talking about an entire different concept - **work**. Work is the effort exerted on something that will change its energy. Work is done on an object when an applied force moves it through a distance. In our everyday language, work is related to expenditure of muscular effort, but this is**not**the case in the language of physics. A person that holds a heavy object does no physical work because the force is not moving the object through a distance. Work, according to the physics definition, is being accomplished while the heavy object is being lifted or pushed as shown in Figure 3.1a but not while the object is stationary Figure 3.1b. Two things enter the picture whenever work is done: (1) application of a force and (2) the movement of something by that force.



Figure 12.4: work is done by pushing the car



When a force F moves its point of application through a distance d, work is said to be done as shown in figure 3.2.
Work = force $\times$ distance

$$W = Fd$$

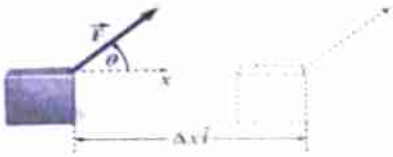
The work done by the force when its point of application undergoes a displacement d is given by

$$W = (F\cos\theta)d \tag{12.13}$$

It is the scalar (dot) product of **F** and **d** as shown in equation 3.2

$$W = Fd\cos\theta \tag{12.14}$$

Where θ is the angle between **F** and **d**.
Thus, work done is the algebraic scalar quantity which can either be positive or negative. The S. I unit of work done is Joule and denoted by J. 1 joule is the energy transferred when a force of 1 Newton is applied to an object and moves it through a distance of 1 meter.



Note: work is positive if $0 < \theta < 90°$
work is positive if $90° < \theta < 180°$
$W = Fd\cos\theta = -Fd\cos\theta$
Work = 0 if $\theta = 90°$
i.e the force is perpendicular to the displacement

Example 12.4 a body of mass $5.0Kg$ is drawn at a constant speed through a distance of $6m$ along a horizontal floor by a rope exerting a constant force of magnitude $10N$ at an angle $30°$ with the horizontal. Calculate;

1. the work done by the rope on the block
2. the work done by friction on the block
3. the total work done
4. the coefficient of kinetic friction between the block and floor

Solution

1. $W = Fd\cos\theta = (10)(6)\cos 30$
$W = 52.0J$

2. $W = F_f dCos180 = (10\cos 30)(6)\cos 180$
$W = -52.0J$

3. Total work done = work done by extension + work done by frictional force
4. $W = (52.0 + (-52.0)) = 0J$
5. $\mu_k = \frac{F_f}{N} = \frac{10\cos 30}{N}$
Vertically $N + \quad in30 - 49$
N= 44 N
$\mu_k = \frac{10\cos 30}{44} = 0.20$

■

12.7 Mechanical Energy

When work is done to wind a spring mechanism, the spring acquires the ability to do work on various gears to run a clock, ring a bell, or sound an alarm. Something has been acquired that enable the spring to do the work. This **"something"** that enables an object to do work is energy.

Energy is the capacity to perform work. Energy and Work are both measured in the same unit, Joule (J). Mechanical energy can be in the form of potential energy (energy due to position of something) or kinetic energy (energy due to movement of something) or the sum of the two.

12.7.1 Potential Energy

An object may store energy by virtue of its position. The energy that is stored and held in readiness is called Potential Energy (PE) because in the stored state it has the potential for doing work. A

stretched rubber band has potential energy because of the relative position of its part. If the rubber band is part of a slingshot, it is capable of doing work as shown in figure 3.3.

Work is required to elevate objects against Earth's gravity. The potential energy due to elevated positions is called gravitational potential energy. The amount of gravitational potential energy possessed by an elevated object is equal to the work done against gravity in lifting it. The upward force required while moving at constant velocity is equal to the weight, mg, of the object, so the work done in lifting it through a height h is

Gravitational potential energy = weight x height

$$PE = mgh$$

12.7.2 Kinetic Energy

If you push an object, you set it in motion. If an object is moving, then it is capable of doing work. It has energy of motion, that energy is kinetic energy. The kinetic energy of an object depend on the mass of the object as well as its speed. $kinetic\&energy = \frac{1}{2}mass\&x\&speed^2$

$$KE = \frac{1}{2}mv^2$$

If a variable force causes a displacement from 0 to point x, then the work done is given as;

$$dW = Fdx$$

$$W = \int_0^x Fdx = \int_0^x \frac{mdv}{dt}dx = \int_0^v mvdv$$

$$KE = \frac{1}{2}mv^2$$

Work - Energy Theorem

Work - Energy theorem states that the work done by the resultant external force acting on a body is equal to the change in kinetic energy of the body.

$$W = K.E_f - K.E_i = \Delta KE$$

Work equals change in kinetic energy. The work in this equation is the net-work (work based on the net force).

Kinetic energy and potential energy are two among many forms of energy, and they underlie other forms of energy, such as chemical energy, nuclear energy, and energy carried by sound and light.

■ **Example 12.5** A proton accelerated at $3.6 \times 10^{15} ms^{-2}$ through a distance of $3.5cm$. If the initial velocity is $2.4 \times 10^7 ms^{-1}$, find the change in kinetic energy at the end of the distance. $(M_o = 1.67 \times 10^{-27} kg)$. **Solution**

$$\Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

Using $v^2 = u^2 + 2as$

$$= (2.4 \times 10^7)^2 + 2x3.6 \times 10^{15} x0.035$$

$$= 3.28 \times 10^{14} ms^{-2}$$

$$\Delta KE = \frac{1}{2}1.67 \times 10^{-27}(3.28 \times 10^{14} - 5.76 \times 10^{14})$$

$$\Delta KE = 2.1 \times 10^{13} J$$

Example 12.6 A body of $2Kg$ has initial velocity of $8ms^{-1}$ in the +X direction, as it passes the origin of the coordinates. It is subjected to a retarding force such that $F_x = -0.5N$. What will be its X coordinate when it stops?

Solution

From Work-Energy theorem

Work done = Change in KE

$$Fd = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$(0.5)d = 0 - \frac{1}{2}2 \times 8^2$$

$$d = 128m$$

Conservation of Energy

We can better understand the processes and changes that occur in nature if we analyze in terms of energy changes (transformation from one form into another, or transfer from one location to another). The study of various form of energy and their transformation from one form to another has led to one of the greatest generalization in physics - the law of conservation of energy.

Energy cannot be created or destroyed; it may be transformed from one form into another, but the total amount of energy never changes.

In a mechanical system, the principle of conservation of energy will imply that in the absence of any dissipated forces like friction, the total energy is constant.

For example, when a body falls from a height, its potential energy is being converted to kinetic energy such that

K.E gained = P. E lost

$$\frac{1}{2}mv^2 = mgh$$

Illustration of conservation of mechanical energy

Suppose a body is released from rest at height h above the floor as shown in the diagram below, the total energy at this height is totally equal to potential energy.

As it falls its potential energy decreases but its kinetic energy increases to compensate so that the sum of the two remain constant (at the midpoint, potential energy equal kinetic energy).

Let us calculate the energies PE and KE when the body has fallen half the distance



$PE = mg\,h/_2$ and $KE = \frac{1}{2}\,m\,v^2$

the total energy at this position is given as:

$$E = \frac{1}{2}\,m\,v^2 + mg\,h/_2$$

By the principle of conservation of energy

$$\frac{1}{2}\,m\,v^2 = mgh$$

Immediately before the body strike the ground, its potential energy is zero and its total energy is

$$E = \frac{1}{2}mv^2$$

$$mgh = \frac{1}{2}mv^2$$

The value of KE is equal to the original PE of the body. Thus for free falling body, PE + KE is a constant.

▪ **Example 12.7** A body uses a rope attached to the top of a ceiling to swing forth and back. The lowest and highest points of the swing is 1.5 m and 2.5 m above the ground respectively. What is the body maximum speed?

Solution

At the highest point the body has only PE (the body is momentarily at rest) $PE = mgh$
At the lowest point, he has both KE and PE, by energy conservation principle

$$\frac{1}{2}mv^2 = mgh$$

$$\frac{1}{2}mv^2 = mg(h_0 - h_1)$$

$$v = \sqrt{2g(h_0 - h_1)}$$

$$v = \sqrt{2x9.8x(2.5 - 1.5)} = 4.4\ \text{ms}^{-1}$$

▪

12.8 Tutorial 11

Problem 12.1 A body whose mass is 50 Kg is raised to a height of 2 m above the ground. What is its potential energy? If the body is allowed to fall, find the kinetic energy (a) when half-way down (b) just before impact with the floor

Problem 12.2 A trailer is pulling a car with a force of 900 N on a level road. The car is moving at 70 Km/h. How much work is done by the trailer on the car in 15 mins.

Problem 12.3 The brake of a $1600Kg$ vehicle traveling at $15ms-1$ on a level road are applied long enough to do 90 KJ work. Find the speed of the vehicle. What is the amount of work required to stop the vehicle?

[1] This book was written and compiled using LaTeX2.0